APPLIED MATERIALS®

Applied AutoMod® 14.0.1

User's Guide

think it. apply it.

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About This Guide

To help you find the right information about using the AutoMod software, this section briefly describes the purpose, intended audience, prerequisites, organization, and typographic conventions used in this guide. It also lists additional documentation that can help you learn more about AutoMod products.

Purpose

This guide explains how to run an existing simulation in the AutoMod software and how to experiment with and analyze the simulation using AutoStat.

This guide is not intended to teach general simulation concepts and practices. Use this book in combination with the other printed and online documentation that accompanies the software and with the training classes offered by Applied Materials, Inc.

Audience

This guide is designed for an analyst or facility administrator who wants to run and analyze an existing simulation in AutoMod. If you want information about how to build a new model in AutoMod, refer to the *AutoMod User's Guide*.

Prerequisites

This book assumes a basic familiarity with personal computers and with the Windows operating system. The book also assumes an understanding of simulation concepts and practices. If you are not familiar with simulation, consult the following books:

Discrete-Event System Simulation, 2nd edition, Jerry Banks, John S. Carson, and Barry L. Nelson, Copyright 1996, Prentice-Hall, Inc. — an introductory text.

Improve Quality and Productivity with Simulation, Thomas J. Gogg and Jack R. A. Mott, Copyright 1992, JMI Consulting Group — an intermediate text.

Simulation Modeling and Analysis, Second Edition, Law and Kelton, Copyright 1990, McGraw-Hill, Inc. — an advanced text.

Organization

This book contains the following chapters:

Chapter	Purpose
"Introduction"	Describes AutoMod's interface controls and options.
"Running a Model"	Provides instructions for running a model and interpreting simulation statistics, as well as guidelines for debugging and verifying a model.
"Business Graphs"	Explains how to create business graphs to illustrate simulation statistics.
"Printing"	Explains how to plot or print a view of simulation graphics from within AutoMod.
"Analyzing a Model with AutoStat"	Describes how to apply proper statistical sampling techniques to your model in order to accurately estimate performance under random conditions. Also explains how to conduct experiments and create graphs of model responses to assist you in analyzing simulation results.
"The AutoMod File System"	Describes the directories that are installed with the AutoMod software and the files and directories that comprise an AutoMod model.

Conventions

This guide uses the following conventions:

Bold text indicates:

- Input you can type using the keyboard.
- Names of dialog box entities.

The `monospace font` indicates:

- Input you can type using the keyboard.
- AutoMod syntax (function names, actions, keywords, and so on).
- Text that the software displays.
- The contents of a file.

<>

The left and right angle brackets (<>) indicate that you substitute the information requested by the word or words enclosed by the brackets. For example, in the following command, you type the `<model_data_file_path>` appropriate for your system.

```
MultiModelSync -me <model_data_file_path>
```

If the path to the model data file on your machine is C:\Models\Mymodel\Data.txt, the command is

```
MultiModelSync -me C:\Models\Mymodel\Data.txt
```

Additional information

Use the following sources for obtaining more information about Applied Materials products:

Reference	Purpose
AutoMod User's Guide	Explains how to build, run, and analyze a model using the AutoMod software. The book also explains how to design model graphics using ACE, animate model graphics using kinematics, create movies of simulation activity using AutoView, and conduct experiments and model analysis using AutoStat. In addition, the book contains a glossary of software terms and appendices that explain the purpose of model files and list the reserved keywords that you cannot use when naming model entities.
AutoMod Help	Explains how to write model logic and use the AutoMod editor. The online help also discusses how to use AutoMod utilities including AutoStat, AutoView, the Model Zip Archiver, the OPC Utility, the Process Server, and the Web tools. The help includes information on how to create a custom model interface using ActiveX controls and provides a reference library for previewing and selecting graphics from the graphic templates installed with the AutoMod software.
AutoMod Tutorials	Provide instructions for building a model in AutoMod, designing a vehicle graphic in ACE, and animating vehicle and robot graphics using kinematics.
Getting Started with AutoMod	Teaches how to use the AutoMod and AutoStat software through examples and chapter exercises.
Model Communications User's Guide	Explains how to run models as a combined simulation on networked machines, control models from third-party applications, and communicate with an OPC server to test a control system by replacing system hardware with an AutoMod simulation.

1 Using the Interface

The AutoMod™ Runtime License has two versions: a version that includes AutoStat™, a statistical analysis package, and a version that does not. This guide contains the information necessary for both versions of the license.

The AutoMod Runtime License only works with AutoMod models; it does not run models built with either AutoSched™ or the Simulator system.

To use a model with a runtime license, you need:

- The executable file (either <modelname> or <modelname>.exe)
- The <modelname>.dir directory
- External data files (if used)
- Optional environment settings file (.asilibrc file).

Note All of these files and directories must have been compiled with the same AutoMod version as the runtime license.

What you will learn

Each chapter in the guide focuses on the different features and options available to you as a runtime user. To get the most information out of your model and gain a complete understanding of the AutoMod simulation environment, you should read through the entire guide.

The chapters are:

- **Introduction** - explains the basic AutoMod interface and view controls.
- **Running a Model** - explains runtime menus, dialog boxes, keyboard commands, statistics and reports.
- **Business Graphs** - explains the types of graphs available, as well as how to create graphs, use statistics, and dictate the frequency of reports.
- **Printing** - explains how to use the printing menus and dialog boxes, how to create/send a data file, and how to get screen captures.
- **Using AutoStat** - explains the basics of using AutoStat to analyze your runtime model. (You only need to read this chapter if you purchased the AutoStat runtime option.)

There is also one appendix: Appendix A, **The AutoMod File System,** explains the files that comprise an AutoMod model.

The mouse

This guide often refers to the main mouse button. Unless you have changed your system defaults, the *main* mouse button is the *left* mouse button.

The main mouse button has multiple functions: selecting entities for editing, clicking buttons, selecting items from menus, placing graphic entities, and controlling the view (as described in the sections below). The vast majority of work done with the mouse involves the main mouse button. When the text of this guide says to "click," it is referring to the main mouse button.

» **Tip** In addition to using the mouse to control the view (as described below), you can also zoom, rotate, and translate the view by using the View Control (see "View Control dialog box" on page 3), or by using shortcut keys (see the Shortcut Keys Help, in the AutoMod Help).

Centering and zooming

Use the main mouse button for zooming the current view:

- Hold down **ALT + CTRL + SHIFT**, while also holding down the main mouse button, and drag the mouse up or down to zoom in or out.

Use the *right* mouse button for centering and zooming in on a specific area:

- To *center* the model graphics at the cursor's position on the grid, place the cursor at the desired location and click the right mouse button.
- To *zoom* in on an area, place the mouse pointer in the AutoMod window, hold down the right mouse button, drag the mouse to "fence" an area, and release the mouse button.

» **Tips**

- You can undo the last zoom (or other view change) with the keyboard command **CTRL+SHIFT+U**.
- If you are using a three-button mouse, clicking the *middle* mouse button changes the view to a Top view.
- If you are using the 64-bit version of the AutoMod software, with a mouse that features a wheel button, you can also zoom the view by holding down the **ALT** key, while scrolling the wheel to zoom in or out.

Rotating and translating

To *rotate* the view, hold down the **ALT** key, while also holding down the main mouse button, and drag the mouse to rotate the view.

To translate the view, hold down **ALT + CTRL**, while also holding down the left mouse button, and drag the mouse to translate the view in any direction.

AutoMod dialog boxes

Two frequently used dialog boxes are View Control and Grid Control. The View Control dialog box allows you to manipulate the view of the model. The Grid Control dialog box allows you to set factors that affect the drawing grid.

Opening systems using keyboard commands

As an alternative to opening systems from the **System** menu, you can also open systems using keyboard commands by pressing CTRL+[0-9] (to open systems with an ID from 1-10) or CTRL+SHIFT+[1-9] (to open systems with an ID from 11-19). The following chart shows system identification numbers and the corresponding keyboard command used to open that system.

System ID number	0	1	2	.	9	10	11	12		19
Keyboard command	------	CTRL +1	CTRL +2		CTRL +9	CTRL +0	CTRL +SHIFT+1	CTRL +SHIFT+2		CTRL +SHIFT+9

System ID numbers and corresponding commands to open a system

Notes

- The numeric pad cannot be used to open a system.
- You cannot use a keyboard command to open a system whose ID is 20.

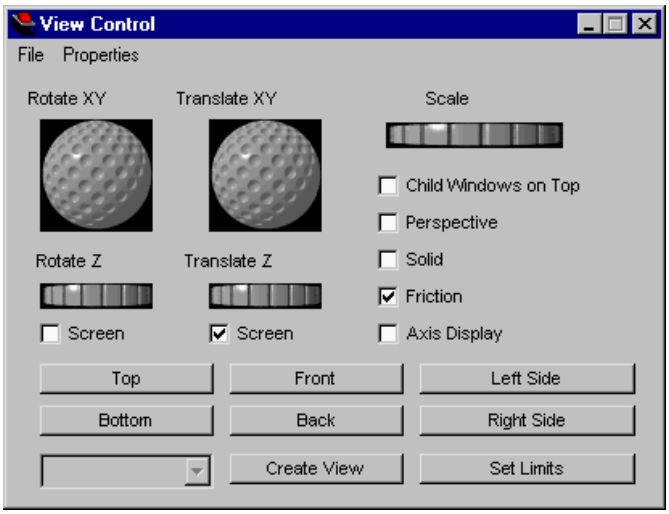
To define system ID numbers

1. On the **Model** menu, click **Display**.
2. Select a system and type a number from **1** to **20** into the **System ID** text box.

You can also use that keystroke to toggle the system's graphic display on and off in either the edit or simulation environment. For more information, see "Toggling system visibility" on page 67.

View Control dialog box

Click **View Control** from the AutoMod window to manipulate your view of a model.



Tip In addition to the View Control, you can also manipulate the view of a model by using the mouse (see "The mouse" on page 2), or by using shortcut keys (see the Shortcut Keys Help in the AutoMod Help).

The following options are available in the View Control dialog box:

Option	Function
Screen	Select **Screen** to control model movement in relation to the screen or world coordinates. When *screen mode* is selected, positive X extends from the center to the right of the screen, positive Y extends from the center to the top of the screen, and positive Z extends from the center "out" toward you. In screen mode, the axes are fixed and do not move, even if you rotate or translate the model. When *world mode* is selected (that is, when screen mode is off), altering the rotate and translate options alters the positions of the axes relative to you and to the screen, but positive X, positive Y, and positive Z remain the same in relation to the center of the model as when the model was first drawn. Therefore, with world mode selected, moving entities along the X axis does not always move them left and right across the screen.
Rotate XY	Rotates the model geometry around the X and/or Y axes. Position the pointer on the golf ball. Hold the mouse button and drag; your model rotates on the X and Y axes, giving you different views. Hold down **SHIFT** to move along one axis only (X *or* Y).
Rotate Z	Rotates the model geometry around the Z axis. No matter at which angle the model geometry appears, the screen axes remain the same (X is left and right across the screen, Y is up and down the screen, and Z is in and out of the screen). Therefore, in screen view, the operative Z axis is like a pole projecting out from the middle of the work window toward you. In world view, the operative Z axis remains in the same relation to X and Y as they exist in the "world" inside the model. If you take a model that was created in top view and switch to front view, the world Z axis extends from the top to the bottom of the screen and the model rotates around that axis.
Translate XY	Slides the drawing along the X and/or Y axes. With screen on, **Translate XY** moves the entire drawing up and down or left and right. With world on, the direction of movement depends on the current positions of the axes. Scroll the translate bar in the direction you want your model to move. Hold down **SHIFT** to move along one axis at a time; for example, along the X axis only.
Translate Z	Changes your viewpoint along the Z axis of your model. In world view, **Translate Z** slides the drawing along the Z axis, but the apparent direction of movement depends on the current positions of the axes.
Scale	Use the **Scale** dial to zoom the view toward you (to see details) or away from you (to see the entire model).

The View Control dialog box offers several predefined views. Click the buttons in the View Control dialog box to view a model from the angles listed here:

View	Function
Top	View the model from the positive Z axis.
Front	View the model from the negative Y axis.
Left Side	View the model from the negative X axis.
Bottom	View the model from the negative Z axis.
Back	View the model from the positive Y axis.
Right Side	View the model from the positive X axis. You can revert to the last view of your AutoMod window by using Undo. The keyboard command is: **CTRL+SHIFT+U**. Undo reverts one step back; that is, if you translated and then rotated, undo undoes the rotation only.

The View Control dialog box displays five check boxes.

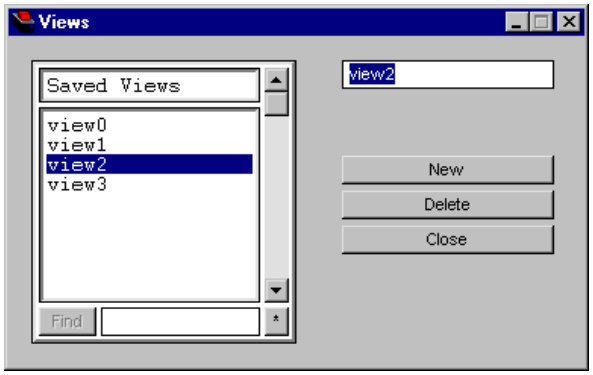
Check Box	Function
Child Windows on Top	Prevents palette and dialog boxes from getting lost behind other windows.
Perspective	Select or clear **Perspective** to toggle the projection type between perspective and orthogonal, respectively. A *perspective* view shows all lines leading to a vanishing point; this is how we naturally view the world. An *orthogonal* view shows all lines at right angles to each other; this is a projection of a 2-D drawing into a 3-D view.
Solid	Select **Solid** to view model geometry as solid entities (the default mode). Clear **Solid** to view model geometry as wireframe entities.
Friction	Toggles continuous movement on and off. Select **Friction** to cause model graphics to move only when you indicate. When **Friction** is cleared, any movement command (for example rotating, translating, or scaling the current view) causes model graphics to move continuously until you stop them by turning friction on or by pressing the space bar.
Axis Display	Select **Axis Display** to display a triad at the model origin. This triad indicates the world coordinate axes.

» **Tip** Many of the viewing changes listed in the preceding can also be accomplished using the keyboard. For a list of keyboard commands, refer to "Using keyboard commands" on page 68.

Note The "Keyboard Commands" help file lists all of the AutoMod keyboard shortcuts. To open the help file, click **Help > Keyboard Help**.

Finally, buttons in the View Control dialog box allow you to create views and set the model's drawing limits.

The **Create Views** button opens the Views dialog box.

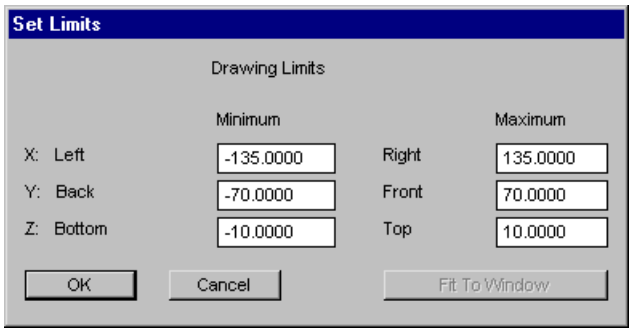


This dialog box allows you to save the currently displayed view, or delete a previously saved view. To save the current view, click **New** to create a saved view; you can then edit the view's name by typing a new name in the text box and pressing **ENTER**. You can also create a view by editing an existing view and clicking **New**.

The **Set Views** list allows you to set the current view to any of the views that you have saved. Previously saved views are defined in a binary file called .amodviews. This file is saved in the current working directory. Models in the same directory share this file.

You can also use a keyboard command to create and retrieve specific model views. For more information on the Default Views keyboard command, see "Using keyboard commands" on page 68.

The **Set Limits** button opens the Set Limits dialog box, which allows you to set the model's drawing limits.



The drawing limits define the size of the drawing grid in the AutoMod window. When you open a model, the drawing limits are automatically resized to encompass the graphics in the model. To increase the size of the drawing grid, set lower minimum and higher maximum values for the drawing limits.

Click the **Fit to Window** button to cause the selected entities to be enclosed by the grid and fit to the window. **Fit to Window** does not fit entities that have been enlarged to such an extent that they have disappeared, but if you have translated an entity out of view, **Fit to Window** returns it to view.

View control menus

The **File** menu allows you to save the current view in the AutoMod window as a graphic file. The supported graphic file formats are described in the following table:

View Control Properties menu

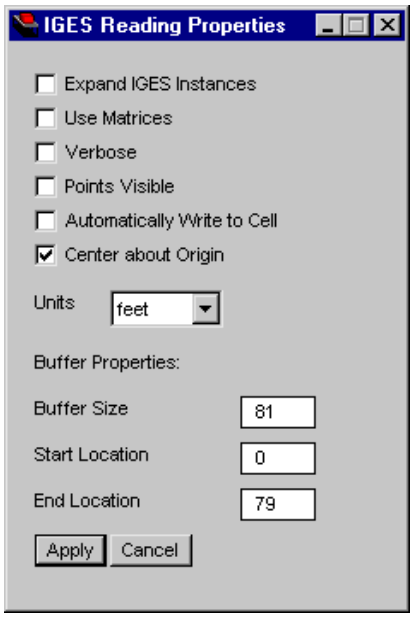
Option	Action
Save Picture to HSF File	Saves the current view in the HOOPS Stream File (HSF). HSF files can be opened and edited using the AutoMod Graphic Viewer.
Save Picture to VRML File	Saves the current view to a VRML 1.0 file. VRML (Virtual Reality Modeling Language) is a specification for displaying 3-D objects on the Web. VRML files have a .wrl extension (short for "world") and can be viewed using a VRML browser or a Web browser with a VRML plug-in.
Save Picture to VRML 2.0 File	Saves the current view to a VRML 2.0 file (also known as a VRML97 file). This is a newer version of the VRML file format.
Save Picture as JPEG File	Saves the current view to a JPEG file. JPEG (Joint Photographic Experts Group) is a popular compression technique for 2-D color images; use this format if you want to present a 2-D image on a Web site. JPEG files are saved with a .jpg extension.
Save Picture as PDF File	Saves the current view to a PDF file. PDF (Portable Document Format) files can be viewed using a third-party PDF file viewer.
Save Picture as PostScript File	Saves the current view to an encapsulated PostScript file. PostScript files have an .eps extension and can be viewed or printed using third-party utilities.
Copy to Windows Clipboard	Copies the current view to the Windows clipboard.

The **Properties** menu allows you to set preferences for reading and writing IGES files.

! **Important** The following table shows the valid IGES data types that can be read in AutoMod. If you read an IGES file that contains a data type not listed here, AutoMod ignores the unrecognized entity.

Entity #	Entity Type
100	Circular arc
102	Composite curve
106	Copious data: - Centerline - Section
110	Line
116	Point
212	General note (text)
308	Subfigure definition entity
408	Singular subfigure instance entity

The **Reading IGES** option in the **Properties** menu opens the IGES Reading Properties dialog box, shown here:



Options in the IGES Reading Properties dialog box are defined as follows:

Option	Action
Expand IGES Instances	When **Expand IGES Instances** is selected, each object in the IGES files is read into AutoMod as its own entity, allowing you to edit each individual shape. If **Expand IGES Instances** is cleared, the original data structures of the IGES file are maintained.

Option	Action
Use Matrices	IGES files can position entities within a 3x3 matrix of rotations and XYZ translations. By default, cell fields hold the corresponding orders as follows: • Rotations in ZYX order • XYZ translations • XYZ scales. The **Use Matrices** option causes a 4x4 matrix representation to be saved in the cell file instead of the preceding decomposed values.
Verbose	Displays a series of periods to indicate the progress of the conversion process.
Points Visible	By default, single point coordinates are *not* read into AutoMod files as visible entities. If you have single point coordinates in your IGES file and want to view them in the AutoMod model, select this box.
Automatically Write to Cell	Automatically converts the IGES file to an AutoMod cell file.
Center About Origin	By default, IGES files are placed at the model origin when imported into a model. **Note** Deselecting **Center About Origin** causes the file placement to be determined by the IGES file, which could result in placement outside of the current grid.
Units	Specifies the unit of measure for the IGES file that you are reading into AutoMod.
Buffer Size	Specifies the length of the line in which the IGES file was written. Lines are usually written in 80 characters. The default for this field is 81.
Start Location	(Used in conjunction with the buffer size) Specifies the first valid column in a line of data. The default is 0, which is the first column.
End Location	(Used in conjunction with the buffer size) Specifies the last valid column in a line of data. The default is 79, which is the last column of data for an 80 character buffer.

The **Writing IGES** option opens the IGES Writing Properties dialog box, shown here:

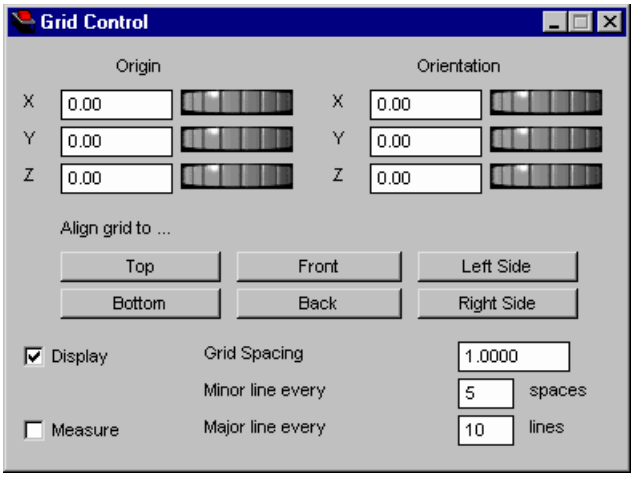
The IGES Writing Properties dialog box contains the following options:

Option	Action
Convert to Meters	Converts the unit of measure to metric units in the IGES file.
Write Invisible	Writes the graphic data for items that have been made invisible to the IGES file.
Verbose	Displays a series of periods to indicate the progress of the conversion process.
Lines Only	Converts all curves, text, and so on to line representations. Curves are represented by a series of small lines. Text is represented at its proper scale and in an approximation of the selected font. Once converted into lines in the IGES file, curves and text lose their meanings as unified shapes or letters, but can be used as guides by the CAD operator.
Start Section	Opens the Start Section Data dialog box. The start section contains text information about the file, such as the author and company.
Write to level	Type a number between 1 and 255 to indicate the layer of the IGES file to which you want to write the current file.

Grid Control dialog box

The grid appears by default whenever you open AutoMod. In the edit environment, the spacing and dimensions of the drawing grid reflect the distance units defined for the system that is currently open. In the runtime environment, the grid uses the distance units defined for the model (the top-level process system).

To edit the drawing grid, click **Grid Control** in the AutoMod window.



Option	Function
Origin/Orientation	The options that appear under **Origin** and **Orientation** enable you to adjust the position of the grid. Use the dials to move the grid origin to any position. Think of the **Origin** as translational grid movement, and **Orientation** as rotational grid movement.
Align Grid to...	Allows you to move the grid to any one of six predefined positions: Top, Bottom, Right Side, Left Side, Front, or Back. This feature enables you to input graphic information on different axes.
Display	Toggles the grid display on and off.
Measure	Opens the Measurement window, which allows you to measure the distance between two points in the AutoMod window.
Grid Spacing	Specifies the number of distance units between successive grid lines: • **Minor line -** the distance between minor lines (fine grid lines). This distance is calculated by multiplying the minor line value by the default grid spacing value by the units of distance selected for the system you are currently editing. **Note** When displaying the drawing grid in the Simulation environment, the distance between grid lines is measured using the default distance units defined for the model (the top-level process system). If you are unsure what default units the model is using, check with the model builder. • **Major line -** the spacing of major lines (heavy grid lines) in terms of the number of minor lines between each major line.

Changing the font for displayed text

By default, AutoMod programs use a sans serif font (sanserif.glf) for displaying on-screen text such as user-defined labels, entity labels (such as resource and queue names), business graph labels, and so on.

Several additional font files are installed in the font directory in the AutoMod installation directory. The name and a sample of each of the available font files is provided here:

italic.glf

The quick brown fox jumped over the lazy dog

oldeng.glf

The quick brown fox jumped over the lazy dog

outline.glf

The quick brown fox jumped over the lazy dog

roman.glf

The quick brown fox jumped over the lazy dog

roman2.glf

The quick brown fox jumped over the lazy dog

sanserif.glf

The quick brown fox jumped over the lazy dog

To use a particular font file in the software, the file must be named with one of the names shown in the following table:

Font name	Affects text in:
automod.glf	AutoMod (excluding business graphs and graphics templates)
autoview.glf	AutoView
ace.glf	ACE
asi.glf	Business graphs, the DTrace utility, and graphics templates

To change the default font

1. In the fonts directory, delete the font file that affects the text whose font you want to change (as shown in the table).

   For example, if you want to change the font used in AutoMod, delete the file automod.glf.

2. Copy the font file you want to use.
3. Rename the copied font file with the name of the font you deleted in step 1.

   For example, if you want to use the Roman font (roman.glf) in AutoMod, you would copy the file roman.glf, then rename the copied file automod.glf.

4. If the program for which you are changing fonts is currently open, restart the program.

The program now uses the selected font.

5.

2 Running a Model

In this chapter, you will learn how to:

- Run your model.
- Use the simulation environment menus.
- Control animation.
- Use keyboard commands and command line options.
- Display statistics.
- View the AutoMod report.
- Create Web output from a model.
- Change the run control (run length, reporting settings, and so forth.)
- Use alarms, breakpoints, and pauses.

Upon completion

When you have read this chapter and understand how to use the information discussed in it, you will be ready to run your model to verify its accuracy and then test your system.

Introduction

In order to run a model using a runtime license, you must have a model that has been built and compiled with a full version of the AutoMod software. Running a model allows you to compare the model's performance against your estimates and expectations. This analysis takes place in the simulation environment.

Running a model

To run a model using a runtime license, you must have a model that has been built with a full version of the AutoMod software. You will need:

- The executable file (either <modelname> or <modelname>.exe)
- The <modelname>.dir directory
- External data files (if used)

All of these files and directories must have been compiled with the same AutoMod version and build as the runtime license.

There are two ways to run a model:

- From within Windows
- From a command prompt

Running a model from within Windows

To run a model from within Windows

1. Navigate to the model directory (the model directory contains the <modelname>.dir directory).
2. Double-click the file <modelname>.exe. The Simulation window opens. By default, the simulation is initially paused.
3. Click **Control > Continue** to start the simulation.

Running a model from a command prompt

When you run a model from a command prompt, you can use command line options to change the default run conditions. For example, you can turn graphics off or create a trace file of simulation events.

To run a model from a command prompt

1. Open a command prompt and change directories to the model directory (the model directory contains the <modelname>.dir directory).
2. Type the following at the command prompt:

   ```
   <modelname> -<command_line_options>
   ```

   For example:

   ```
   forklift -nt
   ```

Note Command line options are case-sensitive.

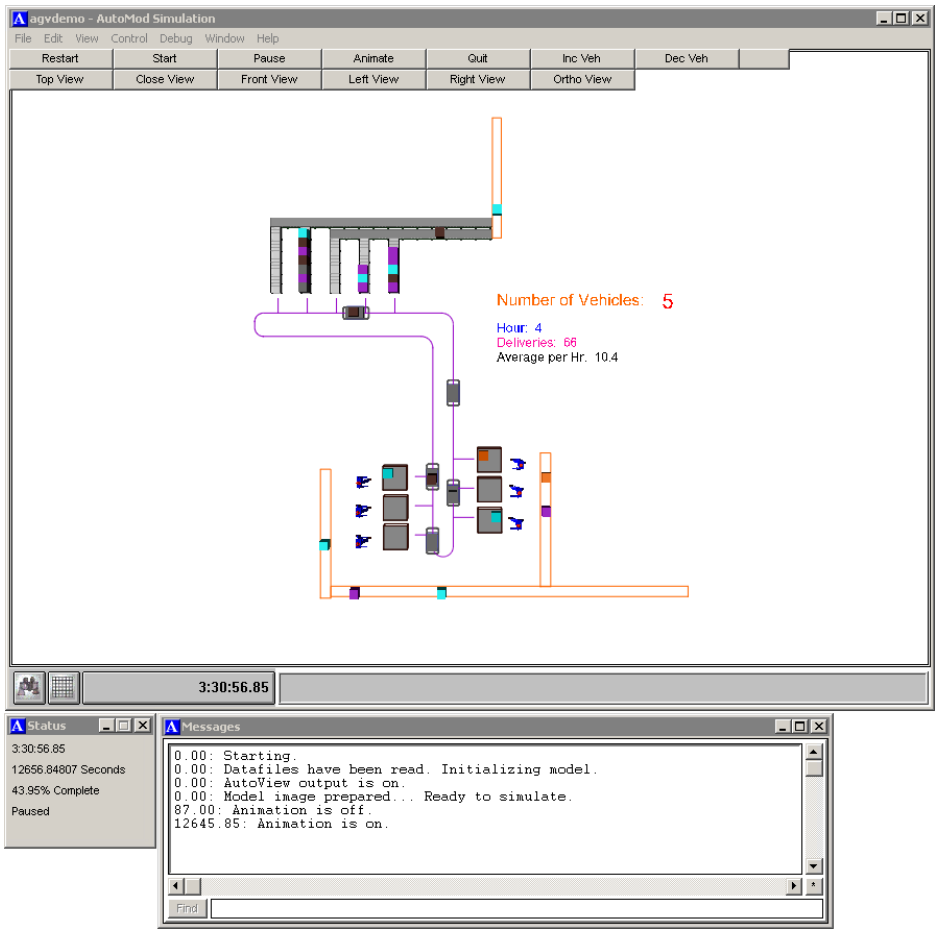
These command line options are available when running a model:

Command line option	Description
B	**(tab-delimited reports)** Generates tab-delimited report files (.rep files) of model statistics in the model directory.
b	**(tab-delimited reports)** Same as the **-B** option, but does not produce a .replist file.
c	**(do not compile pictures)** Turns off graphic compiling when the model is run, resulting in smoother shapes and more detailed graphics. This option results in a slower simulation.
d	**(animation off)** Turns the animation off. The model is paused at the beginning of the simulation; to automatically continue a model with animation off, use the **-n** option.
D	**(debugger)** Turns the animation off until the model reaches the first stop set in the Debugger.
i	**(no interaction)** Starts the simulation with graphics displayed. The simulation runs until the end of the user-defined run control and then closes.

Command line option	Description
L	**(solids off)** Makes all graphics wireframe (even if they are defined as solid).
n	**(no graphics)** Starts the simulation without graphics. The simulation closes at the end of the user-defined run control.
r	**(redirect output files)** Redirects all output files to a specified .nrd directory (named run directory) in the model directory. To run a model named `kitting`, for example, and send output files to a Jan01.nrd directory, you would use this command: `kitting -r Jan01` Redirects the message file, print file, and all report files (.rep, .report, and .replist). Also redirects AutoView files and any files generated by syntax. Generates .rep files, whether or not the **-B** or **-b** options are used. Generates full reports, regardless of the model's report settings.
R	**(tab-delimited reports)** Generates tab-delimited report files (.rep files) of model statistics. Report files (full statistics) are placed in the model directory. (Same as **-b**, but generates full reports, regardless of the model's report settings.)
t	**(trace on)** Creates a trace file of load and movement system events (<modelname>.debug).
w	**(no windows)** Starts the simulation without any visible windows. The simulation runs until the end of the user-defined run control and then closes. This option greatly accelerates performance, but you cannot view or interact with the model during the simulation.

The simulation environment

Models run in the Simulation window.



The Status dialog box shows the simulation time and indicates whether the simulation has been paused or is continuing (running). It also indicates the ratio between simulation time and real time.

Note The Status dialog box displays clock units in the time units defined for the top-level process system in the model. For example, if the top-level process system uses seconds as its default time unit, the clock displays in seconds. If you are not sure which default time units the model is using, check with the model builder.

The Messages dialog box shows errors, warnings, alarms, breakpoints, and single steps for the model run, and also prints messages from the simulation.

To save the location of these windows on the screen, click **File > Save Startup Config(uration)**.

Simulation environment menus

The menus and their options in the simulation environment are different from those in the edit environment. The simulation environment includes the following menus: **File** menu, **View** menu, **Control** menu, **Debug** menu, **Window** menu, and **Help** menu.

Using the File menu

The **File** menu includes options that relate to opening a built (compiled) model.

Menu option	Function
Open	Opens a simulation.
Save Startup Config(uration)	Saves your window configuration. To use Save Startup Config(uration), organize the placement of the windows on the screen in the desired manner (that is, resize or relocate the windows, select options in the View Control, and set the system display preferences). After your preferences are set, select **Save Startup Config**. A new .am2rc file is created in your current directory containing the instructions for setting up simulations on your computer. This means that when you open a simulation within the directory containing the .am2rc file, your configuration is automatically set to the configuration you saved. » **Tip** You can cause all of your models to use the saved configuration by moving the .am2rc file to your HOME directory (you set the HOME directory when you install the AutoMod software).
Run Model Editor	Closes the simulation environment and opens the edit environment so you can edit the model. ■ **Note** You must have purchased and authorized the full version of AutoMod to select this option.
Run AutoStat	Closes the simulation environment and runs the current simulation in the AutoStat software. ■ **Note** You must have purchased and authorized the AutoStat software to select this option.
Exit	Closes the simulation environment and exits the software.

Using the View menu

The **View** menu includes options for configuring the display, viewing business graphs and statistical reports, finding entities, and displaying or hiding the Messages and Status windows.

Menu option	Function
System Display	Opens the Display dialog box, in which you can display or hide system and individual entity graphics. This option also allows you to toggle the display of graphic templates and change the color of system entities, such as guide path, conveyor sections, and so on. For more information, see "Display options" on page 64.
Business Graphs	Opens the Business Graphs dialog box, in which you can create and display graphs of entity statistics. For more information, see Chapter 3, "Business Graphs."

Menu option	Function
Reports	Opens the Reports dialog box, in which you can display the value of entity statistics. For more information, see "Displaying entity reports" on page 75. The **Reports** option is followed by three submenu options that you can use to display statistical information about loads, variables, or events: • **Loads** - Displays status information for loads in the simulation. For more information, see "Load statistics" on page 88. • **Variables** - Displays the value of user-defined variables in the simulation. For more information, see "Variable values" on page 105. • **Events** - Displays the contents of the current and future events lists. For more information, see "Event statistics" on page 86.
Finder	Opens the Finder dialog box, in which you can select entities to find in the Simulation window. This option also allows you to display reports and attributes of selected entities. For more information, see "Finding graphical entities in a simulation" on page 70.
Message Window	Displays the Messages dialog box, which shows model messages, warnings, and errors.
Status Window	Displays the Status dialog box, which shows the current simulation time and indicates whether the simulation is paused or is continuing (running).

Using the Control menu

The **Control** menu contains options that allow you to start and stop the simulation, vary the time between animation updates, set alarms, and single-step through your model.

Menu option	Function
Continue/Pause	Continues or pauses the simulation. Animation, business graphs, and statistics are automatically updated each time the model is paused.
Next Action	Advances the simulation to the next event in the model. By repeatedly choosing **Next Action**, you can single-step through model events.
Set Alarm	Pauses the simulation at a predetermined time. Setting an alarm allows you to check statistics at a specific point during the simulation.
Turn off Animate/Turn on Animate	This is a toggle option; once you have selected **Turn off Animate**, **Turn on Animate** appears, and vice versa. **Turn off Animate** stops the animation, but allows the simulation to continue. **Turn on Animate** displays the animation while the simulation continues.

Menu option	Function
Animation Step	Allows you to change the time between animation updates. The default time between updates is one time unit, so the screen display is updated every second of simulated time. **Note** You can also change the time between animation updates by calling the SetDisplayStep function. This function is useful for overriding the default animation step at the beginning of a simulation. For more information, see "SetDisplayStep function" in the AutoMod Help.
Restart Model	Stops the run and restarts the simulation from the beginning.

Using the Debug menu

The **Debug** menu contains options useful for testing a simulation and finding and resolving model errors. Because fixing model errors requires a full license of the AutoMod software, many of the debugging options are not discussed in this guide.

Menu option	Function
Open Debugger	Opens the Debugger dialog box, which displays the model's logic as it is executed during the simulation. Using the Debugger, you can step through the model logic one action at a time or set breakpoints for pausing the simulation before a specific action occurs. This option is available if your model was compiled with the Debugger.
Entity Tracing	Opens the Entity Tracing dialog box, in which you can add entities to the tracing list. During a simulation, messages are printed to a file that allow you to track simulation events involving the traced entities.
Breakpoints	Opens the Breakpoints List dialog box, in which you can view and delete the breakpoints set for entities in the simulation. To delete one or more breakpoints, select the breakpoint(s) and click **Delete**. For information about how to add breakpoints, see "Setting breakpoints" on page 123.

Controlling animation

To turn the animation off and on, click **Control > Turn On/Off Animate**.

When the Animation Control is turned off, the simulation runs but does not update the graphics until the simulation is paused or ends. Entity statistics continue to accumulate and the AutoMod report is generated, but no vehicle or load movement is shown. The main advantage of turning the animation off is that the simulation runs much faster.

When the Animation Control is turned on, the graphics display in the Simulation window starts immediately and the clock begins to display elapsed time. You can control how often the animation is updated by setting the animation step (see "Setting the animation step" on page 62).

Prioritizing graphic quality or performance

By default, animation graphics are compiled at runtime, resulting in less-detailed shapes and graphics than are visible in the edit environment. If detailed animation is important in your simulation, you can turn off graphic compiling at run time by running the simulation from the command prompt and using the **-c** command line option. For example:

```
<modelname> -c
```

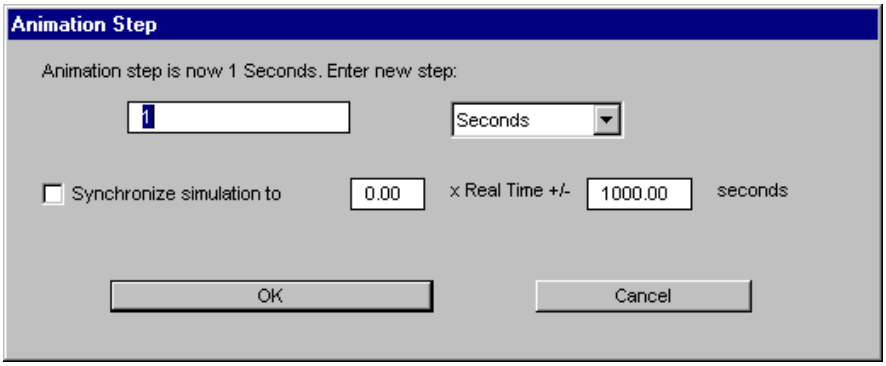
Conversely, if you want even faster performance than the default (at the expense of graphic quality), you can set the environment variable AMOD_USE_OPTIMIZE=TRUE and then restart the AutoMod runtime environment. In this case, the AutoMod software uses a picture optimization algorithm to improve performance, although some VRML textures might not display correctly.

Setting the animation step

The animation step is the period of simulated time between animation updates. The longer the animation step, the faster the simulation. Conversely, setting a shorter animation step slows the simulation because graphics are redrawn more frequently. The default animation step is one second. You can change the animation step before or during a simulation.

▶ **To change the animation step**

1. Click **Control > Animation Step**. The Animation Step dialog box opens.



2. Type a real or integer value, then select the desired time units from the list. For example, type **2.5** and select **seconds** to set the animation step to 2.5 seconds.
3. If you want the simulation to run in real time (or at a fraction or multiple of real time), click **Synchronize simulation to** and type a number in the text box (it defaults to 1 when you select the option). In the +/- seconds text box, define the *synchronization offset*, which has two purposes:
   - It defines how frequently the software compares the simulation clock to the computer's clock to ensure that the simulation is synchronized.
   - It determines how much faster or slower the model can run outside of the synchronization rate before the software makes adjustments to resynchronize the simulation with real time. For more information, see "Synchronizing the simulation to real time" on page 63.

Note For the best performance, set the model's initial animation step to an amount that allows the simulation to run at the synchronized rate with minimal adjustments. (You might need to experiment to find an animation step that requires the least amount of automatic adjustment.) If you set the initial animation step too slow or too fast, the software will frequently adjust the animation step until it finds an appropriate amount, which can degrade performance at the beginning of a simulation.

! **Important** When synchronizing a simulation with real time, define a synchronization offset greater than zero. Decreasing the offset increases the accuracy of the synchronization; however, if you set the offset too small, it can place too much of a computational burden on your machine and cause the model to fall behind the defined synchronization rate.

4. Click **OK** to close the Animation Step dialog box.

   The animation step has now been changed.

   Notes
   - You can double the animation step during a simulation by pressing **D** (uppercase). You can halve the animation step by pressing **d** (lowercase).
   - You can also change the time between animation updates by calling the SetDisplayStep function. This function is useful for overriding the default animation step at the beginning of a simulation. For more information, see "SetDisplayStep function" in the AutoMod Help.

Synchronizing the simulation to real time

If you want a simulation to run in real time (or at a multiple of real time), you can use the **Synchronize simulation to** option in the Animation Step dialog box and specify a multiple of real-time in the text box. For example, to run the simulation in real time, type **1.0**; to run at twice the speed of real time, type **2.0**; to run at half the speed of real time, type **0.50** (see "Setting the animation step" on page 62).

Note When you synchronize a simulation, you explicitly reduce the model's performance by preventing it from running at the fastest possible rate. Synchronizing a simulation with real time is useful when presenting the simulation, especially if the audience might be confused by a simulation

running in non-real time. Synchronization is also useful if the model is used in training, where the simulation speed needs to reflect an actual work environment.

To synchronize a simulation, the AutoMod software slows the animation so that the simulation is running at the defined rate. AutoMod slows the animation by automatically decreasing the animation step, causing the software to update displayed graphics more frequently. For more information, see "Setting the animation step" on page 62. Because the number and type of events that occur at any time during a simulation is continually changing, the simulation might run slower at some times (during computationally-intensive events) than others. The software attempts to compensate for these changes in performance by adjusting the animation step to maintain the defined synchronization rate.

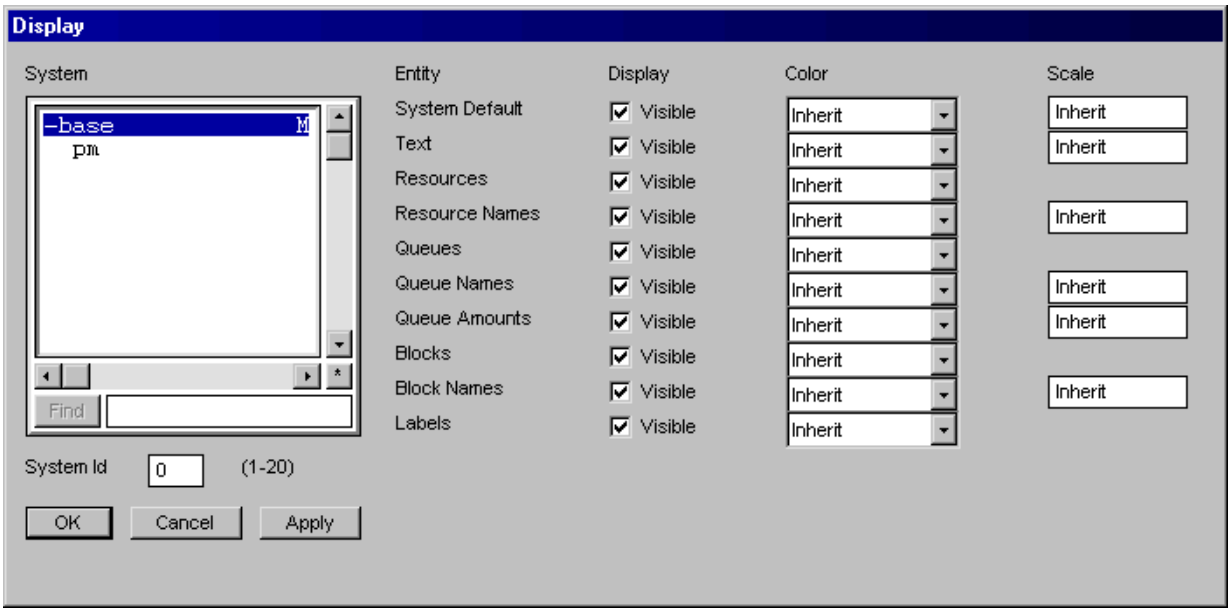
Depending on the value of the synchronization offset, the software periodically compares the simulation clock and the computer clock to ensure that the simulation is still synchronized. When the difference between the clocks is greater than the synchronization offset, the software automatically adjusts the animation step to slow down or speed up the simulation. If the model is running faster than the defined rate, the software might also slow the simulation by pausing for a number of milliseconds that is less than the defined synchronization offset.

» **Tip** To synchronize a simulation by default, call the SetSyncRate function rather than defining synchronization in the Animation Step dialog box. For more information, see "SetSyncRate function" in the AutoMod Help.

Display options

The Display dialog box allows you to display or hide system and entity graphics, display or hide graphics templates, change the color of a system or system entities, and scale the size of certain entities like names. You can also define a system ID number that corresponds to a keyboard shortcut you can use to toggle system graphics on and off.

You can change the Display options both in the edit environment (by clicking **Model > Display**) or in the simulation environment (by clicking **View > Display**). Changes made to the display in the edit environment are permanent; changes made to the display in the simulation environment are used only for the current simulation.



Changing the color of an entity

▶ **To change the color of an entity**

1. Select the name of the system in the **System** list.
2. Choose the color for the entity from the list to the right of that entity.

3. There are three ways to see the entity change color in the AutoMod or Simulation window:
   - Click **Apply**.
   - Select a different system from the list.
   - Click **OK**.

Note Colors set in the Display dialog box can be overwritten by changes made in reports (see "Changing the value of a writable attribute" on page 81).

Inheriting colors

Selecting `Inherit` in an entity's **Color** list causes the entity to inherit the system color. For example, if you select purple as the color for a conveyor system, then entities assigned the color **Inherit** display in purple.

Selecting **Inherit** in both a system and an entity's **Color** list causes the entity's graphic to display in the default colors. These colors vary with each system.

Scaling the size of an entity

Using the Display dialog box, you can scale the size of entities like stations, control points, and displayed names. Scaling the size of a name also causes the name to be repositioned appropriately; for example, if you shrink queue names, the names move closer to their queues.

To scale the size of an entity

1. Select the name of the system in the **System** list.
2. In the **Scale** field to the right of the entity, type `Inherit` or enter a positive number. Entering `0.5`, for example, halves the size of an entity; typing `3` triples its size.
3. There are three ways to see the entity change color in the AutoMod or Simulation window:
   - Click **Apply**.
   - Select a different system from the list.
   - Click **OK**.

Inheriting scale

Typing `Inherit` in an entity's **Scale** list causes the entity to inherit the scale set for **Text**. For example, if you halve the size of **Text** in a system, then the size of entities assigned the scale value **Inherit** is also halved.

Note Station names do not inherit from stations, and control point names do not inherit from control points.

Tip **Scale** values entered here for control points and stations are multiplied by each control point or station's **Scale** attribute value (as defined in syntax or in the edit environment). The same is true of **Scale** values for control point names and station names (as defined in the edit environment). For example, entering `2` in the **Scale** field for stations here, and `2` in the **Scale** field of the Conveyor Station Attributes dialog box multiplies the size of that particular station by `4`.

Toggling entity visibility

To turn an entity's display on or off

1. Select the name of the system in the **System** list.
2. Click the **Visible** select box for the entity display you want to toggle.

Note Paths, sections, and AS/RS racks have a list instead of a select box in the Display column. The list includes options to display the entity's standard graphics, its graphics templates, or no graphic at all (invisible). If graphics templates have not been attached to an entity, the standard graphic will be displayed even if **graphics templates** has been selected for that entity.

3. There are three ways to make the entity appear or disappear in the AutoMod or Simulation window:
   - Click **Apply**.
   - Select a different system from the list.
   - Click **OK**.

Toggling system visibility

To turn a system's graphic display on or off, type the keystroke corresponding to the ID number of the system.

Note You can set a system's ID number in the edit environment by clicking **Model > Display**.

Select a system and type a number from **1** to **20** into the **System ID** text box. You can use that keystroke to toggle the system graphic display on and off in either the edit or simulation environment. The following chart shows the system identification and the corresponding keystrokes. Multiple systems can have the same keystroke assignment. The system identification 0 cannot be toggled.

System ID Number	0	1	2	.	9	10	11	12	13	14		19	20
Keystroke	------	1	2		9	0	Shift 1	Shift 2	Shift 3	Shift 4		Shift 9	Shift 0

System ID numbers and corresponding keystrokes

Note The numeric pad cannot be used to toggle system visibility.

Using keyboard commands

In the simulation environment, you can use any of the following keyboard commands to control a model's animation and to change the current view.

»

! **Important** Keyboard commands are case sensitive; turn off Caps Lock before using keyboard commands.

Action	Keystroke	Description
Help menu	h	Displays a list of the keyboard commands.
Version display	Ctrl+Shift+V	Displays AutoMod version information.
Rotate commands		
Rotate screen	R	Toggles the rotate commands to apply to screen axes.
Rotate world	r	Toggles the rotate commands to apply to world axes.
Rotate X axis clockwise	X	Rotates image clockwise around the X axis.
Rotate X axis counterclockwise	x	Rotates image counterclockwise around the X axis.
Rotate Y axis clockwise	Y	Rotates image clockwise around the Y axis.
Rotate Y axis counterclockwise	y	Rotates image counterclockwise around the Y axis.
Rotate Z axis clockwise	Z	Rotates image clockwise around the Z axis.
Rotate Z axis counterclockwise	z	Rotates image counterclockwise around the Z axis.

Action	Keystroke	Description
Translate commands		
Translate screen	T	Toggles the translate commands to apply to screen axes.
Translate world	t	Toggles the translate commands to apply to world axes.
Negative X axis translate	I	Moves the display along the negative X axis.
Positive X axis translate	i	Moves the display along the positive X axis.
Negative Y axis translate	J	Moves the display along the negative Y axis.
Positive Y axis translate	j	Moves the display along the positive Y axis.
Negative Z axis translate	K	Moves the display along the negative Z axis.
Positive Z axis translate	k	Moves the display along the positive Z axis.
Scale commands		
Scale down	S	Reduces the display.
Scale up	s	Enlarges the display.
Display commands		
Friction stop motion	space bar	Stops graphic movement when friction is off.
System display	(0-9) or Shift-(0-9)	Toggles system graphics on or off; keystrokes must correspond to the ID number of a system included in your model. For more information, see "Toggling system visibility" on page 67.
Friction on/off toggle	f	Toggles the friction on or off.
Grid toggle	G	Displays or hides the grid.
Load a named view	L	Enables access to a saved model view.
Name the current view	l	Saves a particular model view.
Perspective/orthogonal toggle	P	Switches between perspective and orthogonal views.
View undo	Ctrl+Shift+U	Undoes the most recent view change.
View names toggle	V	Displays or hides names of model entities.
View top	v	Resets the picture to top (default) view.
Graphics templates toggle	W	Displays or hides graphics templates.
Wire/solid toggle	w	Switches between wireframe and solid graphics.

Action	Keystroke	Description
Simulation commands (available at runtime only)		
Start animation	A<time> Enter	Turns off graphics, runs the simulation to the specified <time>, pauses the model, and turns graphics back on.
Stop simulation	a<time> Enter	Pauses the simulation at the clock <time> specified. This is similar to the **Set Alarm** option on the **Control** menu.
Absolute/relative clock toggle	c	Toggles between absolute and relative clock time.
Display faster	D	Doubles the display step size (speeds up animation).
Display slower	d	Halves the display step size (slows animation).
Graphics on/off toggle	g	Turns animation on or off.
Next step	N	Simulates one full step size and updates the display.
Next action	n	Stops the simulation at the next event.
Pause/continue	p	Pauses or continues the model. Animation, business graphs, and statistics are updated each time the model is paused.
Show frames/second	Ctrl+Shift+T	Shows the number of frames displayed per second.
Animation step size	u<time> Enter	Changes the animation step to the specified rate. <Time> is the number of seconds to wait between animation updates.

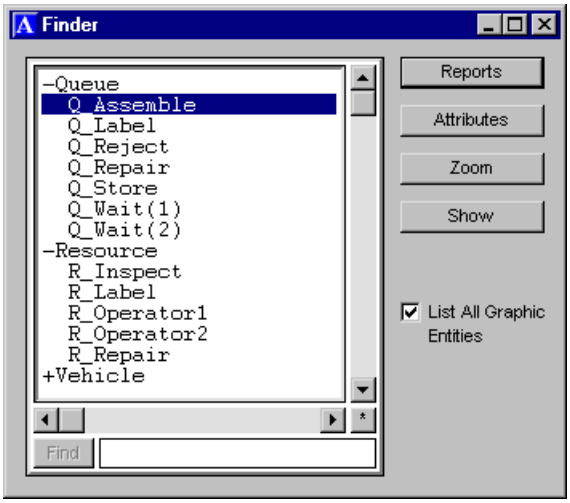
Finding graphical entities in a simulation

Using the Finder, you can quickly locate a graphical entity in a simulation (for example, a piece of equipment or a specific load). You can also display an entity's statistics and attributes.

Note You cannot find some graphic entities using the Finder. Excluded entities include some movement system entities (such as conveyor sections, path mover paths, and so on) and entities whose graphics have been hidden in the simulation. For information about hiding an entity's graphic, see "Display options" on page 64.

To find one or more graphical entities

1. Click **View > Finder**. The Finder dialog box opens, as shown here:



2. In the Finder dialog box, select **List All Graphic Entities** to display a list of the graphical entities currently in the simulation.
3. Click the name of the entity you want to find.

Note You can find multiple entities by holding Shift and clicking the names of the entities in the list.

4. To show the location of the selected entities in the simulation, either:
   - Click **Zoom** to recenter and zoom the view around the selected entities.

     or
   - Click **Show** to temporarily scale the entities' graphics in the Simulation window.

Tip You can display entity statistics or attributes by clicking **Reports** or **Attributes** in the Finder. For more information, see "Displaying entity reports" on page 75 and "Displaying entity attributes" on page 78.

Picking graphical entities in a simulation

In addition to selecting graphical entities by name using the Finder, you can also select graphical entities by picking them in the Simulation window. Picking graphical entities is useful for displaying more information about a specific entity or a group of entities that are located near one another in the Simulation window.

Note You cannot pick some graphics during a simulation (for example, conveyor transfers).

To pick a single graphical entity

Click the entity's graphic in the Simulation window.

A window opens displaying the entity's attributes. For more information, see "Displaying entity attributes" on page 78.

To pick multiple graphics

Drag a selection box around the entities in the Simulation window.

The Finder opens, displaying the selected entities. You can then click either **Reports** or **Attributes** to display more information about each of the selected entities. For more information, see "Finding graphical entities in a simulation" on page 70.

Taking entities down during simulation

You can dynamically take down or bring up resources in a process system and motors in a conveyor or power & free system during a simulation.

To take down a resource or motor during a simulation

1. Display the report for Resources or Motors.

   For more information, see "Displaying entity reports" on page 75.

2. Select the Resource or Motor you want to take down, then click **Edit > Take Down** in the Report dialog box.

   The resource or motor is taken down.

To bring up a resource or motor during simulation

1. Display the report for Resources or Motors.

   For more information, see "Displaying entity reports" on page 75.

2. Select the Resource or Motor you want to bring up, then click **Edit > Bring Up** in the Report dialog box.

The resource or motor is brought up.

Entity statistics and attributes

There are three ways to view simulation statistics during and after a run:

- View the printed AutoMod reports.
- Display entity reports.
- Display entity attributes.

Note Of these three methods for viewing statistics, displaying individual entity attributes provides the most information about an entity in the simulation. In addition to displaying the same statistics that are provided in reports, this option also provides information about the entity's standard and user-defined attributes that can be referenced in the model logic.

Tip For more information, see "Attributes" in the AutoMod Help.

These methods of viewing statistics are discussed in the following sections.

Viewing the printed AutoMod reports

During simulation, each model snap generates statistics that are automatically printed to the <modelname>.report file. The statistics that are generated depend on which entities are included in each system in your model. To view the printed report, open the <modelname>.report file using any text editor.

Note If you pause the model during a run and view the report, the report shows statistics since the beginning of the run, regardless of any run control snaps you might have set.

An example AutoMod .report file is shown here:

```
*** AutoMod 11.0 ***
Model Sim1
Statistics at Absolute Clock = 10:00:00.00, Relative Clock = 10:00:00.00
CPU time: Absolute: 23.784 sec, Relative: 23.784 sec

Resource Statistics
```

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait	State
Since Reset:										
R_Unloader	5187	0	0.29	1	1	0	0.288	2.00	1.77	Idle
Wait	1488	0	0.26	--	4	0	--	6.17	--	
Down	109	0	0.09	1	1	0	0.091	30.00	0.00	Idle

Note Hyphens in a .report file indicate that the statistic is not applicable and was not reported.

In addition to the automatically generated .report file, you can generate *tab-delimited* report files using the **-b**, **-B**, **-r**, or the **-R** command line options (see "Running a model from a command prompt" on page 56). These options generate one file for each model entity type, plus a process system file and, if applicable, a path mover system file. Entity files for process system entities have names of the form <modelname>.<entitytype>.rep. Entity files for systems and submodels have names of the form <modelname>.<systemname>.<entitytype>.rep. A simple path mover model called `packing`, with one path mover system named `pm`, might generate these .rep files:

packing.block.rep	Block statistics
packing.loads.rep	Load statistics
packing.model.rep	Process system data
packing.pm.cp.rep	Control point statistics
packing.pm.path.rep	Path statistics
packing.pm.pathmoversys.rep	Path mover system statistics
packing.pm.vehicle.rep	Vehicle statistics
packing.process.rep	
packing.stream.rep	

The delimited report files provide data on all values found in the standard .report file, as well as several additional entity attributes.

Note In order to format the data efficiently for post-processing analysis (using your own custom text analysis tool), the delimited report files are limited to relative statistics (see "Absolute/Relative" on page 77).

You can also create custom reports by defining the model snap function to print attribute values and statistics to one or more files at the end of each model snap. And you can define the model finished function to write custom reports at the end of the simulation.

Interpreting report header information

The <modelname>.report file begins with version and clock information about the simulation. The first two lines of the report header indicate the version of the AutoMod software that was used to run the model, and the model name. The third line shows the length of the simulation, which, in the preceding example report, is 10 hours (the simulation clock is displayed in the format days:hours:minutes:seconds.hundredths of seconds).

Note The report displays both absolute and relative times. In the example report, the statistics are not reset, so both sets of statistics are the same and span the entire simulation.

The fourth line measures the length of the simulation in CPU time (real time). On the computer that generated the example report, the simulation took 23.784 seconds to run from start to finish. The length of time required to run the model varies depending on the speed of your computer's processor, whether animation is displayed, the animation step (if the animation is displayed) and whether you pause the model during a simulation.

Note This header information is also contained in the process system .rep file.

Interpreting reported statistics

In the example .report file, only statistics for resources are reported. Using the definitions of resource statistics (see "Resource statistics" on page 97), you can interpret the example report to find that:

- Runloader was busy about 28.8 percent of the time (Util statistic).
- Runloader's average processing time was 2.00 seconds (Av_Time).
- The average waiting time for all loads (including loads that didn't wait to use the resource) was 1.77 seconds.
- In total, 1,488 loads waited to claim the resource because the resource was busy. On the average, 0.26 loads were waiting for the resource at any point in time. Up to four loads were waiting over the 10 hours of the simulation.
- The average waiting time for the 1,488 loads that had to wait to use a resource was 6.17 seconds.
- Runloader went down 109 times, with an average downtime of 30 seconds (Av_Time). It was down 9.1 percent of the time (Util statistic).
- Runloader was idle 62.1 percent of the time (calculated by subtracting the sum of busy and down time [28.8 + 9.1] from 100 percent).

Definitions and example statistics for each model entity are provided later in this chapter. For more information, see "Process system statistics" on page 82 and "Movement system statistics" on page 109.

Reported statistics types

Statistics that are included in the reports can be one of two types:

Standard

Standard statistics are individual entity statistics since the last statistics reset. Statistics are reset at the beginning of a snap if you have selected the reset option in the Run Control dialog box

Full

Full statistics are individual entity statistics since the beginning of the model run (absolute time) and since the last reset (relative time). Statistics are reset at the beginning of a snap if you have selected the reset option in the Run Control dialog box.

Note When statistics are reset, the Total is set to equal the Current.

Full statistics for some entities include additional information that is not reported in standard statistics (for example, full block statistics include information about the number of loads or vehicles that waited to claim the block, full resource statistics include downtime information, and so on.) For information about the difference between standard and full statistics for each entity, see "Process system statistics" on page 82 and "Movement system statistics" on page 109.

Statistics types are selected when the model is built and cannot be changed in the simulation environment.

Displaying entity reports

During or after a simulation, you can display entity reports, which often contain the same information as the printed AutoMod reports. Displayed reports are current only at the time you request them. You can update the displayed statistics during a run by clicking **View > Update**. Statistics are automatically updated each time you pause the model.

To display entity reports

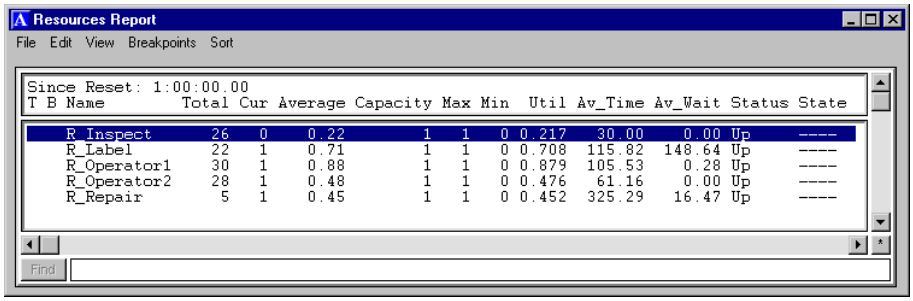
1. Click **View > Reports**.

   The Reports dialog box opens.

2. In the **System** list, select the system containing the entities for which you want to view reports.

3. In the entity type list, select the type of entity for which you want to view reports, then click **Display**.

A report contains statistics for entities of the selected type. For example, the report for resources in a process system is shown here:



For information on how to interpret the reported statistics, see "Process system statistics" on page 82 and "Movement system statistics" on page 109.

Note You can also display reports for graphical entities by selecting one or more entities in the Finder dialog box and then clicking **Reports**. For more information about using the Finder, see "Finding graphical entities in a simulation" on page 70.

Report menus

Each entity report contains the following menus: **File**, **Edit**, **View**, **Breakpoints**, and **Sort**.

Using the File menu

Options on the **File** menu are:

Option	Description
Close	Closes the report dialog box.
Save As	Saves the displayed statistics in a space-delimited text file.
Save Selected As	Saves the selected statistics in a space-delimited text file.
Print	Prints the displayed statistics to the default printer.

Using the Edit menu

Options on the **Edit** menu are:

Option	Description
Copy	Copies the selected statistics to the Windows clipboard.
Select All	Selects all statistics in the dialog box.
Find	Allows you to search a report for entries containing specific text or values. For example, in a lengthy resource report, you can quickly find the reported statistics for `R_Operator` by searching for one of the following values: **R_Oper**, **Operator**, or **rator**. You can also search the report for specific values, such as **87.3**.
Bring Up	Brings up the selected resource or motor. (This option is only available in the Resource and Motor Report windows.)
Take Down	Takes down the selected resource or motor. (This option is only available in the Resource and Motor report windows.)
Add Tracing	Prints entity status messages to a trace file (<modelname>.trace) throughout the simulation. If you are experiencing problems related to a specific entity, tracing the entity is a good way to isolate the events leading up to the error, as well as the time it occurs.

Option	Description
Remove Tracing	Removes the tracing defined for the selected entity; the simulation no longer prints entity status messages to a trace file.

Using the View menu

Options on the **View** menu are:

Option	Description
Update	Updates the displayed statistics to the current simulation time. Use this option when viewing statistics while the simulation is running.
Attributes	Display the attributes for the selected entity or entities. For more information, see "Displaying entity attributes" on page 78.
Absolute/Relative	Click to toggle between displaying absolute and relative statistics. Absolute statistics are gathered from the beginning of a simulation (time zero). Relative statistics are gathered from the beginning of the last run control snap that resets statistics. If no snaps are defined to reset statistics, then absolute and relative statistics are identical.
Zoom	Recenter and zoom the view around the selected entity or entities.
Show	Temporarily scales the selected entities' graphics in the Simulation window.

Using the Breakpoints menu

Breakpoints allow you to automatically pause a simulation whenever certain conditions (usually load or vehicle interactions) occur for a specific entity.

Note The **Breakpoints** menu is not active until you select one or more entities for which breakpoints can be defined.

The first section of the **Breakpoints** menu includes a list of the conditions for which you can define a breakpoint for the selected entities. For example, the **Breakpoint** menu for processes includes the following conditions:

- Entering (The simulation pauses whenever a load enters the process.)
- Leaving (The simulation pauses whenever a load leaves the process.)
- Can't Enter (The simulation pauses when a load attempts to enter the process but can't; for example, because of limited process capacity.)

To create breakpoints for the currently selected entities, select the appropriate conditions in the list. For example, to automatically pause the simulation when a load leaves a process or when it attempts to enter a process but can't, select **Leaving**, then select **Can't Enter** in the **Breakpoints** menu.

For more information, see "Setting breakpoints" on page 123.

The **Breakpoints** menu includes the following additional options:

Option	Description
Add All Breakpoints	Sets breakpoints for all conditions for the selected entities.
Remove All Breakpoints	Removes breakpoints for all conditions from the selected entities.

Using the Sort menu

Options on the **Sort** menu are:

Option	Description
Ascending/ Descending	Sorts the listed entities in either ascending (lowest first) or descending (highest first) order; the list is sorted by the statistic selected in the **Sorting on** list discussed next.
Sorting on	Determines which statistic is used for sorting the entities in the displayed report. By default, entities are sorted by Name in alphabetical (ascending) order. You can sort entities by any statistic in the report, including utilization, capacity, and so on. After selecting the statistic on which you want to sort the entities, you can select either ascending or descending to change the sorting order.

Displaying entity attributes

Displaying an entity's attributes allows you to view the entity's reported statistics, as well as the value of the entity's standard and user-defined attributes (the same entity attributes that you can reference in a model's logic).

To display an entity's statistics and attributes, do any one of the following:

- In the Finder, select one or more entities, then click **Attributes**. For more information, see "Finding graphical entities in a simulation" on page 70.
- Pick a single entity during simulation. For more information, see "Picking graphical entities in a simulation" on page 71.
- From the **View** menu of an entity report, click **Attributes**. For more information, see "Displaying entity reports" on page 75.
- In the upper list of any attributes dialog box, double-click an entity pointer expression to open a new attributes dialog box for the selected entity.

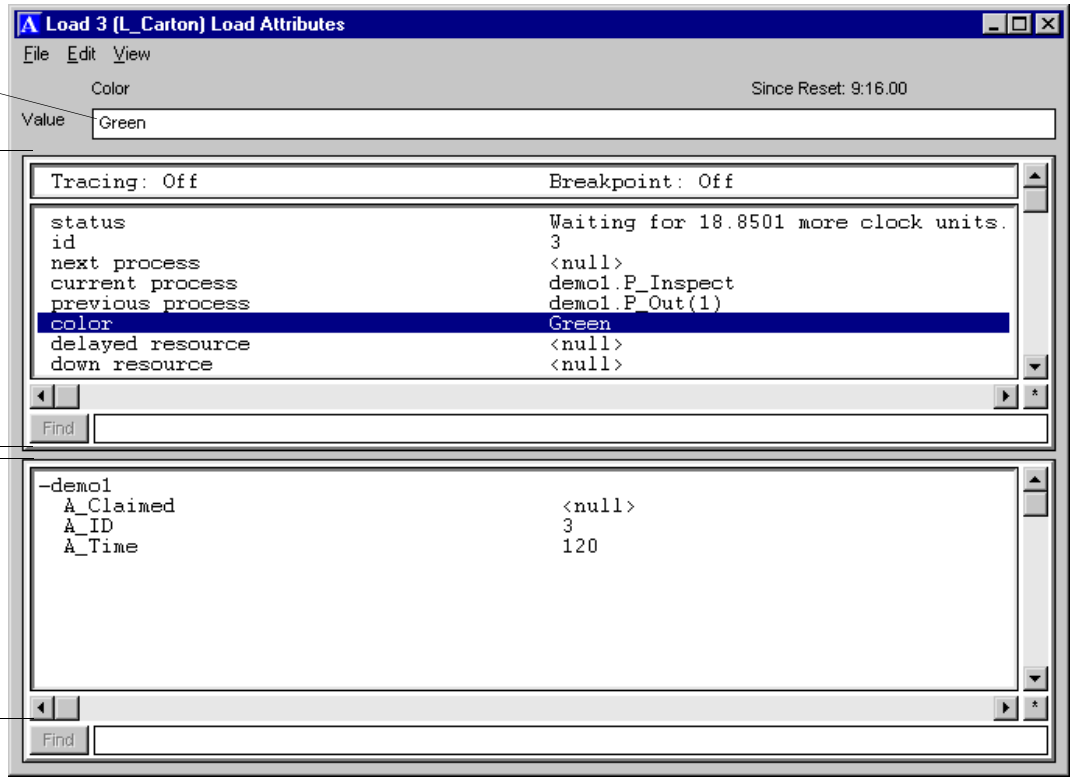
After using any of these methods, an attribute dialog box opens displaying the attribute values and statistics for that entity. The attribute dialog box for a load is shown here:

» **Tip** For a definition of the standard entity attributes listed in the dialog box, see "Attributes" in the AutoMod Help.

Using the **Value** box, you can change the value of any writable attribute during a simulation.

The upper list contains statistics and standard entity attributes.

The lower list contains user-defined attributes (for loads and vehicles), frequency statistics (for tables), or state statistics (for resources).



In addition to standard entity attributes, the attributes dialog box for some entities contains the same statistics that are provided in the entity reports. For a definition of these statistics, see "Process system statistics" on page 82 and "Movement system statistics" on page 109.

Attribute dialog box menus

Each entity attribute dialog box contains the following menus: **File**, **Edit**, **View**, and **Breakpoints**.

Using the File menu

Options on the **File** menu are:

Option	Description
Save As	Saves the displayed attributes in a space-delimited text file.
Save Selected As	Saves the selected attributes in a space-delimited text file.
Print	Prints the displayed attributes to the default printer.
Close	Closes the attributes dialog box.

Using the Edit menu

Options on the **Edit** menu are:

Option	Description
Copy	Copies the selected attributes to the Windows clipboard.
Find	Allows you to search the current list for entries containing specific text or values. For example, in a lengthy attribute list, you can quickly find the `remaining space` attribute by searching for one of the following values: **rem**, **remaining**, or **space**. You can also search an attribute list for specific values, such as **87.3**.
Bring Up	Brings up a down resource or motor.
Take Down	Takes down a resource or motor.
Add Tracing	Prints entity status messages to a trace file (<modelname>.trace) throughout the simulation. If you are experiencing problems related to a specific entity, tracing the entity is a good way to isolate the events leading up to the error, as well as the time it occurs.
Remove Tracing	Removes the tracing defined for the selected entity; the simulation no longer prints entity status messages to a trace file.

Using the View menu

Options on the **View** menu are:

Option	Description
Update	Updates the displayed attribute values to the current simulation time. Use this option when viewing attributes while the simulation is running.
Report	Display the report for the current entity. For more information, see "Displaying entity reports" on page 75.
Absolute/Relative	Click to toggle between displaying absolute and relative values. Absolute values are gathered from the beginning of a simulation (time zero). Relative values are gathered from the beginning of the last run control snap that resets statistics. If no snaps are defined to reset statistics, then absolute and relative values are identical.
Zoom	Recenter and zoom the view around the selected entity.
Show	Temporarily scales the entity's graphic in the Simulation window.

Using the Breakpoints menu

The options on the **Breakpoints** menu in the Attributes dialog box are the same as those in the Reports dialog box. For information about this menu, see "Using the Breakpoints menu" on page 77.

Changing the value of a writable attribute

At any point during a simulation, you can change the value of a writable entity attribute. You can use this feature to dynamically test scenario changes during a simulation.

» **Tip** For information about setting attribute values in a procedure or function, see "set" in the AutoMod Help.

▶ **To change the value of an attribute**

1. Display an entity's attributes.

   For more information, see "Displaying entity attributes" on page 78.

2. Select the attribute whose value you want to change.
3. In the **Value** box, delete the old value, then type a new value for the attribute.

   ■ **Note** If the **Value** box is unavailable, the attribute you have selected is either read-only or an object list (you cannot change the members of a list from the Attributes dialog box).

4. Press **ENTER**.

The attribute value has now been updated to the new value.

Formulas for calculating reported statistics

Average and utilization statistics are calculated using the following formulas:

Formula	Description
Average	Average statistics are calculated by multiplying the total (claims, loads, and so on) by the average time (per claim, per load, and so on), and dividing by the total simulation time of the model: $\text{Average} = \frac{\text{Total} \times \text{Av_Time}}{\text{total simulation time}}$
Utilization	Utilization statistics represent the fraction of an entity's capacity that was utilized, and can be converted to percentages by multiplying by 100. Utilization is calculated using the following formula: $\text{Util} = \frac{\text{Total} \times \text{Av_Time}}{\text{Capacity} \times \text{total simulation time}}$ which is the same as $\text{Util} = \frac{\text{Average}}{\text{Capacity}}$

■ **Note** If an entity has a capacity of zero, the entity's utilization statistic reports a value of 1.

Process system statistics

You can view statistics for the following process system entities:

- Blocks
- Counters
- Events (Display only)
- Loads
- Order lists
- Processes
- Queues
- Random number streams (Printed only)
- Resources
- States (resources and state monitors)
- Tables
- Variables (Display only)

Block statistics

There are three categories of block statistics:

- Block attributes
- Block usage statistics
- Block wait statistics

Block attributes

Block attributes are *not* included in the printed AutoMod reports or in displayed reports; they only appear when displaying attribute values (see "Displaying entity attributes" on page 78).

» **Tip** For a list of block attributes and their definitions, see "Block attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Block usage statistics

Block usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed block attributes. Block usage statistics provide information about the number of claims on a block. A *claim* is the use of one unit of a block's capacity by a load or vehicle during a simulation; loads or vehicles claim a block by executing a `claim` action or by moving into a block's graphic in a movement system.

Block usage statistics are defined as follows:

Statistic	Description
Name	The name of the block.
Total	The total number of claims on the block.
Current	The number of current claims on the block.
Average	The average number of claims on the block.
Capacity	The defined maximum number of simultaneous claims on the block.
Maximum	The maximum number of simultaneous claims on the block.
Minimum	The minimum number of simultaneous claims on the block.

Statistic	Description
Utilization	The fraction of the block's capacity that loads or vehicles utilized simultaneously. **Note** If the block has a capacity of 0, the utilization statistic reports a value of 1.
Average time	The average time of a claim on the block.
Average wait time	The average time a claim waited for the block. This is the average of *all* claims on the block, including those claims that did not wait.

Block wait statistics

Block wait statistics are full statistics that appear in the printed AutoMod reports and in displayed block attributes. Block wait statistics provide information about claims that waited for a block; a claim must wait for a block when the block has no available capacity.

Block wait statistics are defined as follows:

Statistic	Description
Total	The total number of claims that waited for the block.
Current	The number of claims that are currently waiting for the block.
Average	The average number of claims that waited for the block.
Maximum	The maximum number of claims that simultaneously waited for the block.
Minimum	The minimum number of claims that simultaneously waited for the block.
Average time	The average time that a waiting claim spent waiting for the block. This is the average of only those claims that waited for the block.

Example reports

Example 1—Standard block statistics

An example .report file containing standard block statistics is shown here:

Block Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
B_Block1	478	2	2.00	2	2	1	1.00	15.06	4.54
B_Block2	477	1	1.00	1	1	0	1.00	7.55	3.40

Block statistics are summarized for each block in the model.

Example 2—Full block statistics

An example .report file containing full block statistics for one block is shown here:

```
Block Statistics
Name            Total   Cur  Average   Capacity   Max  Min  Util    Av_Time    Av_Wait
Since Reset:
B_Block1        478     2    2.00      2          2    1    1.00    15.06      4.54
Wait            477     1    0.60      --         1    0    --      4.56       --
Since Beginning:
B_Block1        955     2    2.00      2          2    1    1.00    15.06      4.54
Wait            954     1    0.60      --         1    0    --      4.55       --
```

Block statistics are provided since the last statistics reset and since the beginning of the model run.

Counter statistics

There are three categories of counter statistics:

- Counter attributes
- Counter usage statistics
- Counter wait statistics

Counter attributes

Counter attributes are *not* included in the printed AutoMod reports or in displayed reports; they only appear when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of counter attributes and their definitions, see "Counter attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Counter usage statistics

Counter usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed counter attributes. Counter usage statistics provide information about the number of claims on a counter. Counters can be claimed by any entity that executes a procedure (for example, loads or vehicles). A counter is claimed each time it is incremented, and a claim is released each time the counter is decremented. For example, an entity that increments a counter by three claims the counter three times.

Counter usage statistics are defined as follows:

Statistic	Description
Name	The name of the counter.
Total	The total number of claims on the counter.
Current	The number of current claims on the counter.
Average	The average number of claims on the counter.

Statistic	Description
Capacity	The defined maximum number of simultaneous claims on the counter. **Note** If the counter has a capacity of 0, the utilization statistic reports a value of 1.
Maximum	The maximum number of simultaneous claims on the counter.
Minimum	The minimum number of simultaneous claims on the counter.
Utilization	The fraction of the counter's capacity that was utilized simultaneously.
Average time	The average time between an increment and a subsequent decrement of the counter.
Average wait time	The average time that a claim waited for the counter. This is the average of *all* claims, including those that did not wait for the counter.

Counter wait statistics

Counter wait statistics are full statistics that appear in the printed AutoMod reports and in displayed counter attributes. Counter wait statistics provide information about claims that waited for a counter; a claim must wait for a counter when the counter has no available capacity.

Counter wait statistics are defined as follows:

Statistic	Description
Total	The total number of claims that waited for the counter.
Current	The number of claims that are currently waiting for the counter.
Average	The average number of claims that waited for the counter.
Maximum	The maximum number of claims that simultaneously waited for the counter.
Minimum	The minimum number of claims that simultaneously waited for the counter.
Average time	The average time that a waiting claim spent waiting for the counter. This is the average of only those claims that waited for the counter.

Example reports

Example 1—Standard counter statistics

An example .report file containing standard counter statistics is shown here:

```
Counter Statistics

Name        Total  Cur  Average   Capacity   Max  Min  Util   Av_Time   Av_Wait
C_Count1    224    198  94.93     Infinite   199  0    --     1525.65   --
```

Counter Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
C_Count2	66	50	36.22	50	50	0	0.724	1975.41	259.04

Counter statistics are summarized for each counter in the model. Util and Av_Wait statistics are not reported for counters with infinite capacity.

Example 2—Full counter statistics

An example .report file containing full counter statistics for one counter is shown here:

Counter Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
Since Reset:									
C_Count2	50	50	50.00	50	50	50	1.000	3600.00	1503.02
Wait	88	88	57.62	--	88	27	--	2357.01	--
Since Beginning:									
C_Count2	66	50	43.11	50	50	0	0.862	4702.68	1503.30
Wait	91	88	32.15	--	88	0	--	2544.04	--

Counter statistics are provided since the last statistics reset and since the beginning of the model run.

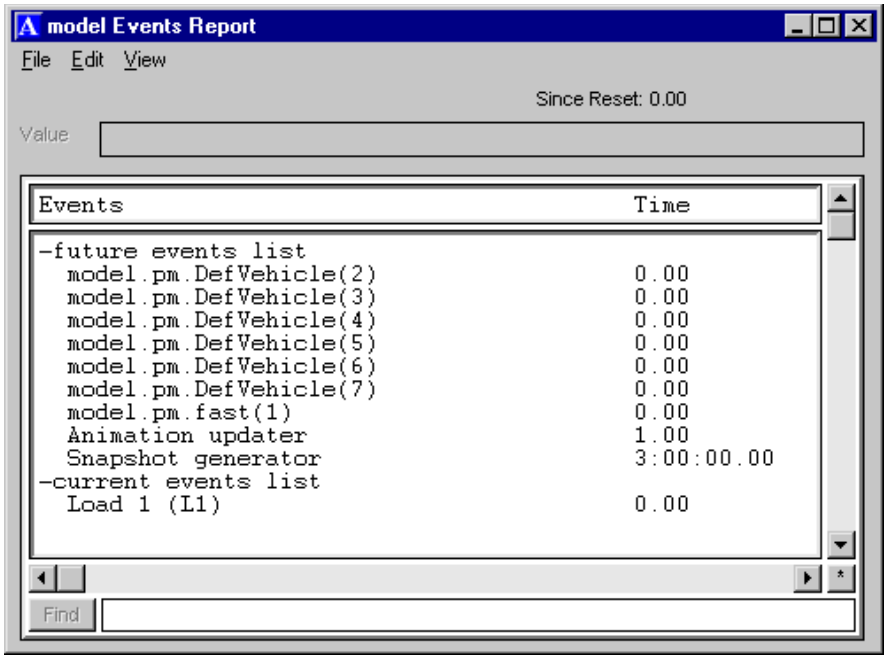
Event statistics

Event statistics appear only in the displayed event report. Displaying events can be useful when debugging modeling errors.

▶ **To display the event report during a simulation**

Click **View > Events**.

The Events Report opens:



An AutoMod simulation is driven by an active entity (such as a load or vehicle) performing actions defined in model logic. Although several events can occur at the same simulation clock time (for example, 4:28:46.07 on the simulation clock), one event must occur before the other in the software. Consequently, at any time during a simulation, only one entity is active while all other entities delay. When an entity becomes active, the simulation processes as many events for that entity as possible, until the entity is delayed for some reason. Once delayed, the entity waits on one of several lists. The current events list and future events list are both lists on which entities can delay while waiting to become the active entity.

In general, a load or vehicle's `priority` determines where the load is inserted on a list when it is delayed, and in what order it becomes the active entity. Loads or vehicles with a lower numeric value defined for their `priority` attribute (higher priority, for example, 1) are inserted above loads or vehicles with larger `priority` values (lower priority, for example, 4). Loads and vehicles get off the lists in order of priority.

Note In addition to loads and vehicles, internal entities that schedule and complete events also appear in the events lists. For example, the Animation updater (shown in the preceding illustration) is scheduled to update the animation at time intervals defined by the current animation step. The Snapshot generator is also scheduled to update the AutoMod report at the end of each snap defined in the run control.

The events report contains information about two events lists in the simulation:

- The current events list
- The future events list

This section describes these events lists in more detail.

The current events list

The current events list is the list of entities waiting to perform actions during the current simulation clock time. Entities on the future events list are placed on the current events list when the simulation clock is updated to the time associated with their next event. For example, if two loads are both waiting on the future events list to complete events scheduled at time 3:16:00.00, the loads delay until the simulation advances to this time and then both loads move from the future events list to the current events list. One load then moves off the current events list, becomes the active load, and begins completing its scheduled events; the other load delays on the current events list to complete its events as soon as the active load completes its events for the current time. Newly created loads, including cloned loads, get on the current events list at the instant they are created. (In the illustration in the previous section, you can see that Load 1, which is created at time 0.0, is waiting on the current events list.)

An entity leaves the current events list when it becomes the active entity in the simulation. The entity then executes as many actions as it can during the current simulation clock time. Each entity in the current events list takes its turn (based on priority) to become the active entity and perform all possible actions during the current simulation clock time. When all entities have been removed from the current events list, the simulation clock advances to the next event time.

The future events list

The future events list is the list of entities that are waiting to become active at a known future simulation clock time. The future events list is sorted by simulation clock time with the lowest value (earliest time) first. If two or more entities are waiting on the future events list for the exact same simulation time, the entities are sorted by priority.

Loads and vehicles get on the future events list by executing an action with a known time delay, for example, a `wait for` or `use` action. When the simulation time advances to the time associated with one or more entities on the future events list, those entities move from the future events list to the current events list, where they wait to become the active entity in the simulation.

Load statistics

There are two categories of load statistics:

- Load attributes
- Load statistics

Load attributes

Load attributes are *not* included in the .report file or, for the most part, in displayed reports. They *are* included in the loads.rep file, and they appear when you display attribute values (see "Displaying entity attributes" on page 78). For a list of load attributes and their definitions, see "Load attributes" in the AutoMod Help.

Load statistics

Load statistics appear in displayed reports, as well as in the loads.rep file and in displayed load attributes. They do not appear in the .report file.

▶ **To display the loads report**

Click **View > Loads**.

The loads report opens, containing statistics for the current loads in the simulation.

Load statistics provide information about the status, process, and location (territory) of loads during a simulation.

Load statistics are defined as follows:

Statistic	Description
id #	The sequential number assigned to the load at the load's creation.
Load Type	The load's type.
From Process or Current Process	The load's current process, if applicable; otherwise, the previous process.
To Process	The load's next process, if it has been determined; else `null`.
Territory	The location currently occupied by the load. Valid territories for loads in a simulation include: • Sitting in a queue. • Onboard a vehicle segment. • On a conveyor section.
Status	The current state of the load, such as "Waiting in load waiting list to be picked up by <vehicle>" or "Waiting to move into queue."

Order list statistics

There are two categories of order list statistics:

- Order list attributes
- Order list usage statistics

Order list attributes

Order list attributes are *not* included in the printed AutoMod reports or in displayed reports; they only appear when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of order list attributes and their definitions, see "Order list attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Order list usage statistics

Order list usage statistics are standard statistics that appear in the printed AutoMod reports, displayed reports, and in displayed order list attributes. Order list usage statistics provide information about the number of loads or vehicles on an order list.

Order list statistics are defined as follows:

Statistic	Description
Name	The name of the order list.

Statistic	Description
Total	The total number of loads or vehicles on the order list.
Current	The number of loads or vehicles that are currently on the order list.
Average	The average number of loads or vehicles on the order list.
Maximum	The maximum number of loads or vehicles that were on the order list simultaneously.
Minimum	The minimum number of loads or vehicles that were on the order list simultaneously.
Average time	The average time a load or vehicle spent on the order list.
Total Backordered	The total number of backorders placed on the order list.
Current Backordered	The number of backorders on the order list currently waiting to be filled.
Average Backordered	The average number of backorders waiting to be filled at any time.
Maximum Backordered	The maximum number of backorders waiting to be filled at any time.
Minimum Backordered	The minimum number of backorders waiting to be filled at any time.

Example reports

Example 1—Standard order list statistics

An example .report file containing standard order list statistics is shown here:

```
Order List Statistics

Name       Total  Cur  Average   Max  Min  Av_Time   Tot_Back_Ordered
OL_List1   754    34   25.22     34   16   120.40    0
OL_List2   1080   180  133.70    189  77   445.67    0
```

Order list statistics are summarized for each order list in the model.

Example 2—Full order list statistics

An example .report file containing full order list statistics for one order list is shown here:

```
Order List Statistics

Name        Total   Cur   Average   Max   Min   Av_Time   Tot_Back_Ordered
Since Reset:
OL_List1    754     34    25.22     34    16    120.40    0
Since Beginning:
OL_List1    1432    34    16.65     34    0     83.69     42
```

Order list statistics are provided since the last statistics reset and since the beginning of the model run.

Process statistics

There are five categories of process statistics:

- Process attributes
- Process usage statistics
- Traffic limit statistics
- Process wait statistics
- Arriving and leaving statistics

Process attributes

Process attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of process attributes and their definitions, see "Process attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Process usage statistics

Process usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed process attributes. Process usage statistics provide information about the number of loads in a process. The amount of time loads spend in a process depends on the procedures the load executes while in the process and the amount of time the load must wait in the process (for example, waiting for the traffic count of the next process to fall below the traffic limit).

Process usage statistics are defined as follows:

Statistic	Description
Name	The name of the process.
Total	The total number of loads in the process.
Current	The number of loads that are currently in the process.
Average	The average number of loads in the process.
Maximum	The maximum number of loads in the process simultaneously.

Statistic	Description
Minimum	The minimum number of loads in the process simultaneously.
Average time	The average time a load spent in the process.

Traffic limit statistics

Traffic limit statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed process attributes.

A process' *traffic limit* sets the maximum number of loads that can be sent to a process simultaneously. Traffic limit statistics provide information about the number of loads sent to a process. Traffic limit statistics are useful when using input and output connections to make loads travel between processes .

Traffic statistic values differ from process statistic values when a load must wait for a destination process' traffic count to fall below the traffic limit before it can be sent to the process. A process' *traffic count* refers to the number of loads that are currently claiming the process (loads claim a process when they are sent to it and do not release it until after executing the process' leaving procedure).

Traffic limit statistics are defined as follows:

Statistic	Description
Total	The total number of loads sent to the process.
Current	The number of loads that are currently in the traffic count.
Average	The average number of loads that were sent to the process.
Capacity	The defined maximum number of loads that can execute the process simultaneously; this statistic is the traffic limit. If the number of loads currently claiming the traffic count is equal to the capacity, loads attempting to send to the process are prevented from doing so.
Maximum	The maximum number of loads that were in the traffic count simultaneously.
Minimum	The minimum number of loads that were in the traffic count simultaneously.
Utilization	The fraction of the process' capacity that loads utilized simultaneously. **Note** If the process has a capacity of 0, the utilization statistic reports a value of 1.
Average time	The average time a load spent in the traffic count.
Average wait time	The average time a load spent waiting to be sent to a process with a full traffic count. This is the average of *all* loads, including those loads that were sent to a process without waiting.

Full traffic limit statistics also include a total for each load type sent to a process and a total for each load type leaving a process. Totals are categorized by the load's previous process and destination process.

Process wait statistics

Process wait statistics are full statistics that appear in the printed AutoMod reports and in displayed process attributes. Process wait statistics provide information about the number of loads that waited to be sent to a process. Loads must wait to be sent to a process when the process has reached its traffic limit.

Process wait statistics are defined as follows:

Statistic	Description
Total	The total number of loads that waited to be sent to the process.
Current	The number of loads that are currently waiting to be sent to the process.
Average	The average number of loads that waited to be sent to the process.
Maximum	The maximum number of loads that were simultaneously waiting to be sent to the process.
Minimum	The minimum number of loads that were simultaneously waiting to be sent to the process.
Average time	The average time a load spent waiting to be sent to a process. This is an average of only those loads that waited to be sent to the process.

Arriving and leaving statistics

Arriving and leaving statistics are full statistics that appear only in the printed AutoMod reports. The arriving and leaving statistics provide information about the number of loads of each type that entered the process from any other process in the simulation and the number of loads of each type that left the process to any other process in the simulation. A <null> **from** value indicates the loads were newly created (they were not executing a previous process before they were sent to the current process). A <null> **to** value indicates the loads ended the process without being explicitly sent to die or to another process in the simulation.

Example reports

Example 1—Standard process statistics

An example .report file containing standard process statistics is shown here:

Process Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
P_Start	97	74	55.63	--	74	36	--	2064.67	--
P_Drill	25	2	2.00	--	2	1	--	288.00	--
P_Clean	24	2	0.76	--	3	0	--	114.13	--

Process Traffic Limit Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
P_Start	97	74	55.63	Infinite	74	36	--	2064.67	--
P_Drill	25	2	2.00	2	2	1	1.000	288.00	1989.83
P_Clean	24	2	0.76	Infinite	3	0	--	114.13	--

Process and traffic statistics are summarized for each process in the process system. Util and Av_Wait traffic limit statistics are not reported for traffic limits with infinite capacity.

Example 2—Full process statistics

An example .report file containing full process statistics for one process is shown here:

Process Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
Since Reset:									
P_Drill	25	2	2.00	--	2	1	--	288.00	--
Limit	25	2	2.00	2	2	1	1.000	288.00	1989.83
Wait	95	72	53.61	--	72	34	--	2031.72	--
Since Beginning:									
P_Drill	46	2	1.94	--	2	0	--	304.14	--
Limit	46	2	1.94	2	2	0	0.972	304.14	2112.92
Wait	116	72	34.63	--	72	0	--	2149.35	--
L_Start from P_Start				Since Reset:		23		Start	46
L_Start to P_Clean				Since Reset:		23		Start	44

Process Traffic Limit Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
P_Drill	25	2	2.00	2	2	1	1.000	288.00	1989.83

Process and traffic statistics are provided since the last statistics reset and since the beginning of the model run. Traffic limit statistics are also summarized at the end of the report.

Queue statistics

There are three categories of queue statistics:

- Queue attributes
- Queue usage statistics
- Queue wait statistics

Queue attributes

Queue attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of queue attributes and their definitions, see "Queue attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Queue usage statistics

Queue usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed queue attributes. Queue usage statistics provide information about the number of claims on a queue. A *claim* is the use of one unit of a queue's capacity by a load or vehicle during a simulation; loads or vehicles claim a queue by executing a `move` action to move into the queue.

Queue usage statistics are defined as follows:

Statistic	Description
Name	The name of the queue.
Total	The total number of claims on the queue.
Current	The number of current claims on the queue.
Average	The average number of claims on the queue.
Capacity	The defined maximum number of simultaneous claims on the queue.
Maximum	The maximum number of simultaneous claims on the queue.
Minimum	The minimum number of simultaneous claims on the queue.
Utilization	The fraction of the queue's capacity that loads or vehicles utilized simultaneously. **Note** If the queue has a capacity of 0, the utilization statistic reports a value of 1.
Average time	The average time of a claim on the queue.
Average wait time	The average time that a claim waited for the queue. This is the average of *all* claims on the queue, including those claims that did not wait.

Queue wait statistics

Queue wait statistics are full statistics that appear in the printed AutoMod reports and in displayed queue attributes. Queue wait statistics provide information about claims that waited for a queue; a claim must wait for a queue when the queue has no available capacity.

Queue wait statistics are defined as follows:

Statistic	Description
Total	The total number of claims that waited for the queue.
Current	The number of claims that are currently waiting for the queue.
Average	The average number of claims that waited for the queue.
Maximum	The maximum number of claims that simultaneously waited for the queue.
Minimum	The minimum number of claims that simultaneously waited for the queue.
Average time	The average time that waiting claims spent waiting for the queue. This is the average of only those claims that waited for the queue.

Example reports

Example 1—Standard queue statistics

An example .report file containing standard queue statistics is shown here:

Queue Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
Space	41	1	1.00	Infinite	2	0	--	87.80	--
Q_Queue1	2	1	0.88	1	1	0	0.883	1590.00	1467.09
Q_Queue2	2	1	0.77	1	1	0	0.767	1380.00	1122.73

Queue statistics are summarized for each queue in the model. The first row shows the statistics for `space`, the territory where all loads are created. Util and Av_Wait statistics are not reported for queues with infinite capacity.

Example 2—Full queue statistics

An example .report file containing full queue statistics for space and queue1 is shown here:

Queue Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
Space	46	1	1.00	Infinite	2	1	--	78.26	--
Since Reset:									
Q_Queue1	3	1	1.00	1	1	0	1.000	1200.00	2369.74
Wait	13	11	9.22	--	12	7	--	2552.02	--

Queue Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
Since Beginning:									
Q_Queue1	4	1	0.94	1	1	0	0.942	1695.00	3189.82
Wait	14	11	6.65	--	12	0	--	3417.66	--

Queue statistics are provided since the last statistics reset and since the beginning of the model run. The first row shows the statistics for `space`, the territory where all loads are created. Util and Av_Wait statistics are not reported for queues with infinite capacity.

Random number stream statistics

Random number stream statistics are standard or full statistics that appear only in the printed AutoMod reports. Random number stream statistics provide information about the number of samples taken from random streams in the simulation.

Random number stream statistics are defined as follows:

Statistic	Description
Name	The name of the random stream.
Total	The total number of samples that were drawn from the stream.

Example report

Example 1—Random number stream statistics

There is no difference between standard and full random number stream statistics. An example of random number stream statistics is shown here:

Random Number Streams

Name	Total
stream0	609
stream1	140

Resource statistics

There are four categories of resource statistics:

- Resource attributes
- Resource usage statistics
- Resource wait statistics
- Resource down statistics

For information about the state of a resource in the simulation, see "State statistics" on page 101.

Resource attributes

Resource attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of resource attributes and their definitions, see "Resource attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Resource usage statistics

Resource usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed resource attributes. Resource usage statistics provide information about the number of claims on a resource. A *claim* is the use of one unit of a resource's capacity by a load or vehicle during a simulation; loads or vehicles claim a resource by executing a `get` or `use` action.

Resource usage statistics are defined as follows:

Statistic	Description
Name	The name of the resource.
Total	The total number of claims on the resource.
Current	The number of current claims on the resource.
Average	The average number of claims on the resource.
Capacity	The defined maximum number of simultaneous claims on the resource.
Maximum	The maximum number of simultaneous claims on the resource.
Minimum	The minimum number of simultaneous claims on the resource.
Utilization	The fraction of the resource's capacity that loads or vehicles utilized simultaneously. If the **Utilization by Availability** check box is clear, resource downtime is considered idle time when computing utilization. If the **Utilization by Availability** check box is selected, the utilization calculation excludes resource downtime. Resource utilization by availability is calculated using the following equation: $\text{Util} = \dfrac{\text{total} \times \text{average time}}{\text{capacity} \times (\text{absolute clock} - (\text{average down time} \times \text{total times down}))}$ **Note** • You can use resource attributes to calculate a resource's utilization in a model. To calculate the absolute utilization (since the beginning of the simulation), use absolute attributes in the formula. To calculate the relative utilization (since the last statistics reset), use relative attributes and the relative clock (instead of the absolute clock) in the formula. • If the resource has a capacity of 0, the utilization statistic reports a value of 1.
Average time	The average time of a claim on the resource.

Statistic	Description
Average wait time	The average time that a claim waited for the resource. This is the average of *all* claims on the resource, including those claims that did not wait for the resource.
State	The name of the current state of the resource.

Resource wait statistics

Resource wait statistics are full statistics that appear in the printed AutoMod reports and in displayed resource attributes. Resource wait statistics provide information about claims that waited for a resource. A load or vehicle must wait to claim a resource when the resource has no available capacity.

Resource wait statistics are defined as follows:

Statistic	Description
Total	The total number of claims that waited for the resource.
Current	The number of claims that are currently waiting for the resource.
Average	The average number of claims that waited for the resource.
Maximum	The maximum number of claims that simultaneously waited for the resource.
Minimum	The minimum number of claims that simultaneously waited for the resource.
Average time	The average time that a waiting claim spent waiting for the resource (this is the average of only those claims that waited for the resource).

Resource down statistics

Resource down statistics are full statistics that appear in the printed AutoMod reports and in displayed resource attributes. Resource down statistics provide information about resource downtimes.

Resource down statistics are defined as follows:

Statistic	Description
Total	The total number of times the resource was taken down.
Current	The current status of the resource: 0 indicates the resource is up, 1 indicates the resource is down.
Average	The fraction of time that the resource was down.
Capacity	This number is always one, indicating that a resource can be taken down.
Maximum	The maximum number of times the resource was taken down simultaneously.
Minimum	The minimum number of times the resource was taken down simultaneously.
Utilization	The fraction of time that the resource was down.

Statistic	Description
Average time	The average time a resource was down.
Average wait time	The average time that a load or vehicle waited to take down a resource (this number is always zero, because `take down` actions are instantaneous).
State	The name of the current state of the resource; the state statistic is only reported if a state set was defined in the model and attached to the resource.

Example reports

Example 1—Standard resource statistics

An example .report file containing standard resource statistics is shown here:

Resource Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait	State
R_Op1	47	2	1.77	2	2	0	0.883	135.32	738.14	Working
R_Res1	3	1	0.85	1	1	0	0.847	1015.84	2293.46	Processing
R_Res2	2	1	0.48	1	1	0	0.484	870.51	1224.02	Processing

Resource statistics are summarized for each resource in the model. Util and Av_Wait statistics are not reported for resources with infinite capacity.

Example 2—Full resource statistics

An example .report file containing full resource statistics for one resource is shown here:

Resource Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait	State
Since Reset:										
R_Res2	3	0	0.99	1	1	0	0.991	1189.74	2842.29	Failed
Wait	8	6	7.11	--	8	6	--	3197.58	--	
Down	1	1	0.01	1	1	0	0.009	30.79	0.00	Failed
Since Beginning:										
R_Res2	4	0	0.74	1	1	0	0.738	1327.56	3782.08	Failed
Wait	9	6	5.25	--	9	0	--	4202.31	--	
Down	1	1	0.00	1	1	0	0.004	30.79	0.00	Failed

Resource statistics are provided since the last statistics reset and since the beginning of the model run.

State statistics

State statistics are standard or full statistics that appear in the printed AutoMod reports. State statistics also appear in displayed reports for state monitors and in displayed attributes for resources. State statistics provide information about the state of a resource or state monitor.

State statistics are reported for each individual resource or state monitor in the model, and inherit the statistics type defined for the resource or state monitor. For example, if a resource is using standard statistics, state statistics for that resource also use the standard format.

State statistics are defined as follows:

Statistic	Description
Name	The name of the resource or state monitor.
States	The name of the state.
Total	The total number of times the state has been used by the resource or state monitor.
Current	The current status of the state: 0 indicates the state is inactive, 1 indicates this is the active state of the resource or state monitor.
Average time	The average time the resource or state monitor spent in the state.
Percent time	The percent of time the resource or state monitor spent in the state.
Color	The color defined for the state.

Example reports

Example 1—Standard state statistics

An example .report file containing standard state statistics for two resources is shown here:

```
Resource State Statistics
```

Name	States	Total	Cur	Av_Time	%_Time	Color
R_Res1						
	Processing	23	0	1143.14	30.43	Green
	Idle	0	0	0.00	0.00	Blue
	Warmup	2	1	276.24	0.64	LtGreen
	Failed	6	0	325.87	2.26	Red
	Shutdown	1	0	57600.00	66.67	Black
R_Res2						
	Processing	24	0	1069.34	29.70	Green
	Idle	0	0	0.00	0.00	Blue
	Warmup	2	1	599.11	1.39	LtGreen
	Failed	7	0	276.79	2.24	Red
	Shutdown	1	0	57600.00	66.67	Black

State statistics are summarized for each resource in the model.

Example 2—Full state statistics

An example .report file containing full state statistics is shown here:

```
Resource State Statistics
```

Name	States	Total	Cur	Av_Time	%_Time	Color
R_Res1						
Since Reset:						
	Processing	21	0	1260.16	30.63	Green
	Idle	0	0	0.00	0.00	Blue
	Warmup	2	1	900.00	2.08	LtGreen
	Failed	2	0	268.33	0.62	Red
	Shutdown	1	0	57600.00	66.67	Black
Since Beginning:						
	Processing	44	0	1198.99	30.53	Green
	Idle	0	0	0.00	0.00	Blue
	Warmup	3	1	784.16	1.36	LtGreen
	Failed	8	0	311.48	1.44	Red
	Shutdown	2	0	57600.00	66.67	Black

State statistics are provided since the last statistics reset and since the beginning of the model run.

Table statistics

Table statistics include two categories of statistics:

- Table entry statistics
- Frequency statistics

Table entry statistics

Table entry statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed table attributes. Table entry statistics provide information about tabulated values.

Table entry statistics are defined as follows:

» **Tip** You can display tabulated data in a histogram by defining a bar chart business graph.

Statistic	Description
Name	The name of the table.
Entries	The number of tabulated values.
Mean	The mean tabulated value.
Standard deviation	The standard deviation of the tabulated values.
Maximum	The largest tabulated value.

Statistic	Description
Minimum	The smallest tabulated value.

Frequency statistics

Frequency statistics are full statistics that are included in the printed AutoMod reports and in displayed table attributes. Frequency statistics provide a frequency distribution of all entries in the table.

Frequency statistics are defined as follows:

Statistic	Description
Frequency class	The range of tabulated values being measured.
Frequency	The number of tabulated values in this class.
Percent of total	The percent of total tabulated values in the frequency class.
Cumulative percent	The cumulative percent of total tabulated values since the initial frequency class.

Example reports

Example 1—Standard table statistics

An example .report file containing standard table statistics for two tables is shown here:

```
Table Statistics Summary

Name        Entries   Mean      Std Dev   Max       Min
T_Table1    63        36.8851   8.2197    59.3198   28.3879
T_Table2    3589      4.1335    2.4386    13.0000   1.000
```

Table statistics are summarized for each table in the model.

Example 2—Full table statistics

An example .report file containing full table statistics for one table is shown here:

```
Table Statistics Summary

Name        Entries   Mean       Std Dev    Max         Min
T_Table1    70        123.9581   273.3033   1020.7929   28.6941

Full Table Statistics for              T_Table1
Since Reset:      09/16/97   09:00:00

Obs:   70   Mean:   123.96   StdDev:   273.30   Max:   1020.79   Min:   28.69

Frequency class                    Frequency    % Of Total        Cumulative %
                  <     0.000      0            0.00              0.00
```

...						
=>	28.000	<	30.000	10	14.29	14.29
=>	30.000	<	32.000	11	15.71	30.00
=>	32.000	<	34.000	2	2.86	32.86
=>	34.000	<	36.000	4	5.71	38.57
=>	36.000	<	38.000	4	5.71	44.29
=>	38.000	<	40.000	1	1.43	45.71
		>=	40.000	38	54.29	100.00
Average value of entries >				40.000 :	201.358	

Since Beginning: 09/17/97 09:00:00

Obs: 70 Mean: 123.96 StdDev: 273.30 Max: 1020.79 Min: 28.69

Frequency class				Frequency	% Of Total	Cumulative %
		<	0.000	0	0.00	0.00
...						
=>	28.000	<	30.000	28	21.05	21.05
=>	30.000	<	32.000	22	16.54	37.59
=>	32.000	<	34.000	4	3.01	40.60
=>	34.000	<	36.000	6	4.51	45.11
=>	36.000	<	38.000	7	5.26	50.38
=>	38.000	<	40.000	8	6.02	56.39
		>=	40.000	58	43.61	100.00
Average value of entries >				40.000 :	148.215	

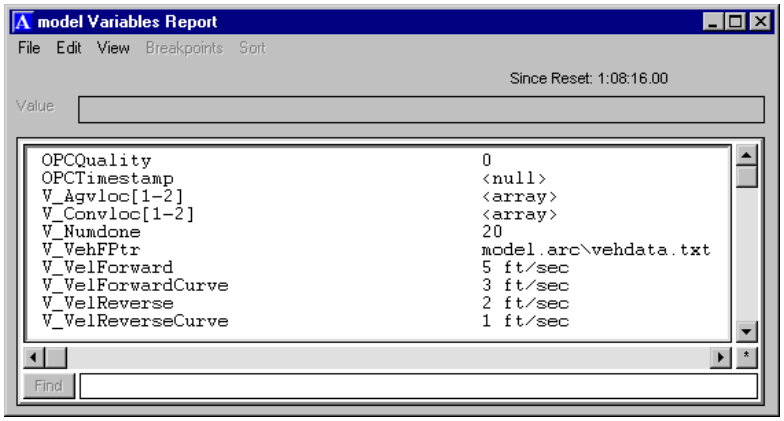
Frequency statistics are provided since the last statistics reset and since the beginning of the model run.

Variable values

Variable values appear only in the displayed variables report. Displaying variable values is frequently useful when debugging modeling errors.

To display the model's variable values during a simulation

Click **View > Variables**. The Variables Report opens:



The Variables Report contains a list with an entry for every variable in the selected model or submodel. If the variable is arrayed, the dimensions and range of the array appear in brackets to the right of the variable name. You can display the values of array elements by double-clicking the name of the arrayed variable in the dialog box. For more information, see "Sorting and displaying arrayed variable values" on page 107.

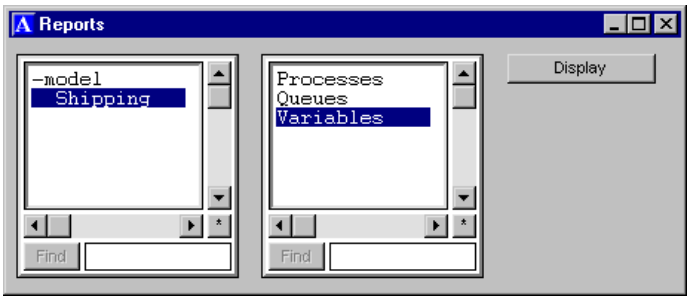
Clicking **View > Variables** lists only those variables defined in the top process system in a model. To display the value of variables defined in a submodel, you must display the submodel's variable report. For more information, see "Displaying variables defined in a submodel" on page 106.

Displaying variables defined in a submodel

To display the value of submodel variables

1. In the Simulation window, click **View > Reports**.

   The Reports dialog box opens, as shown here:



2. In the model list, select the name of the submodel for which you want to display variable values.

   For example, the submodel `Shipping` is selected in the previous dialog box.

3. In the entity list, select **Variables**.
4. Click **Display**.

The Variables Report opens displaying a list of values for variables defined in the selected submodel.

Sorting and displaying arrayed variable values

To display the value of arrayed variable elements

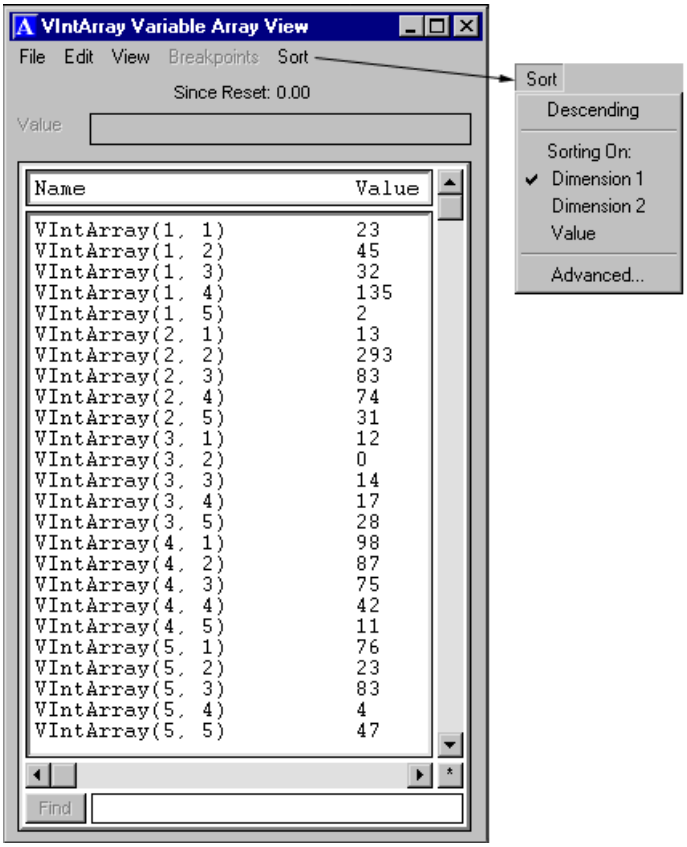
In the Variables Report, double-click the name of the arrayed variable for which you want to display element values. (For information about how to display the Variables Report, see "Variable values" on page 105.)

The Variable Array View dialog box opens, listing the value of each variable in the array.

By default, the Variable Array View lists arrayed variables sequentially, according to dimension. For example, the variable `V_IntArray` is a 5x5 integer array with the following values:

```
23      45      32      135     2
13      293     83      74      31
12      0       14      17      28
98      87      75      42      11
76      23      83      4       47
11      24      16      0       0
```

The Variable Array View lists the array elements in ascending order starting with the first dimension (the rows), followed by the second dimension (the columns), as shown here:



For convenient access to large arrays, you might want to sort variables in a different order (for example descending order) or by a different variable dimension. You can sort the variable values using the following procedure.

To sort arrayed variable values

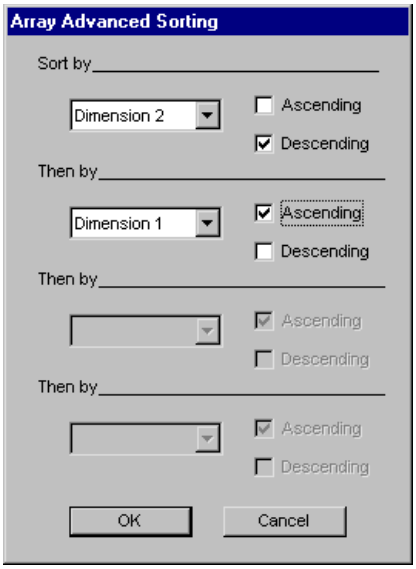
In the Variable Array View dialog box, click **Sort** then select a dimension number (to sort by a specific dimension) or select **Value** (to sort by the variable's values).

You can also select either **Ascending** or **Descending** to change the current sort order.

To sort arrayed variable values using advanced sorting

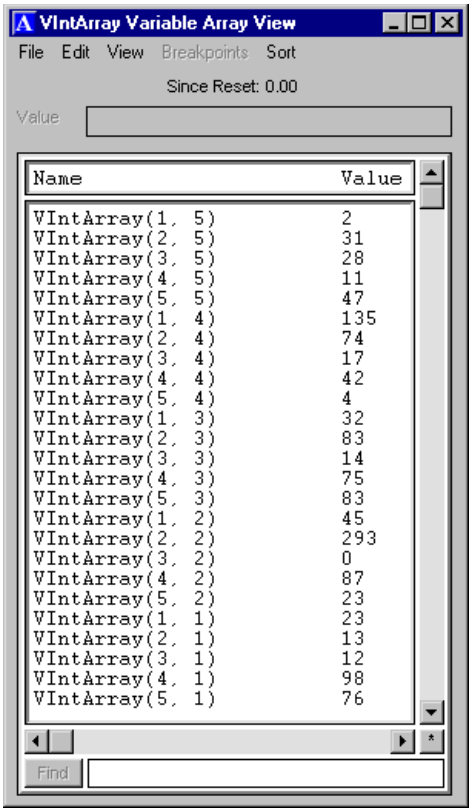
1. Click **Sort > Advanced**.
2. In the Array Advanced Sorting dialog box, select the dimension(s) and order you want to use for sorting the array.

   For example, the following dialog box sorts the selected array in descending order in the second dimension, followed by ascending order in the first dimension:



3. Click **OK**.

The Variable Array View report opens, with the arrayed values sorted according to the dimension and order you specified in step 2.



Name	Value
VIntArray(1, 5)	2
VIntArray(2, 5)	31
VIntArray(3, 5)	28
VIntArray(4, 5)	11
VIntArray(5, 5)	47
VIntArray(1, 4)	135
VIntArray(2, 4)	74
VIntArray(3, 4)	17
VIntArray(4, 4)	42
VIntArray(5, 4)	4
VIntArray(1, 3)	32
VIntArray(2, 3)	83
VIntArray(3, 3)	14
VIntArray(4, 3)	75
VIntArray(5, 3)	83
VIntArray(1, 2)	45
VIntArray(2, 2)	293
VIntArray(3, 2)	0
VIntArray(4, 2)	87
VIntArray(5, 2)	23
VIntArray(1, 1)	23
VIntArray(2, 1)	13
VIntArray(3, 1)	12
VIntArray(4, 1)	98
VIntArray(5, 1)	76

In this example, the arrayed variable `V_IntArray` is sorted by descending order in the second dimension, followed by ascending order in the first dimension.

Movement system statistics

You can view statistics for the following movement system entities:

- Chains
- Control points
- Motors
- Paths
- Photoeyes
- Sections
- Stations
- Vehicles

Chain statistics

There are three categories of chain statistics:

- Section attributes
- Chain entry statistics
- Chain usage statistics

Section attributes

Section attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of section attributes and their definitions, see "Section attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Chain entry statistics

Chain entry statistics are standard or full statistics that appear only in the printed AutoMod reports. Chain entry statistics provide information about the number of carriers that traveled on a chain in a power & free system.

Chain entry statistics are defined as follows:

Statistic	Description
Chain	The name of the chain.
Entries	The total number of carriers that traveled on the chain.
Average Time/Entry	The average time a carrier spent on the chain.
Contents	There are three categories: • **Average** - The average number of carriers on the chain. • **Current** - The current number of carriers on the chain. • **Maximum** -The maximum number of carriers that were on the chain simultaneously. **Note** When a power & free chain transfers to itself, the carrier is in the same section twice for a period of time, which can affect statistics. The maximum statistic becomes one higher than expected. Average contents, average time, and so on are also affected. To avoid this, draw two sections instead of one for continuous or connected power & free chains.

Chain usage statistics

Chain usage statistics appear in displayed reports and in displayed section attributes. Chain usage statistics provide information about the number of carriers that traveled on a chain in a power & free system.

Chain usage statistics are defined as follows:

Statistics	Description
Name	The name of the section.
Total	The total number of carriers that traveled on the section.
Current	The current number of carriers on the section.
Average	The average number of carriers on the section.

Statistics	Description
Maximum	The maximum number of carriers on the section simultaneously.
Minimum	The minimum number of carriers on the section simultaneously.
Utilization	Utilization is not reported for chains; this value is always 0.
Average time	The average time a carrier spent on the section.
Average wait time	Average wait time is not reported for chains; this value is always 0.

Example report—chain statistics

There is no difference between standard and full chain statistics. An example of a printed .report file containing chain statistics is shown here:

```
Power and Free Chain Statistics
```

Chain	Entries	Average Time/Ent	Contents Ave	Current	Max
chain31	79	40.56	0.89	1	3
chain32	81	44.23	1.00	0	3
chain33	81	28.84	0.65	1	3
chain34	80	28.75	0.64	1	3

Chain statistics are summarized for each chain in the power & free system.

Control point statistics

There are two categories of control point statistics:

- Control point attributes
- Control point usage statistics

Control point attributes

Control point attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of control point attributes and their definitions, see "Control point attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Control point usage statistics

Control point usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed control point attributes. Control point usage statistics provide information about the number of claims on a control point in a path mover system. A *claim* is the use of one unit of a control point's capacity by a vehicle during a simulation. As a vehicle moves through a path mover system, it claims the control point at the end of

each segment on which it is traveling. A vehicle releases a claim according to the release specification of the claimed control point . A vehicle always has at least one control point claimed.

Control point statistics are defined as follows:

Statistic	Description
Name	The name of the control point.
Total	The total number of claims on the control point.
Current	The number of current claims on the control point.
Average	The average number of claims on the control point.
Capacity	The defined maximum number of simultaneous claims on the control point.
Maximum	The maximum number of simultaneous claims on the control point.
Minimum	The minimum number of simultaneous claims on the control point.
Utilization	The fraction of the control point's capacity that vehicles utilized simultaneously. High numbers can indicate excessive vehicle congestion at this location. **Note** If the control point has a capacity of 0, the utilization statistic reports a value of 1.
Average time	The average time of a claim on the control point.
Average wait time	The average time that each claim waited for the control point. This is the average of *all* claims on the control point, including those claims that did not wait.

Example report—Control point statistics

There is no difference between standard and full control point statistics. An example of a printed .report file containing control point statistics is shown here:

Control Point Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
cp1	12	1	0.24	1	1	0	0.238	71.34	4.64
cp2	40	1	0.97	1	1	0	0.974	87.67	49.94
cp3	6	0	0.23	1	1	0	0.229	137.45	10.59
cp4	3	0	0.12	1	1	0	0.120	143.49	0.00

Control point statistics are summarized for each control point in the path mover system.

Motor statistics

There are two categories of motor statistics:

- Motor attributes
- Motor usage statistics

Motor attributes

Motor attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of motor attributes and their definitions, see "Motor attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Motor usage statistics

Motor usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed motor attributes. Motor usage statistics provide information about the state of a motor (either up or down) in a conveyor or power & free system.

Motor statistics are defined as follows:

Statistic	Description
Name	The name of the motor.
Status	The current status of the motor; either `up` or `down`.
Total down	The total number of times the motor was taken down.
Percent up	The percent of time the motor was up.
Average time up	The average time the motor was up.

Example report—Motor statistics

There is no difference between standard and full motor statistics. An example of a printed .report file containing motor statistics is shown here:

```
Motor Statistics

Name      Status   Total Down   %_Up   Av_Time Up
M_sec1    Up       0            1.00   3600.00
M_sec2    Up       1            0.94   1689.02
M_sec3    Up       1            0.90   1624.19
M_sec4    Down     30           0.50   58.06
```

Motor statistics are summarized for each motor in the movement system.

Path statistics

There are two categories of path statistics:

- Path attributes
- Path usage statistics

Path attributes

Path attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of path attributes and their definitions, see "Path attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Path usage statistics

Path usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed path attributes. Path usage statistics provide information about the number of claims on a path in a path mover system.

Path statistics are defined as follows:

Statistic	Description
Name	The name of the path.
Total	The total number of claims on the path.
Current	The number of current claims on the path.
Average	The average number of claims on the path.
Capacity	The defined maximum number of simultaneous claims on the path.
Maximum	The maximum number of simultaneous claims on the path.
Minimum	The minimum number of simultaneous claims on the path.
Utilization	Utilization is not reported for paths; this value is always 0.
Average time	The average time of a claim on the path.
Average wait time	The average time that each claim waited for the path. This is the average of *all* claims on the path, including those claims that did not wait.

Example report—Path statistics

There is no difference between standard and full path statistics. An example of a printed .report file containing path statistics is shown here:

```
Guide Path Statistics

Name    Total  Cur  Average  Capacity   Max  Min  Util  Av_Time   Av_Wait
path11  6136   0    0.96     Infinite   4    0    --    13.56
path12  4587   1    1.09     Infinite   4    0    --    20.53
path13  4619   1    0.98     Infinite   4    0    --    18.24
path15  3066   0    0.12     Infinite   1    0    --    3.45
```

Path statistics are summarized for each path in the path mover system.

Photoeye statistics

There are two categories of photoeye statistics:

- Photoeye attributes
- Photoeye usage statistics

Photoeye attributes

Photoeye attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of photoeye attributes and their definitions, see "Photoeye attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Photoeye usage statistics

Photoeye usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed photoeye attributes. Photoeye statistics provide information about the state of a photoeye (either cleared or blocked) in a conveyor or power & free system.

Photoeye usage statistics are defined as follows:

Statistic	Description
Name	The name of the photoeye.
Status	The current status of the photoeye; either `cleared` or `blocked`.
Total blocked	The total number of times the photoeye was blocked.
Percent blocked	The percent of time the photoeye was blocked.
Average time blocked	The average time the photoeye was blocked.

Example report—Photoeye statistics

There is no difference between standard and full photoeye statistics. An example of a printed .report file containing photoeye statistics is shown here:

```
Photoeye Statistics
```

Name	Status	Total Blocked	%_Blocked	Av_Time Blocked
photoeye1	Cleared	237	0.31	4.65
photoeye2	Blocked	239	0.33	4.99
photoeye3	Blocked	8	0.14	62.76
photoeye4	Cleared	73	0.15	7.33

Photoeye statistics are summarized for each photoeye in the movement system.

Section statistics

There are three categories of section statistics:

- Section attributes
- Section entry statistics
- Section usage statistics

Section attributes

Section attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of section attributes and their definitions, see "Section attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Section entry statistics

Section entry statistics are standard or full statistics that appear only in the printed AutoMod reports. Section entry statistics provide information about the number of loads that traveled on a section in a conveyor system. Section entry statistics are defined as follows:

Statistic	Description
Entries	The total number of loads that traveled on the section.
Average Time/Entry	The average time a load spent on the section.
Contents	There are three categories: • **Average** - The average number of loads on the section. • **Current** - The current number of loads on the section. • **Maximum** - The maximum number of loads that were on the section simultaneously.

Section usage statistics

Section usage statistics appear in displayed reports and in displayed section attributes. Section usage statistics provide information about the number of loads that traveled on a section in a conveyor system.

Section usage statistics are defined as follows:

Statistic	Description
Name	The name of the section.
Total	The total number of loads that traveled on the section.
Current	The current number of loads on the section.
Average	The average number of loads on the section.
Maximum	The maximum number of loads on the section simultaneously.
Minimum	The minimum number of loads on the section simultaneously.

Statistic	Description
Utilization	Utilization is not reported for sections; this value is always 0.
Average time	The average time a load spent on the section.
Average wait time	Average wait time is not reported for sections; this value is always 0.

Example report—Section statistics

There is no difference between standard and full section statistics. An example of a printed .report file containing section statistics is shown here:

```
Conveyor Section Statistics

                    Average    Contents
Section   Entries   Time/Ent   Ave     Current   Max
sec_1     63        604.17     10.57   14        15
sec_2     62        457.69     7.88    10        13
sec_3     60        843.52     14.06   17        24
sec_4     137       27.41      1.04    2         5
```

Section statistics are summarized for each section in the conveyor system.

Station statistics

There are two categories of station statistics:

- Station attributes
- Station usage statistics

Station attributes

Station attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of station attributes and their definitions, see "Station attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Station usage statistics

Station usage statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed station attributes. Station usage statistics provide information about the number of claims on a station in a conveyor or power & free system. A *claim* is the use of one unit of a station's capacity by a load or vehicle during a simulation.

Station usage statistics are defined as follows:

Statistic	Description
Name	The name of the station.
Total	The total number of claims on the station.
Current	The number of current claims on the station.
Average	The average number of claims on the station.

Statistic	Description
Capacity	The defined maximum number of simultaneous claims on the station.
Maximum	The maximum number of simultaneous claims on the station.
Minimum	The minimum number of simultaneous claims on the station.
Utilization	The fraction of the station's capacity that loads or vehicles utilized simultaneously. **Note** If the station has a capacity of 0, the utilization statistic reports a value of 1.
Average time	The average time of a claim on the station.
Average wait time	The average time a claim waited for the station. This is the average of *all* claims on the station, including those claims that did not wait.

Example report—Station statistics

There is no difference between standard and full station statistics. An example of a printed .report file containing station statistics is shown here:

Station Statistics

Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
sta1	57	0	0.01	Infinite	1	0	--	0.50	--
sta2	5	2	1.96	2	2	0	0.980	1410.51	575.67
sta3	9	2	1.86	2	2	0	0.928	742.33	317.79
sta4	57	57	28.10	100	57	0	0.281	1774.63	0.00

Station statistics are summarized for each station in the movement system.

Vehicle statistics

Vehicle statistics are available for all movement systems except conveyors. There are five categories of vehicle statistics:

- Vehicle attributes
- Vehicle trip statistics
- Vehicle congestion statistics
- Additional displayed status statistics
- Current status summary

Vehicle attributes

Vehicle attributes are *not* included in the printed AutoMod reports or in displayed reports; they appear only when displaying attribute values (see "Displaying entity attributes" on page 78). For a list of vehicle attributes and their definitions, see "Vehicle attributes" in the AutoMod Help. Attributes that provide the same information as reported statistics are not displayed.

Vehicle trip statistics

Vehicle trip statistics are standard or full statistics that appear in the printed AutoMod reports and in displayed vehicle attributes. Vehicle trip statistics are divided into four categories:

- Delivering
- Retrieving
- Going to park
- Parking

The categories list the following statistics for each vehicle (subdivided by vehicle type) and for all vehicles of each type:

Statistic	Description
Percent Of Total Time	The percent of total simulation time that vehicles spent delivering loads, retrieving loads, going to park, or parking.
Trips Made	The number of trips made to deliver loads, retrieve loads, or go to park.
Average Time/Trip	The average time spent per trip made to deliver loads, retrieve loads, or go to park. (This statistic is not available for parking.)

Vehicle congestion statistics

Vehicle congestion statistics appear only in the printed AutoMod reports. The statistics provide the average capacity and total capacity lost due to vehicle congestion.

Statistic	Description
Average Capacity	The percentage of vehicle capacity lost due to congestion. The statistic represents the percent of time that vehicles were attempting to move but were blocked, either by a preceding vehicle or by the limited capacity of a block or control point in the vehicle's path. The statistic does not include decelerations, accelerations, or time spent going more slowly than defined.
Total Capacity	The total vehicle capacity lost due to congestion. This statistic represents the total amount of vehicle capacity that was available, but was not used because vehicles were blocked, either by a preceding vehicle or by the limited capacity of a block or control point in the vehicle's path. The total capacity is calculated by multiplying the average capacity by the total number of vehicles participating in the simulation. The statistic does not include decelerations, accelerations, or time spent going more slowly than defined.

Current status of vehicles

Vehicle current status statistics are standard or full statistics that appear in the printed AutoMod reports, displayed reports, and in displayed vehicle attributes. The following statistics are provided for the current status of each vehicle:

Statistic	Description
Vehicle	The vehicle's type and ID number. Vehicles' ID numbers are sequentially assigned to each vehicle at the beginning of a simulation.
Current activity	A description of the action the vehicle is performing at the current simulation time.
Status	The current state of the vehicle; possible values are: • **Deliver** - The vehicle is delivering a load. • **Deliver Setdown** - The vehicle is setting down a delivered load. • **Deliver WaitForLoad** - The vehicle has arrived at the destination of an onboard load, but the load has not yet moved off the vehicle (for example, the load can be using a resource or delaying while onboard the vehicle). • **Idle** - The vehicle is currently parked and has no scheduled jobs to perform. • **Move** - The vehicle is moving to a location not associated with a load (for example, a parking location). • **No State** - The vehicle has not yet been assigned a task; the vehicle is in No State only at the beginning of a simulation, before finding work on a scheduling list or selecting a job from the vehicle's list of scheduled jobs. • **Retrieve** - The vehicle is retrieving a load. • **Retrieve Pickup** - The vehicle is picking up a retrieved load. • **Retrieve WaitForLoad** - The vehicle has been reserved and is waiting for a claimed load to arrive at the vehicle's current location.
Time in Cur Status	The time since the vehicle last changed status.
Current Location	If the vehicle is stopped, this is the current location of the vehicle. If the vehicle is moving, this is the vehicle's most recently occupied location.
Destination Location	The location to which the vehicle is traveling. If the vehicle has arrived at its destination and has not yet selected another destination location, then the vehicle's current location and destination location are the same.

Statistic	Description
Load On Or Claimed By Veh	(Appears only in the printed AutoMod reports and displayed reports.) If a vehicle is delivering a load (one or more loads are currently onboard), this is the ID number and the load type of the load that the vehicle is currently delivering. If an empty vehicle is moving to pick up a load, this is the ID number and the load type of the load that the vehicle is picking up. **Note** If a vehicle is delivering or picking up a vehicle from another movement system, this is the vehicle type and ID number of the vehicle from the other system. For more information about moving vehicles into another movement system, see "Moving a vehicle" in the notes on using the `move` action in the AutoMod Help.

Additional displayed status statistics

Displayed vehicle attributes include the following additional status statistics:

Statistic	Description
Number of loads claimed by vehicle	The number of loads the vehicle is currently claiming.
Loads claimed by vehicle	A list of the loads the vehicle is currently claiming.
Current configuration	The vehicle's current kinematic configuration (this value is null for vehicles using graphics without kinematic configurations).
Destination configuration	When a vehicle is changing from one configuration to another, this statistic indicates the destination configuration (this value is null for vehicles using graphics without kinematic configurations).

Current status summary

Current status summary statistics appear only in the printed AutoMod reports. The following statistics are summarized for each vehicle type:

Statistic	Description
Type	The name of the vehicle type.
Average Time In Cur State	The average time that vehicles of this type have spent in their current state. This amount is figured by adding the length of time that each vehicle has spent in its current state and dividing the sum by the number of vehicles of this type.
Maximum Time In Cur State	The maximum length of time that any vehicle of this type spent in its current state.
Minimum Time In Cur State	The minimum length of time that any vehicle of this type spent in its current state.

Example report—Vehicle statistics

There is no difference between standard and full vehicle statistics. An example of a printed .report file containing vehicle statistics is shown here:

Vehicle Statistics

Delivering			Retrieving			Going To Park			Parking
Percent Of Total Time	Trips Made	Average Time/ Trip	Percent Of Total Time	Trips Made	Average Time/ Trip	Percent Of Total Time	Trips Made	Average Time/ Trip	Percent Of Total Time
DefVehicle:									
0.668	245	78.48	0.127	131	27.97	0.148	95	44.81	0.057
0.547	260	60.64	0.183	164	32.15	0.191	126	43.72	0.061
0.579	250	66.65	0.166	155	30.85	0.178	115	44.69	0.055
All DefVehicle:									
0.598	755	68.59	0.159	450	30.32	0.173	336	44.41	0.058

The Average Capacity Lost Due To Congestion is 7.6 Percent
Of Total Capacity Which is 0.2 Vehicles

Current Status Of Vehicles

Vehicle	State	Time In Cur State	Current Location	Destination Location	Load On or Claimed By Veh
DefVehicle(1)	Deliver	88.76	cprepair	cprepair	Load 386 (L_Cart)
DefVehicle(2)	Deliver Setdown	1.36	cplabelin	cplabelin	Load 388 (L_Cart)
DefVehicle(3)	Retrieve	36.98	cpin2	cplabelout	Empty

Current Status Summary

Type	Average Time In Cur State	Maximum Time In Cur State	Minimum Time In Cur State
DefVehicle	42.36	88.76	1.36

Vehicle statistics are summarized for each vehicle and vehicle type in the movement system.

Changing the run control

The run control determines the length of time the run simulates, how statistics are gathered, and so on. The run control cannot be changed with a runtime license, only with a full version of the AutoMod software. However, if your runtime license includes AutoStat, you can define different run control parameters for a set of runs within AutoStat. For more information, see "Analysis information" on page 5.17

Automatically pausing a model during a simulation

There are two ways to automatically pause a model during a simulation:

- Setting alarms
- Setting breakpoints

Setting alarms

You can pause the model at a specific simulation time by setting an alarm.

To pause the model at a specific time

1. Click **Control > Set Alarm**.
2. In the Set Alarm dialog box, type the time at which you want to pause the simulation, then select either **Absolute** or **Relative** from the list. An absolute time is defined relative to the start of the simulation (time 0 + n); a relative time is defined relative to the current clock time (current time + n).

Setting breakpoints

By setting breakpoints, you can automatically pause a simulation whenever certain conditions (usually load or vehicle interactions) occur for a specific entity. For example, setting breakpoints for a queue causes the simulation to automatically pause when:

- A load or vehicle enters a queue.
- A load or vehicle leaves a queue.
- A load or vehicle attempts to enter a queue but can't (for example, because of limited queue capacity).

Each time the simulation pauses because of a breakpoint, the software prints a message to the Messages dialog box indicating the event that caused the simulation to pause.

There are two ways to set breakpoints for an entity: from the window that displays entity reports or from the window that displays entity attributes.

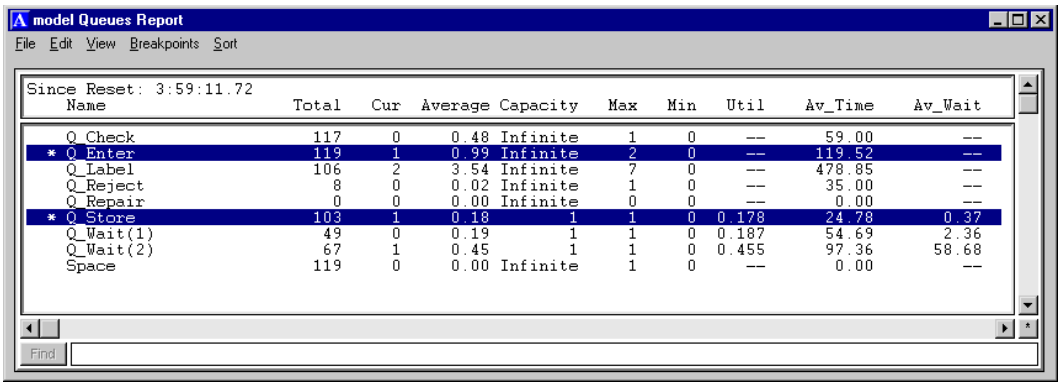
Setting breakpoints in an entity's report

1. Display the report for the type of entity for which you want to create a breakpoint (for example, if you want to set a breakpoint for a queue, display the queues report; if you want to set a breakpoint for a resource, display the resources report, and so on).

   For more information on how to display an entity report, see "Displaying entity reports" on page 75.
2. In the report, select the entity or entities for which you want to create a breakpoint.
3. From the **Breakpoints** menu, select one or more conditions for which you want to define a breakpoint.

   » **Tip** You can cause the simulation to pause for all conditions by selecting **Add All Breakpoints**.

An asterisk appears next to the name of the selected entity or entities. The asterisks next to the queue names indicate that breakpoints are set for two queues: `Q_Enter` and `Q_Store`.



Name	Total	Cur	Average	Capacity	Max	Min	Util	Av_Time	Av_Wait
Q_Check	117	0	0.48	Infinite	1	0	--	59.00	--
* Q_Enter	119	1	0.99	Infinite	2	0	--	119.52	--
Q_Label	106	2	3.54	Infinite	7	0	--	478.85	--
Q_Reject	8	0	0.02	Infinite	1	0	--	35.00	--
Q_Repair	0	0	0.00	Infinite	0	0	--	0.00	--
* Q_Store	103	1	0.18	1	1	0	0.178	24.78	0.37
Q_Wait(1)	49	0	0.19	1	1	0	0.187	54.69	2.36
Q_Wait(2)	67	1	0.45	1	1	0	0.455	97.36	58.68
Space	119	0	0.00	Infinite	1	0	--	0.00	--

An asterisk next to an entity name indicates that one or more breakpoints have been defined for the entity. To see the specific conditions for which breakpoints are defined, select the entity and click **Breakpoints** to display the menu. A check mark next to a condition name indicates that a breakpoint has been defined for that condition.

You have now set breakpoints for the entity.

Setting breakpoints in an entity's attributes dialog box

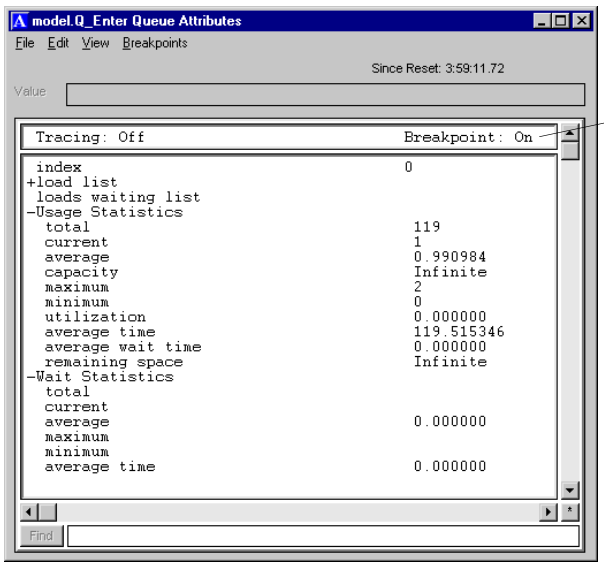
1. Display the attributes for the entity for which you want to add breakpoints.

   For more information, see "Displaying entity attributes" on page 78.

2. From the **Breakpoints** menu, select one or more conditions for which you want to define a breakpoint.

   » **Tip** You can cause the simulation to pause for all conditions by selecting **Add All Breakpoints**.

The Breakpoint heading changes to **On**, as shown here:



Breakpoints have been set for this queue.

The Breakpoint heading **On** indicates that one or more breakpoints have been defined for the entity. To see the specific conditions for which breakpoints are defined, click **Breakpoints** to display the menu. A check mark next to a condition name indicates that a breakpoint has been defined for that condition.

You have now set breakpoints for the entity.

Removing breakpoints

There are two ways to remove breakpoints for an entity: from the window that displays entity reports or from the window that displays entity attributes.

▶ To remove breakpoints in an entity's report

1. Display the report for the type of entity for which you want to remove breakpoints.

   For more information on how to display an entity report, see "Displaying entity reports" on page 75.

2. In the report, select the entity or entities from which you want to remove breakpoints.
3. From the **Breakpoints** menu, select a condition for which a breakpoint has been set to remove the breakpoint.

When you remove a breakpoint from a condition, the check mark next to the condition in the **Breakpoints** menu disappears.

To remove all breakpoints defined for the selected entity or entities, click **Breakpoints > Remove All Breakpoints**. The asterisks next to the entity names in the Report dialog box disappear, indicating that no breakpoints are defined for the selected entities.

▶ Removing breakpoints in an entity's attributes dialog box

1. Display the attributes for the entity for which you want to remove breakpoints.

   For more information, see "Displaying entity attributes" on page 78.

2. From the **Breakpoints** menu, select a condition for which a breakpoint has been set to remove the breakpoint.

When you remove a breakpoint from a condition, the check mark next to the condition in the **Breakpoints** menu disappears.

To remove all breakpoints defined for the selected entity or entities, click **Breakpoints > Remove All Breakpoints**. The Breakpoint heading in the attributes dialog box changes to **Off**, indicating that no breakpoints are defined for the current entity.

3 Business Graphs

In this chapter, you will learn how to:

- Create different types of graphs.
- Define a graph's statistical content.
- Define the frequency of graph updates.
- Make changes to an existing graph.

Before you begin

Before using the information in this chapter, review the statistics that are written to the AutoMod report. For more information, see Chapter 2, "Running a Model."

Introduction

The Business Graphics feature enables you to create dynamic charts and graphs of statistics, including tabulated statistics. Bar graphs, line graphs, and pie charts change appearance to reflect the current status of the values they represent. You can define business graphs in either the edit or simulation environment.

Graph types

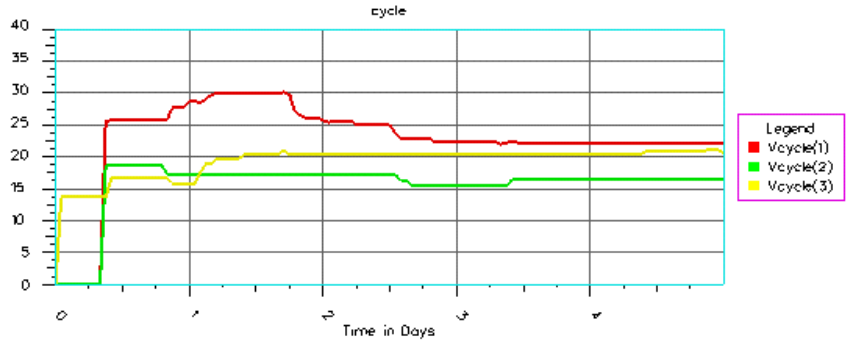
You can create the following types of graphs:

- Time line
- Bar chart
- Pie chart
- Data watch
- File

All graphs are visually updated at a user-defined frequency.

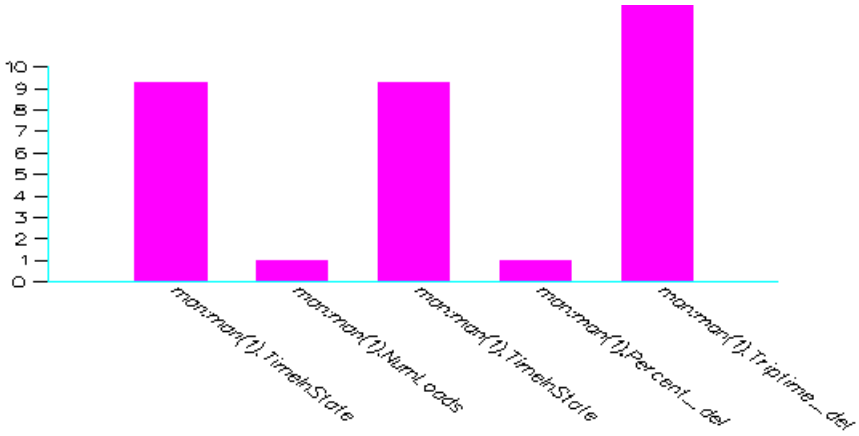
Time line

The lines on the time line are drawn across the graph as data accumulates.



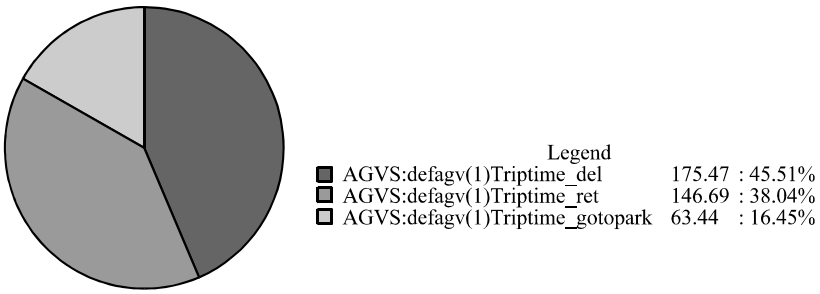
Bar chart

The bars on the bar chart rise and fall to display data as current values are updated.



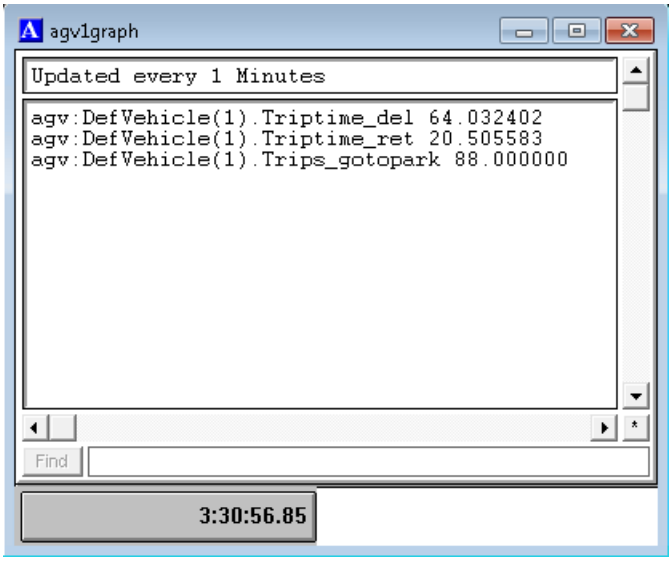
Pie chart

The sections of a pie chart narrow and widen interactively to display data as current values are updated.



Data watch

The data watch is a dynamic list of information. Messages are printed in the DataWatch window.



File

The file option collects the same types of data that are collected for line graphs, but prints that data to an outside file. You can then import this file in a business graphics package or spreadsheet.

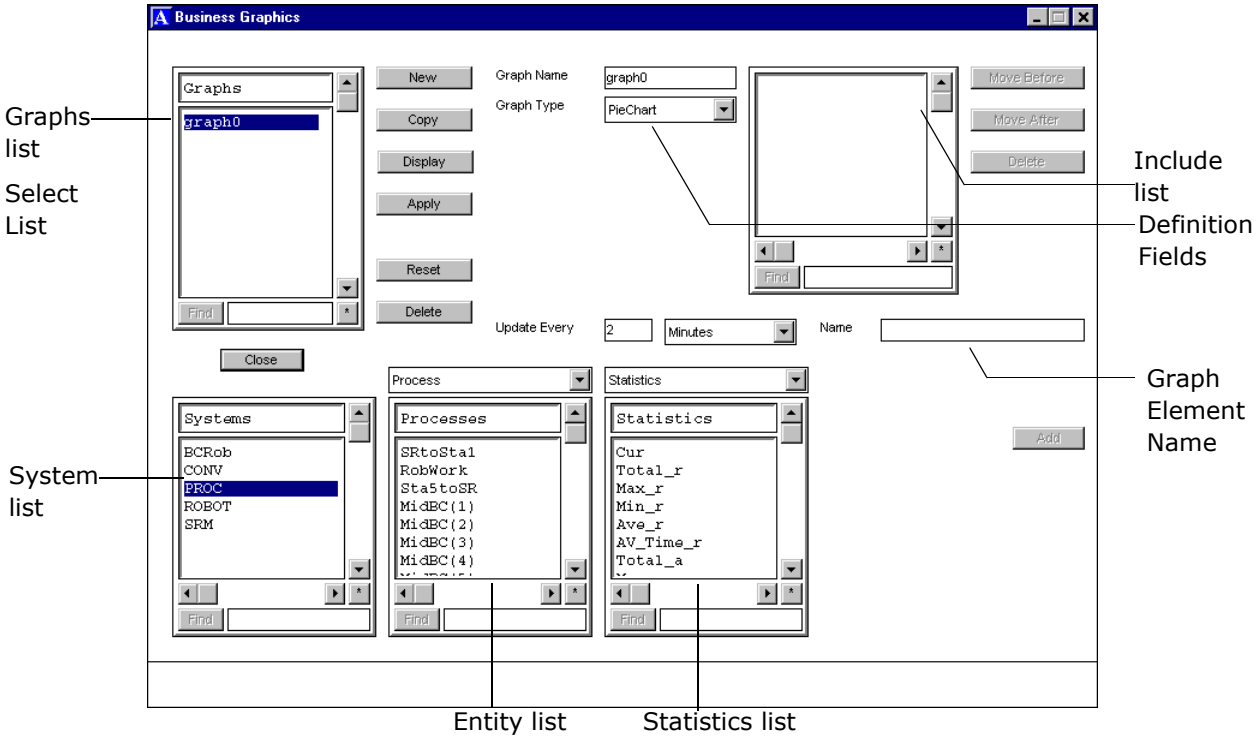
Displaying business graphs

Business graphs are defined in either the edit environment or the simulation environment.

- **Note** Business graphs defined by the model's creator with the full AutoMod license appear in the list every time you run the model. Graphs you define with the runtime license are only saved for the current run.

To display Business Graphs in the simulation environment

1. Click **View > Business Graphs** to display the Business Graphics dialog box.



2. Highlight the graph name in the **Graphs** list.
3. Click **Display**.

4 Printing

In this chapter, you will learn how to:

- Create a picture of your model suitable for printing
- Export business graphs automatically

Introduction

AutoMod does not allow you to print a view of your model directly to a printer. You can, however, save the current view of a model layout as a graphic file, then read that file and print it using an appropriate application outside of AutoMod.

» **Tip** Print the layout of the facility to use as a reference while you develop your model's process procedures.

Saving a picture to a file

AutoMod can save an image of the current view in the following popular graphical formats:

- JPEG
- Bitmap
- TIFF

For more information about each of these formats and also other non-graphical data formats that are available, see "View control menus" on page 6.

▶ **To save a picture to a file**

1. Click **View Control** from the AutoMod window to manipulate your view of the model.
2. Use the View Control check boxes, controls, and buttons to adjust the current view as needed to compose the picture.
3. Click **File** and select the type of graphical format in which to save the picture.
4. Optionally, alter the name of the file being generated and the directory where it will be saved.

■ **Note** The default file name is "screen.<ext>" and the default directory is the model directory.

5. Click **Save**.

   The output file is generated and saved in the directory.

Printing a graphic image file

Numerous applications are available that can read, edit and print graphic image files created by AutoMod. For information about how to print image files, see the documentation that accompanies the application you choose to use.

5 Analyzing a Model with AutoStat

In this chapter, you will learn how to:

- Open the AutoStat™ software.
- Define analyses.
- Make model runs.
- Display analysis results.
- Debug a model.

Before you begin

Before using AutoStat, you must be familiar with the basic functionality of the AutoMod product and your model.

For a discussion of *statistical techniques*, refer to *Discrete-Event System Simulation*, 3rd ed. (Banks, Carson, Nelson, and Nicol, 2000) or *Simulation Modeling & Analysis,* 3rd ed. (Law and Kelton, 1999).

For information about *simulation optimization*, refer to the following books and papers:

- Dennis E. Smith (1973), "Requirements of an 'Optimizer' for Computer Simulations," In *Naval Research Logistics Quarterly*, Volume 20, Number 1, pp. 161-179.
- Yolanda Carson and Anu Maria (1997), "Simulation Optimization: Methods and Applications," In *1997 Winter Simulation Conference Proceedings*, pp. 118-126.
- Royce O. Bowden and John D. Hall (1998), "Simulation Optimization Research and Development," In *1998 Winter Simulation Conference Proceedings*, pp. 1693-1698.

Smith's paper is the foundational work that describes requirements and proposed elements of future optimizers that will be used with simulation models to find good solutions for system design parameters. Carson and Maria present an overview of optimization methods that are currently being used or considered in the simulation community. Bowden and Hall review selected simulation optimization research and development topics.

For information about *evolutionary algorithms*, refer to the following books and papers:

- Hans-Paul Schwefel (1995), *Evolution and Optimum Seeking*. Wiley, New York.
- Thomas Bäck (1996), *Evolutionary Algorithms in Theory and Practice: Evolution Strategies, Evolutionary Programming, Genetic Algorithms*. Oxford University Press, New York.

The AutoStat optimization algorithm is based on evolution strategies, a type of evolutionary algorithm. Schwefel and Bäck are leaders in the research, development, and application of these search heuristics. We recommend Bäck's book for a current state-of-the-art review on the subject.

Overview of AutoStat

The AutoStat program provides run management and statistical analysis for models. The enhanced statistical analysis capabilities help manage your report files. When experimenting with a model, AutoStat helps you determine the warmup period and calculates confidence intervals for model statistics.

When comparing alternative system designs, AutoStat manages the data and statistical analyses for all the alternatives, reducing the time required for making decisions about optimal system design.

Section 1: Why use AutoStat?

The key to a successful simulation study is proper analysis. However, in most simulation studies, much of the time is spent developing the model, while little time is spent actually analyzing the model results. Sometimes decisions are based on just one run of the model. However, when random samples are used as inputs, a single simulation run might not be representative of the true behavior of a model. You could make erroneous inferences about the system being modeled if you only make one run. The AutoStat program helps you apply the proper statistical sampling techniques to your model in order to accurately estimate performance under random conditions.

AutoStat also allows you to get the most out of your hardware because you can develop your model incrementally and make model runs with AutoStat overnight to test and verify the model under a range of inputs.

Running a model early in the development process using AutoStat helps determine the model inputs (factors) and outputs (responses). Also, the model's sensitivity to factor changes is determined earlier, which can eliminate certain experiments. Modeling errors that only occur in long runs or when using different random numbers are detected early in the model-building process.

Finally, AutoStat helps you to determine which results and alternatives are statistically significant, which means that with high probability, the observed results are not caused by random fluctuations.

Although AutoStat takes the drudgery out of conducting experiments and provides you with many graphical and statistical means of analyzing the output, exercise caution when drawing conclusions from the results. AutoStat might report statistical significance, but you must determine whether a change is realistic and practical. For example, AutoStat might indicate that adding one more piece of equipment will improve product throughput by one unit per hour, but depending on the value of the product and the cost of the equipment, the new equipment might not be justified.

Deterministic and stochastic models

The need for statistical analysis depends upon whether the simulation model is deterministic or stochastic. A *deterministic* model does not contain random input. Thus, there is no variability in output from run to run. Only a single run is necessary when all the input is deterministic, but few simulation models fit into this category.

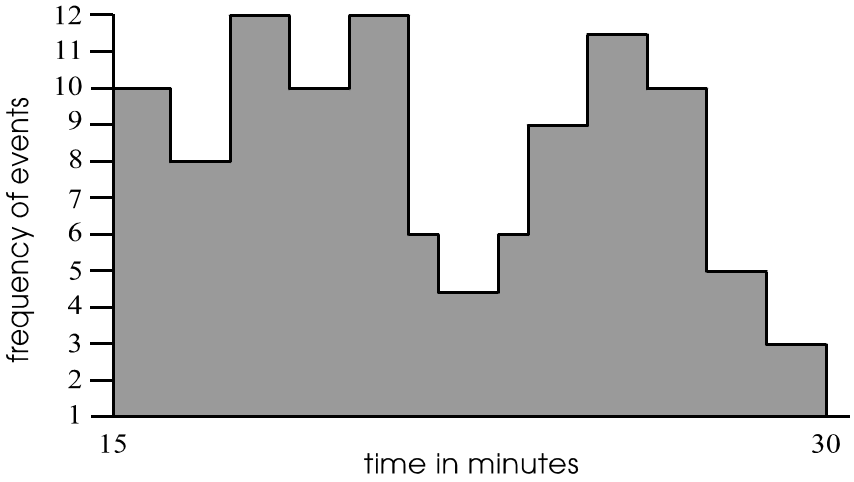
Stochastic means random. A stochastic model contains one or more random sets of input that produce random output. Two runs using different random numbers provide two sets of output. Each set is different from the other, and each is equally representative of system behavior. For this reason, apply proper statistical sampling techniques to the analysis of a stochastic model in order to accurately estimate performance under all random conditions.

AutoStat allows you to easily create complex experiments that produce multiple representative samples of system behavior. AutoStat provides automated procedures for measuring and interpreting the output.

The need for statistical analysis

Suppose you travel to work on a regular, reliable rapid-transit system. Trains arrive at your station every 10 minutes beginning at 6:00 A.M. The trip to work takes 20 minutes every time. This is an example of a system that can be modeled deterministically. Whatever time you arrive at the station, you know when the next train will arrive and how long it will take to get to work. If you arrive at 7:15 A.M., the train will come at 7:20 and you'll be at work at 7:40. In fact, any day you arrive at 7:15, the train drops you off at 7:40.

Now, suppose that same rapid-transit system is not so reliable or regular. Trains arrive every eight to 15 minutes and travel time can vary between 15 and 30 minutes. You cannot be sure of the exact time you'll arrive at work on a particular day because the input is stochastic. By recording the travel time of trains each day, you get a distribution similar to the one shown in the input distribution graph that follows. The travel time has a minimum of 15 minutes and a maximum of 30 minutes. The following graph gives the most accurate picture of the distribution of travel times and can be used as model input for a simple model of train travel. Two factors involved might be the chance that a train breaks down and the time of day.



A stochastic model is much more common than a deterministic model in material handling and manufacturing systems. A deterministic model assumes perfect knowledge of all system characteristics, including arrival times of parts, service times, and down times.

Sources of randomness in material handling and manufacturing systems are abundant. The complexity of the interactions between these random factors is virtually impossible to estimate with mathematical models. Simulation, and the subsequent output analysis it provides, can accommodate the randomness found in:

- Part arrivals
- Time to failure
- Time to repair
- Inventory levels
- Product batch size
- Order picking size and sequence
- Inspection outcome (good, rework, or reject)
- Number of available forklifts, AGVs, and so on
- Number of available employees
- Product physical size

The following table shows a representation of a simulation experiment. For deterministic models, the input and output are constants. In a stochastic model, some of the input is a random variable and some, or all, of the output is based on random variables. AutoStat refers to the input as *factors* and the output as *responses*.

Input (factors)	**Output (responses)**
Examples of Factors:	**Examples of Responses:**
Part interarrival time	Part cycle time
Machine service time	Average queueing time
Number of machines	Conveyor utilization
Types of material handling	Percent downtime
Inventory space	Throughput

The purpose of simulation is to obtain a useful answer. Thus, you must understand how much variability is in a simulation's output. As the randomness of the output increases, a greater number of runs is needed to obtain accurate results. For this reason, statistics is a necessary tool. An average is not always enough information to explain a random variable.

For example, suppose you make one run of a model. In that run there is an average of 3.9 busy forklifts handling order picking. Are 3.9 busy forklifts typical? Or was this particular run representative of an extreme case that occurs infrequently and that arose due to some odd combination and timing of arrival times, downtimes, and other modeled events?

Too often, stochastic models are run once and the output is given as "the answer." As illustrated, that answer might represent a typical day or an unusual day. The only way to determine if the day was typical or unusual is to run a sufficient number of replications. Many runs are usually necessary to get an accurate picture of system behavior.

Case study

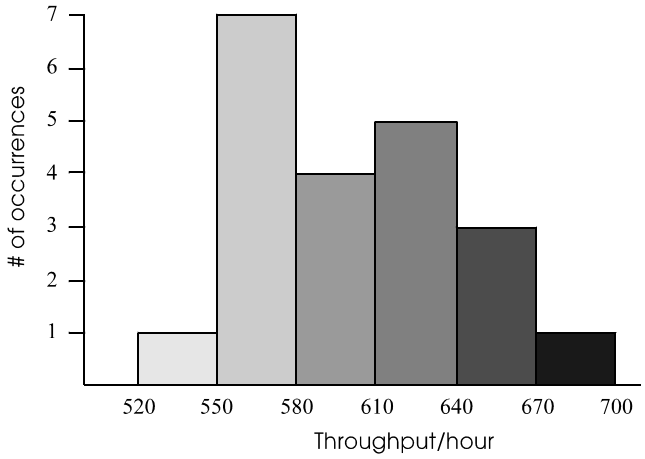
Slide Easy Corporation manufactures slide bearings for a variety of commercial uses. Slide Easy's business is expanding to the point where it will not be able to meet its customer's demand if changes do not occur. To help identify a way to meet this demand, Slide Easy's managers made a simulation model of their factory to test several new material handling configurations and process changes.

To illustrate the random variability in some models, one of the proposed configurations was simulated for 20 runs, each with different random seeds. This gave 20 statistically independent runs. The resulting average hourly throughputs of slide bearings per hour are tabulated here:

Replication Average Number		**Replication Average Number**	
Hourly Throughput		**Hourly Throughput**	
1	685.86	11	567.74
2	659.05	12	638.89
3	555.24	13	605.10
4	658.53	14	624.26
5	560.40	15	614.62
6	599.18	16	570.58
7	618.11	17	576.32
8	534.31	18	613.67
9	599.31	19	565.06
10	605.53	20	568.22

Notice the variability of the mean hourly throughput from run to run. Over all 20 replications, the average hourly throughput is 601.00 slide bearings/hour. Hourly results range from a minimum of 534.31 to a maximum of 685.86 bearings/hour. Each response in the table is equally as legitimate as another, and each is representative of system performance under certain circumstances.

The runs that produced these results were identical except for different random numbers. The following histogram illustrates the distribution of average throughput/hour over the 20 replications.



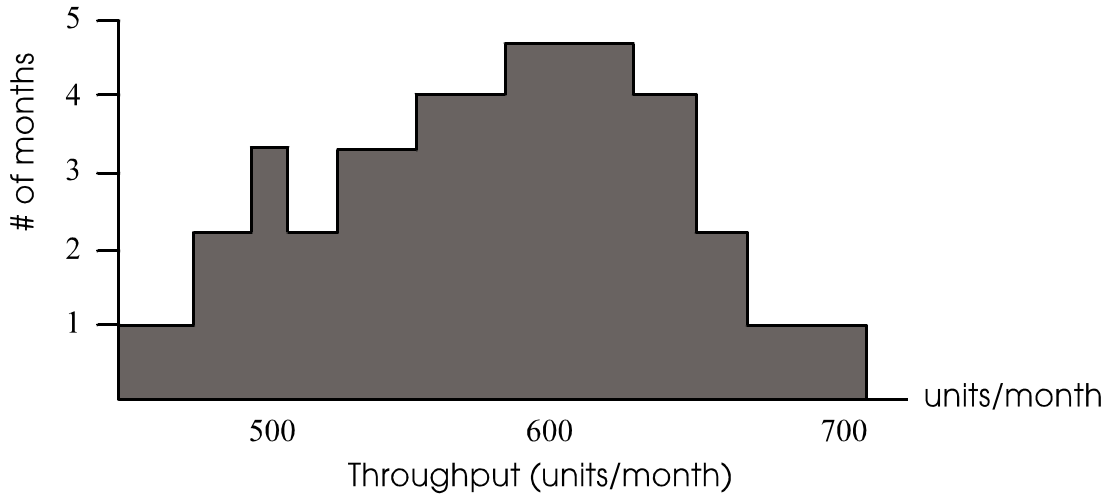
Running the model for only one replication ("one-run syndrome") and using the results as the answer can be dangerous. If that particular run provides the minimum response, or the maximum response, the estimate is far from the true mean response.

Output analysis

If a simulation is run long enough, random values from the input distributions eventually generate virtually every possible combination of events. If there is some probability, no matter how small, of every conveyor going down or all forklift drivers calling in sick, it eventually happens in the model.

Each output from a run contains information on *one* instance of what can happen. The output does not contain information on the likelihood of it happening. Suppose the response from a run shows a throughput of 475 units/month. This is the throughput given a particular set of random values from the input distributions. This might be a low value, a high value, or close to the average. In reality, true system throughput might average 400 units/month.

The result of this one run is just one of many possible throughputs. One data point cannot give you a clear idea of what is happening. It takes many data points to get an accurate estimate of system response. The data points from several runs can be combined in a histogram such as shown here. This histogram illustrates the resulting response distribution.



The goal of a simulation is to estimate mean or long-run average system responses. This can be accomplished by using a point estimate, which is a single number representing the best guess of the actual long-run average system response. Also, a confidence interval estimate is a range of numbers that can be used to represent the uncertainty in the point estimate. Based on the variance of the observations, a range of values is given which contains, with some level of confidence, the actual long-run average system response.

Outputs from many simulation runs are combined to give a point estimate of each response. A point estimate is usually the average, or mean, value from each run. Other examples of point estimates are the median, which is the middle value from an ordered list of numbers, and the mode, which is the most frequent value. (All the analysis with the AutoStat tool, and all discussions in this chapter, deal with the *estimation of means*. The estimation of medians, modes, quantiles, and other statistical parameters is beyond the scope of this guide.)

A point estimate is not the complete answer. An estimate is only as good as the amount of data used to generate it. A point estimate can be generated from one run as the result from that run. Additional runs are necessary to compute a standard error or *confidence interval*. Confidence intervals are used with stochastic (random) models.

- **Note** If you use the AutoStat tool to generate a confidence interval for a deterministic model, the results have no meaning, because all the results for a deterministic model are the same and the width of the confidence interval is zero.

A confidence interval gives a range of values around a point estimate. The point estimate is at the midpoint of the confidence interval as long as there are no restrictions on the response values. For example, suppose the average number of jobs in process is estimated at 5.55, based on the average of several runs. You cannot conclude, with 100 percent confidence, that 5.55 is the average number of jobs in process in the long run. However, based on the outputs from each run, AutoStat can compute a confidence interval. For example, you might

be able to say you are 95 percent confident that the mean number of jobs in process is between 4.96 and 6.13. Here, 95 percent is called the confidence level.

Confidence intervals are widened by increasing the *confidence level*; more values must be included in the range to reduce the chance of an error. To tighten the interval, you can reduce the confidence level or you can make more runs. In general, to reduce the width of the confidence interval by approximately half, you need to have 4N total runs, where N is the number of runs you have already made.

The following table shows the effect of a different number of runs and confidence levels on the confidence interval for mean jobs in process. Notice that the width of the confidence intervals shrinks as the number of runs increases and as the confidence level decreases.

	Number of Runs					
Confidence Level	**5**		**10**		**20**	
	High	**Low**	**High**	**Low**	**High**	**Low**
90 percent	6.775	4.305	6.01	5.07	6.09	5.43
95 percent	7.05	4.63	6.12	4.96	6.15	5.37
99 percent	7.84	3.84	6.38	4.70	6.29	5.23

Case study, continued

Slide Easy Corporation is considering adding two or three grinding machines to meet the anticipated demand of 40,000 bearings/day. Ninety-five percent confidence intervals on mean throughput for two and three additional machines are: 40,300 +/- 2100 and
43,700 +/- 2200 bearings/day, respectively. The point estimate, from the addition of both two and three machines, is greater than the anticipated demand. However, the confidence interval for the addition of two machines implies that the true mean throughput can be anywhere between 38,200 and 42,400.

The way to determine whether this configuration can handle the demand of 40,000 bearings/day, is by making more runs to use more data. By using more data, you can reduce the width of the confidence interval and be more certain that you will have the required 40,000 bearings/day. Since the midpoint of the confidence interval is so close to the goal, the number of additional runs might prove to be quite large. You do know, with 95 percent certainty, that the addition of three machines will easily accommodate the demand, since the complete confidence interval (41500, 45900) lies to the right of Slide Easy's goal of 40,000 bearings/day. So, you can either decide on three additional machines, or decide on two additional machines and accept the possibility of as few as 38,200 bearings/day on average, or make the necessary extra runs to determine if two more machines are sufficient.

How long should I run the simulation?

One of the decisions simulation analysts face in every project is "How long should my model runs be?" If the model is not run long enough, it could be biased by the initial conditions of the model, causing the analyst to draw inaccurate conclusions.

The importance of run length and initial conditions varies depending on whether your model is terminating or non-terminating.

Terminating and non-terminating simulations

The difference between a terminating and non-terminating simulation is the stopping point. A *terminating simulation* has a predetermined, though possibly random, stopping point or event. The stopping point might be a specific period of time, such as a week of production. Other possible stopping points might be the completion of a specified number of parts, or the length of a queue reaching a specified level.

A *non-terminating simulation* measures the long-time performance of the system. In this case, there is no predetermined stopping point that must be incorporated in the model. The simulation is intended to provide an estimate of how the system performs over an indefinitely long period of time, not just a week or year. In a terminating simulation, the nature of the problem predetermines the stopping point or *run length*. In a non-terminating, or steady state, simulation, the analyst determines the run length based on a desire to obtain accurate estimates of long-term system performance.

For example, a manufacturer might be interested in determining whether a facility can fill all the weekly orders under various scenarios. A terminating simulation can be used in this case. Orders arrive on the first day of the week and those orders determine the week's schedule. There can be two stopping points in this model. First, it makes sense to simulate until all orders are filled and stop when there are no more orders. Given a 10-hour work day, six days per week, another stopping point is 60 hours into the simulation. If the orders have not been filled by this time, the facility was unable to handle the demand under the conditions that arose during the week. Several replications under each scenario give a clearer estimate of how well each scenario handles orders.

The *initial conditions* are important in a terminating simulation. If one scenario is initialized with the system empty and idle but all machines in operating order, and another scenario has some machines under repair and a full inventory level, the results cannot be adequately compared. Therefore, be sure the initial conditions in your models represent realistic starting points. Also, use comparable initial conditions in all the scenarios you are studying. One way to simulate a terminating system is to set the warmup period to zero time units and the run length to the time that the simulation terminates. Then make multiple runs like this to determine the random effects.

As another example, if your goal is to estimate the average time it takes to build an automobile, you'll probably consider this a non-terminating simulation. There is no initial condition that must be set for each day. The start of each day looks like the end of the previous day. The cars are left in the same place overnight and work continues the next day, or the plant might operate 24 hours per day. There is no predetermined stopping point. The analyst can specify any

run length. Conditions in the plant vary over time and cause differences in the time it takes to build each car. However, over a long period of time, the system can be treated as having a steady state.

In the long run, initial conditions might have no effect on a non-terminating simulation. However, this is only true as the length of the simulation increases indefinitely. Poor initial conditions can bias a non-terminating simulation if the model is not run long enough to dilute the effects of the initial conditions. For this reason, if possible, choose initial conditions that do not overly influence the results, because simulations cannot be run for an infinite amount of time. Also, data analysis methods such as Replication/Deletion can help to eliminate the initialization bias.

Replication/Deletion for non-terminating simulations

Replication/Deletion seeks to remove initial bias by only using data obtained after a warmup period in each replication.

Note AutoStat can help you determine the warmup period (see "Setting a warmup time for a model" on page 160).

The replication/deletion method strives to use steady-state data in the formation of point estimates and confidence intervals for the various responses, which is accomplished by obtaining the average level of the response for each replication after the warmup period. These averages can be shown to be independent and approximately normal random variables. Thus, they can be used to construct a confidence interval for the steady-state mean value of the response.

It is difficult to be sure you have chosen a sufficiently long warmup period. Therefore, choose a replication length that is significantly longer than your best estimate of the warmup period. This will virtually eliminate bias in the point estimate.

The results from the Replication/Deletion method are relatively easy to understand. However, problems can occur if the warmup period is not properly chosen. This introduces a certain amount of bias in the point estimate for each replication. This bias can be reduced by choosing replication lengths that are long compared to the warmup period. For further discussion, see "Setting a warmup time for a model" on page 160.

For an example of the Replication/Deletion method, consider the average steady-state work-in-process at a job shop. The system starts out empty at the beginning of each replication, but it is determined that the system reaches steady state after about 2 hours. These first two hours are used as a warmup period and statistics are not collected during this time. A total run length of 80 hours is chosen, which is much longer than the warmup period. Ten replications are run to obtain a point estimate and confidence interval for the average work-in-process. The results of the first three replications are shown in the following table.

Replication	Average work-in-process (window =3)
1	7.1
2	7.6
3	7.3

Replication	Average work-in-process (window =3)
*	*
*	*

The point estimate for average work-in-process over the last 78 hours of each replication is found to be 7.3 jobs. A 95 percent confidence interval on the values obtained from the 10 replications is constructed. The resulting confidence interval is found to be 7.3 +/- 0.4 jobs. Thus, you can be 95 percent confident that the average work-in-process at steady state will be between 6.9 and 7.7 jobs.

Section 2: Getting started in AutoStat

The AutoStat product is a statistical analysis and run management program for models.

You use AutoStat to make numerous runs of your model to find errors that only occur in certain situations.

Once you are ready, you can experiment with a model by varying model parameters and varying random numbers to see what effect those changes have on output.

Basic terms and concepts

There are four main concepts in AutoStat:

factors	A factor is a model parameter that you want to vary, such as the number of resources, the speed of a conveyor section, or the processing time of a machine.
responses	A response is a statistic or output that you want to measure, such as throughput, cycle time, or machine utilization.
analyses	An analysis is an experiment that determines information such as warmup time, a model's reaction to varying one or more factors, DOE results, or sensitivity to input data. An analysis uses specific factor values to determine the effect upon the responses. For example, you can vary the number of resources from seven to nine and determine the effect on throughput.
results	Each type of analysis shows results in several forms: graphs, tables of statistics, confidence intervals, comparisons of scenarios, and so on.

Using AutoStat

To analyze a model using AutoStat

1. Open a model in AutoStat (see "Opening a model in the AutoStat program" on page 144).
2. Define an analysis (see "Defining analyses" on page 147).
3. Define factors (see "Defining factors" on page 179).
4. Make your runs (see "Section 5: Making runs" on page 190).
5. Define responses (see "Defining responses" on page 194).
6. Display results (see "Section 7: Displaying analysis results" on page 198).

Opening a model in the AutoStat program

To open a model in the AutoStat program

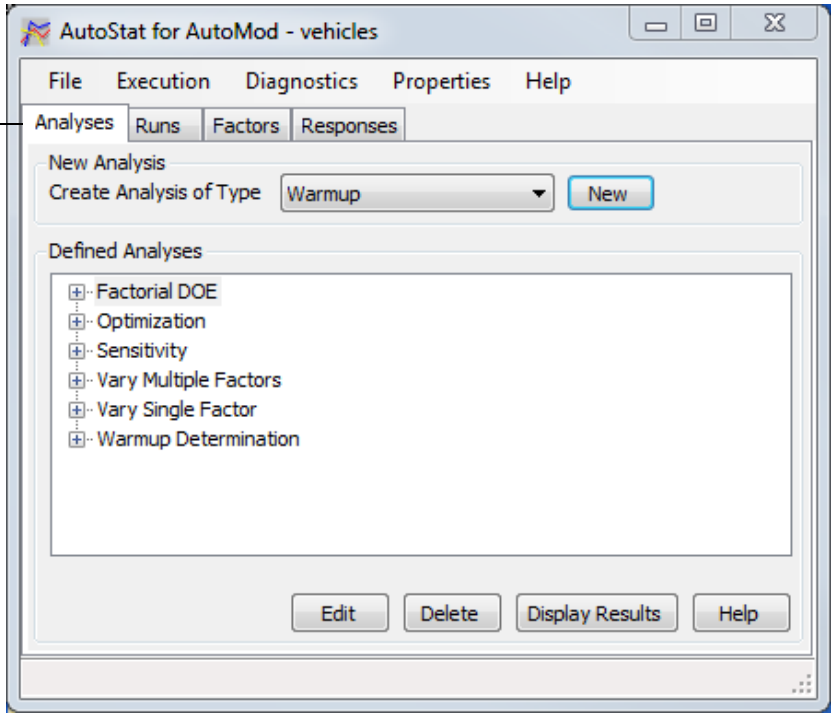
1. From the Start menu, select **AutoStat**.

   The Open a File dialog box opens.

   Note You can also open AutoStat from within a model by clicking **Run > Run AutoStat**.

2. Select the modelname.mod file from the modelname.dir directory (you must have a compiled model with a dir directory to open an AutoMod model in AutoStat).
3. The first time you open each model in AutoStat, use the Model Setup wizard to set up your model (see "Using the Setup wizard" on page 145

Each tab allows you to define, edit, and delete various items.



» **Tip** You can also open a model in AutoStat from the SimController. For more information, see "SimController menu bar" on page 355.

Using the Setup wizard

Whenever you open a model for the first time in the AutoStat program (that is, there is no <modelname>.sta directory), the Setup wizard asks you to:

- Determine whether the model contains randomness.
- Stop models that are potentially in an infinite loop.
- Estimate the model's warmup length.
- Define how long to run the model to collect statistics.
- Indicate whether to use Common Random Numbers (CRN) when conducting analyses that compare configurations.AutoStat uses the information in the wizard to set up the default model properties.

» **Tip** Use the context-sensitive help on each screen of the wizard if you have questions.

Setting model properties

AutoStat uses the information in the wizard to set up default model properties. If you need to change a model property after you have run the Setup wizard, use the Model Properties dialog box.

▶ **To edit model properties without using the wizard**

1. Click **Properties > Edit Model Properties**.

   The Model Properties dialog box opens.

2. Edit the information as needed.

Option	Definition
The model is random	Select this box if the model contains any sources of random input, such as varying part arrivals, varying processing times, and so on.
The model takes time to warm up	Select this box if the model requires a warmup before reaching a steady-state and type the time (see "Defining the run control for a warmup analysis" on page 158 if you are unsure of the time to type). This time can also be set using the warmup graph. ■ **Note** Changing the default warmup time in this dialog box changes any analyses using the default run control. New runs might be required. Also, if you change the warmup value and then try to audit responses, the runs will not be audited; the warmup length of the run must match the value in the dialog box for audits to work correctly.
Default sample time	Define the sample time (default snap length). For more information, see "Analysis information" on page 148. ■ **Note** Changing the **Default sample time** in this dialog box changes any analyses using the default run control. New runs might be required.

Option	Definition
Use Common Random Numbers	Select this box to use the same random numbers for related runs of an analysis that compares two or more model configurations. When this option is selected, the **Vary One Factor** and **Vary Multiple Factors** analyses use common random numbers by default. » **Tip** You can override the default setting for using common random numbers for a specific analysis by editing that analysis and setting the **Use Common Random Numbers** option in the analysis dialog box. For more information, see "Using Common Random Numbers (CRN)" on page 150.
Seed increment	This value is not used.
Stop runs that appear to be running long	Select this box to automatically stop long runs that might indicate an infinite loop. » **Tip** See the Help regarding infinite loops.
Error run limit	The maximum number of error runs that can occur before AutoStat automatically stops making runs. For more information about error runs, see "Making error runs" on page 215.
AutoSave while executing runs	If desired, AutoStat can save run results at a regular time interval. This ensures that runs made overnight are not lost due to power outages, and so on. If you do not choose AutoSave, runs are not saved until you click **File > Save**.
Message File Viewer	Type the executable name of the application you want to use to view message files (text files). If you do not know the name of the executable, use the **Browse** button to locate the desired application.

Section 3: Defining analyses

Analyses are statistical tests performed on your model. AutoStat has several types of analyses.

Defining analyses

There are several analysis types, including warmup determination, varying one or more factors, optimization, and designs of experiments.

To define an analysis

1. In the AutoStat window, select the **Analyses** tab.

   Note The Analyses tab shows a list of all defined analyses and their results. It also contains the run information (the number of runs and estimated time it will take to complete them) for the selected analysis.

2. From the **Create New Analysis of Type** list, select the desired analysis type.
3. Click **New**.
4. Define the information necessary for that type of analysis.

Analysis types

You can define several types of analyses using the **Analysis** tab.

Type	Definition
warmup	*Determine the amount of time it takes the model to warm up (reach a steady state)*. This is useful for models that start empty and need the system to fill up before statistics can be gathered. Models that are initialized with WIP do not need a warmup analysis. For more information, see "Warmup analysis" on page 156.
vary one factor	*Analyze how the model behaves when you change one factor*, such as the number of resources or a machine's processing time. For more information, see "Vary one factor analysis" on page 162.
vary multiple factors	*Analyze how the model behaves when you change more than one factor*, such as the number of resources or a machine's processing time, or other combinations of factors. For more information, see "Vary multiple factors analysis" on page 162.
single scenario	*Determine confidence intervals* for a single set of factor values. For more information, see "Single scenario analysis" on page 164.

Type	Definition
optimization	Let AutoStat *determine the best set of factor values for your model*. Define a set of factors to vary and the values for each of the factors. Then define a fitness function for your model that lists factors and responses to either maximize or minimize, and their relative importance. For example, you can define a function that minimizes the number of machines (within a defined range) while maximizing throughput. AutoStat automatically manages the runs to find the best combination possible. For more information, see "Optimization analysis" on page 165.
sensitivity analysis	*Quickly estimate how sensitive the model is to changes in one or more input parameters*. Use this analysis when you have rough data and want to determine whether you need more exact data. If the model is not sensitive to the input, the existing data can be used. If the model is sensitive to the input, you need more accurate data before you can draw conclusions from the model. For more information, see "Sensitivity analysis" on page 175.
factorial DOE	Perform a design of experiments (DOE) to *quickly estimate the effects of factors and combinations of factors* on the model. For more information, see "Factorial DOE analysis" on page 176.

Analysis information

When defining analyses, you must provide the following information:

- Name (either default or user-defined)
- Number of replications

Information	Definition
Name	The names of factors, analyses, responses, and other objects can be any length. All characters may be used but the TAB and ENTER (new line) keys may not be. The only other restriction is that two or more objects of the same type cannot have the same name. Objects of different types may have the same name; however, it is not recommended. For example, do not name both an AutoMod factor and a warmup analysis "ABC."

Information	Definition
Number of Replications	The number of runs to make using the same factor value or factor value combination. Each replication uses different random seeds to provide variation in the model. For example, suppose you want to vary one factor in an analysis. You assign the factor two values (4 and 7), and set the number of replications to 5. AutoStat will make 10 runs: 5 runs using the factor value 4, and 5 runs using the factor value 7. » **Tip** For all analyses except optimization, use at least 3 replications (optimizations might not require that many). *If you defined your model as deterministic, the number of replications field is not an option,* because you only need to make one replication. For more information, see "Deterministic and stochastic models" on page 135.
Run Control	**Run Control** defines how long the warmup is and how long the sample time (snap length) is. The *sample time* (snap length) is the time to simulate in each run, excluding warmup time. Shorter sample times make the runs finish faster, but the results are more variable, which causes you to need more runs to get enough data for small confidence intervals. Longer runs produce less variability in the runs and therefore fewer runs are required, but the runs take longer to complete. The default sample time can be set in the Model Properties dialog box. It can be overridden for an analysis by using a custom run control. There are two choices when defining run control for an analysis: • **Default -** The values set using either the Model Setup wizard or the Model Properties dialog box (see "Setting model properties" on page 145). • **Custom -** You can override the default warmup and sample time for an analysis by defining a custom run control. Type the length of time you want to simulate during each run. Regardless of whether you use the default or custom run control, the *run control defined in AutoStat overrides the model's run control.*

! **Important** If you change the default warmup or sample time, runs for analyses using the default time are no longer valid and need to be remade.

Using Common Random Numbers (CRN)

If you are analyzing a simulation to compare multiple system configurations (scenarios), you can reduce the required number of replications by analyzing the model using common random numbers.This method of analysis is also known as correlated sampling. Using common random numbers allows you to analyze configuration changes, while duplicating the experimental conditions for each configuration (that is, random events do not change from one configuration to the next).

For example, consider conducting an analysis to evaluate two different conveyor speeds in an assembly operation. In this case, the model consists of two possible configurations (one with the conveyor operating at a slower rate and the other with the conveyor operating at a faster rate). Using common random numbers, the analysis conducts a simulation for each possible conveyor speed, while using the same random numbers for events such as processing times, load arrival rates, , machine failures, and so on. By testing each configuration using common random numbers, you can ensure that the differences observed between simulations are not caused by random fluctuations, but are due to the changes in model configuration (in this example, the change in conveyor speed).

Statistically, using common random numbers can reduce the variance of the estimator of the difference between configurations, thus reducing the width of comparison confidence intervals for a given number of runs. As a result, you can save time by achieving a tighter confidence interval with fewer replications than would be necessary without common random numbers.

- **Note** You can only use common random numbers when analyzing a model built in AutoMod 11.0 or higher. Previous versions of the AutoMod software used either the LCG or Tausworthe random number generators; AutoStat does not support common random numbers with these generators.

For a model to benefit from common random number analysis, the person who built the model must have assigned separate random number streams to each source of random events in the model.

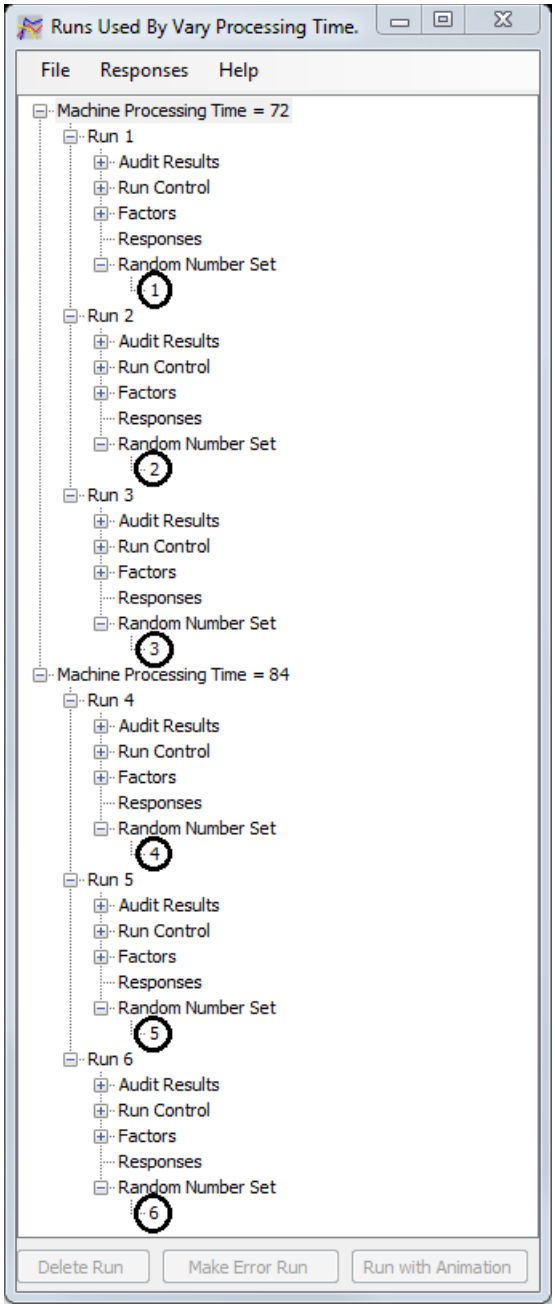
- **Note** To use CRN, make sure the `USE_CMRG_RANDOM_NUMBERS(Y/N)` option in the options file is set to `Y (Yes)`. You'll also need to define common random number stream names in the various data files where you want to use them.

To illustrate how AutoStat implements common random numbers, the following sections discuss the difference between an analysis that uses common random numbers, and one that does not.

How AutoStat varies random numbers in a non-CRN analysis

When you conduct an analysis that does *not* use common random numbers, AutoStat automatically increments the random number set for each run in a replication. As a result, each random stream uses a unique sequence of random numbers for each run in the analysis.

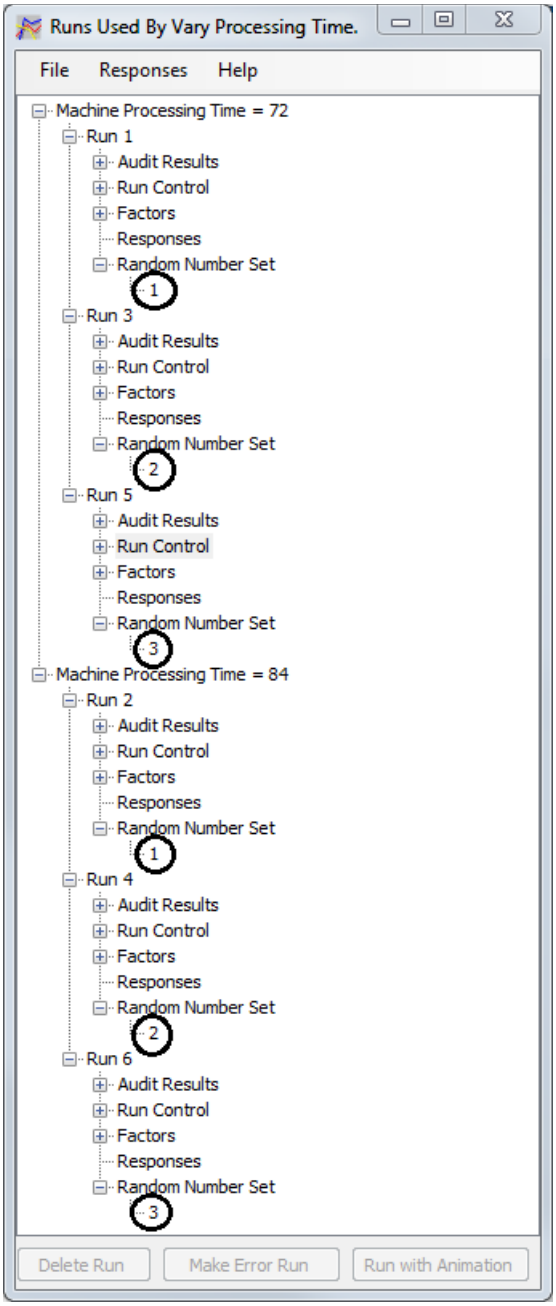
For example, a Vary One Factor analysis simulates two possible machine processing times: 72 seconds or 84 seconds. Assuming the analysis runs 3 replications, AutoStat increments the random number set for each run in the analysis as shown here:



How AutoStat varies random numbers in a CRN analysis

When you conduct an analysis using common random numbers, AutoStat increments the random number set for each replication, but uses the same sequence of random number sets for each factor value.

For example, when common random numbers are used in the Vary One Factor analysis discussed in the previous section, AutoStat uses the same sequence of random number sets for corresponding replications of each factor value as shown here:



Each replication consists of a group of correlated runs, which share the same random numbers but differ by the factor values you are varying (for example, run 1 and run 4, run 2 and run 5, and run 3 and run 6).

AutoStat ensures that related runs for each configuration use the same sequence of random numbers. However, when designing the model, you must ensure that the random numbers are used for the same purpose in each run, regardless of configuration changes.

Creating a CRN analysis for a model in AutoStat

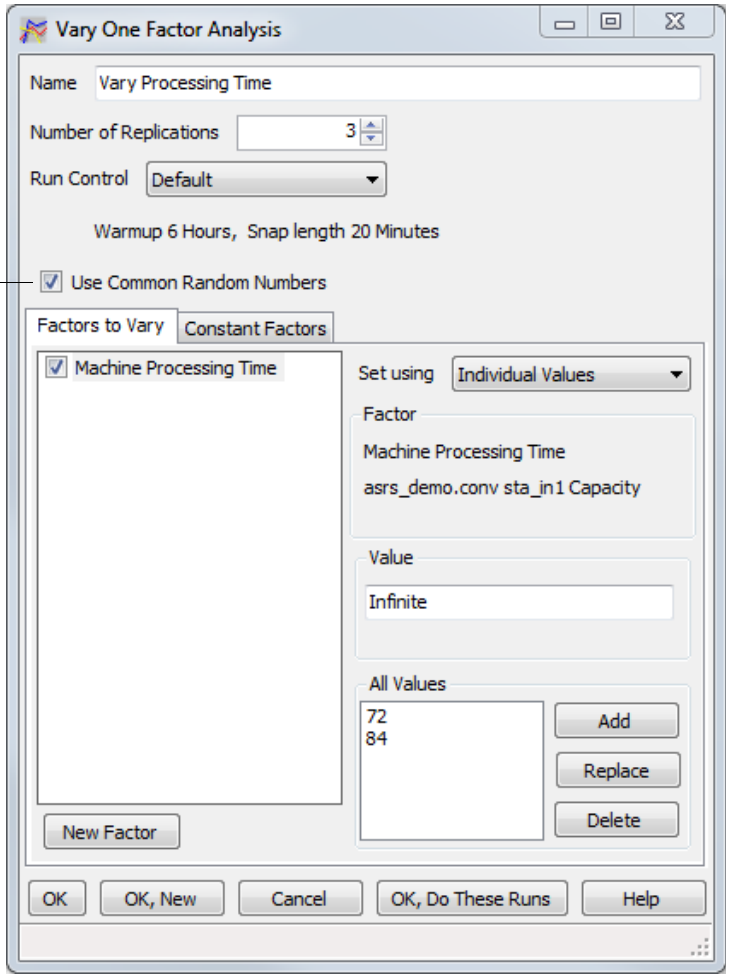
Note For a model to benefit from common random number analysis, the person who built the model must have assigned separate random number streams to each source of random events in the model.

You can use common random numbers with two types of analyses:

- Vary One Factor
- Vary Multiple Factors

To conduct the analysis using common random numbers, select the option **Use Common Random Numbers** in the analysis dialog box. (The model properties determine the default value of this option; it might already be selected based on the settings defined in the Setup Wizard. For more information, see "Setting model properties" on page 145.) The following illustration shows how to enable the option for a Vary One Factor analysis:

Selecting this option causes the analysis to use common random numbers.



When you enable common random numbers, AutoStat automatically manages all the runs to benefit from using common random numbers. After the analysis is complete, you can display comparison confidence intervals to indicate the differences between the compared configurations. For more information, see "Compare all to all" on page 203 and "Compare all to one" on page 209.

Note If an error occurs during a run for a common random number analysis, the related runs for the same replication (although good runs) are also not used for the current analysis.

Example—Using Common Random Numbers

In this example, a simulation evaluates different machines for use in an assembly process. Loads entering the system first travel on a conveyor to three operators. Each operator processes a load for a random amount of time described by a triangular distribution. Each load then travels to an assembly machine. Depending on which machine is used in the system, processing at the machine lasts for one of the following times:

- 72 seconds
- 84 seconds
- 96 seconds
- 108 seconds
- 120 seconds
- 132 seconds

Analyzing the model in AutoStat allows you to compare several system configurations, each using a different machine. To support analysis using common random numbers, each source of random data in the model is assigned a separate random stream, as shown here:

```
begin P_process arriving /* Starting procedure */
   move into Q_geton
   move into conv.get_on
   send to P_OperAssemble(1)
end

begin P_OperAssemble arriving /* Operator assembly */
   travel to conv.sta_asmb(procindex)
   use R_Oper(procindex) for streamOper(procindex)
      triangular 20, 80,
      140 sec
   if procindex = 3 then send to P_MachAssemble
   else send to P_OperAssemble(procindex+1)
end

begin P_MachAssemble arriving /* Machine processing */
   travel to conv.work_area
   use R_Processor for V_proctime
   /* Machine times varied as a factor
      in the AutoStat analysis*/
   travel to conv.get_off
   send to die
end
```

To evaluate each system configuration, a Vary One Factor analysis is defined to conduct 10 replications of 10 days each. The analysis uses average processing time (from the time loads are created until they are removed from the system) as a response. When the Vary One Factor analysis is conducted in AutoStat

without using common random numbers, 90 percent confidence intervals comparing all machine processing times to the base (72 seconds) are reported as follows:

Compare To Machine processing time 72

Machine processing time		**84**	**96**	**108**	**120**	**132**
Average P_ MachAssemble	Ave. Diff.	20.82	62.84	128.60	213.66	332.65
	CI Low	20.50	60.84	124.80	207.83	327.10
	CI High	21.14	64.84	132.40	219.49	338.20
Average P_ OperAssemble(1)	Ave. Diff.	-4.16	1.17	2.22	26.44	440.80
	CI Low	-7.39	-2.77	-1.39	13.56	353.34
	CI High	-0.93	5.11	5.83	39.32	528.26
Average P_ OperAssemble(2)	Ave. Diff.	0.31	1.42	9.65	52.28	301.44
	CI Low	-0.96	0.10	6.01	42.54	267.01
	CI High	1.58	2.74	13.29	62.02	335.87
Average P_ OperAssemble(3)	Ave. Diff.	0.20	5.83	35.11	112.57	338.72
	CI Low	-0.41	4.33	29.35	101.73	318.97
	CI High	0.81	7.33	40.87	123.41	358.47
Average P_Process	Ave. Diff.	-0.01	0.06	0.01	0.97	392.65
	CI Low	-0.06	-0.05	-0.06	0.08	223.24
	CI High	0.05	0.17	0.07	1.87	562.06
Average Processing Time	Ave. Diff.	17.01	71.15	175.45	405.65	1800.40
	CI Low	12.38	63.80	160.78	369.99	1495.93
	CI High	21.64	78.50	190.12	441.31	2104.87

Using common random numbers, 90 percent confidence intervals for the same comparison are shown here:

Comparing to Machine processing time 72

Machine processing time		**84**	**96**	**108**	**120**	**132**
Average P_ MachAssemble	Ave. Diff.	20.94	60.33	125.78	214.18	329.51
	CI Low	20.64	58.85	123.72	211.62	323.57
	CI High	21.24	61.81	127.84	216.74	335.45

Comparing to Machine processing time 72

Machine processing time		**84**	**96**	**108**	**120**	**132**
Average P_ OperAssemble(1)	Ave. Diff.	0.00	0.00	1.10	25.55	412.54
	CI Low	-1.12	-1.12	-0.23	17.89	334.37
	CI High	1.12	1.12	2.43	33.21	490.71
Average P_ OperAssemble(2)	Ave. Diff.	0.00	0.25	6.28	50.62	279.20
	CI Low	-0.12	-0.06	4.06	42.38	253.95
	CI High	0.12	0.56	8.50	58.86	304.45
Average P_ OperAssemble(3)	Ave. Diff.	0.06	3.30	28.33	109.56	323.70
	CI Low	-0.03	2.47	24.94	103.84	306.17
	CI High	0.15	4.13	31.72	115.28	341.23
Average P_Process	Ave. Diff.	0.00	0.00	0.01	0.39	299.72
	CI Low	-0.07	-0.07	-0.07	0.00	162.12
	CI High	0.07	0.07	0.08	0.78	437.32
Average Processing Time	Ave. Diff.	20.99	63.85	161.42	399.98	1641.55
	CI Low	19.20	61.12	153.67	377.52	1400.73
	CI High	22.78	66.58	169.17	422.44	1882.37

The analysis that uses common random number results in significantly narrower confidence intervals for the same number of replications. For example, the width of the confidence interval comparing a machine processing time of 84 seconds to 72 seconds for the Average Processing Time (of all processes) is reduced from 9.26 (21.64 – 12.38 = 9.26) in the non-CRN analysis to 3.58 (22.78 – 19.20 = 3.58) in the CRN analysis. Likewise the width of the interval comparing a machine processing time of 96 seconds to 72 seconds is reduced from 14.70 (78.50 – 63.80 = 14.70) in the non-CRN analysis to 5.46 (66.58 – 61.12 = 5.46) in the CRN analysis, and so on. Given information about the cost of each machine, you can use this data to determine which machine provides the best cost/performance in the system.

Warmup analysis

Use a warmup analysis *to estimate how long it takes a system to reach a steady state* (see "Determining when the model reaches a steady state" on page 160). If statistics are gathered during the warmup period, they might not truly reflect the steady state of the system. See "Reducing the bias of initial conditions" on page 161.

A model's initial conditions determine whether it needs to warm up.

If your model starts empty and must reach a loaded (or steady) state before representing the real system, then the model requires a warmup time.

If the model contains WIP status at the beginning of the simulation or if the system has a period of time in which all work is started and finished, the model does not require a warmup. For example, a distribution center might fill all orders each day and no orders are ever carried over to the next day. In this case, you are interested in the day's cycle and want to look at statistics for the entire day. No warmup time is necessary.

For information about other types of analyses, see "Analysis types" on page 147.

Defining a warmup analysis

You define an analysis by using the Warmup Analysis dialog box to:

- Name the analysis (see "Analysis information" on page 148).
- Edit the number of replications (see "Analysis information" on page 148).
- Define the run control (see "Defining the run control for a warmup analysis" on page 158).
- Define factors to hold constant (if necessary) and assign them values (see "Defining factors" on page 179 and "Setting a constant value" on page 187).
- Make the runs for the analysis (see "Section 5: Making runs" on page 190).
- View the warmup graph (see "Viewing the warmup graph" on page 159).

Note You can also use the Setup wizard (see "Using the Setup wizard" on page 145) to define a warmup analysis.

Defining the run control for a warmup analysis

To edit the run control for a warmup analysis, use the Warmup Analysis dialog box to edit the following:

Option	Definition
Number of Snaps	The number of times you want to collect statistics for each run. Use a large number of snaps (at least 20, but 40 or 50 are better) to determine how the model changes over time. The more snaps, the more options you have when viewing the warmup graph (see "Viewing the warmup graph" on page 159).
Snap Length	How long each snap is. Use in conjunction with the number of snaps as described in the following.

Determining the number of snaps and snap length

Use the following guideline when deciding on the number of snaps and the snap length:

(3 x cycle time*) <= (# of snaps) x (snap length)

Three times the cycle time should be less than or equal to the number of snaps times the snap length.

*For cycle time (time in system), use either the time it takes the first part of the slowest part type to get all the way through the system or the time it takes for at least one of each part type to get through the system. Multiplying the cycle time by 3 helps ensure that you have enough time to determine when the warmup is really complete.

For example, suppose it takes 24 hours for at least one of each part type to finish in your model. Assuming you are using 40 snaps, the formula would be:

```
3 x 24 <= (40) x (snap length)
```

To determine the snap length, you need to divide 72 hours by 40 snaps:

```
72/40 = 1.8 hours
```

» **Tip** Snap length and the number of snaps do not have to be exactly three times the cycle time, so round up to combinations that are easy to calculate, such as 40 snaps of 2 hours (for 80 hours).

Viewing the warmup graph

▶ **To view a warmup graph to estimate a warmup time**

1. Display the graph for the warmup analysis using the **Analysis** tab (see "Section 7: Displaying analysis results" on page 198).

   The snap information appears in the upper left of the graph.

2. Select the response(s) for which you want to view data. Define a new response if necessary.

   » **Tip** Time in system and WIP at critical resources are good responses for determining the warmup period. For more information, see "Useful responses for determining warmup" on page 159.

3. Select the desired averaging window to smooth the graph so a trend is more easily observed.

   The *Averaging window* is used in warmup graphs to determine how many adjacent snaps are averaged together. For example, if your average is five, snaps 1-5 are averaged together and graphed, then snaps 2-6 are averaged together and graphed, and so on.

   Adjusting the averaging window can smooth the graph and make a trend toward a steady state more apparent.

   Adjust the averaging value several times to see if one gives a clearer picture of when responses are warming up.

   If you make the average too small, the graph can be noisy and hard to interpret.

   If you make the average too large, the graph might smooth out too much, masking all change.

   » **Tip** The ability to average snaps together is one reason why a large number of snaps in the run control is suggested for a warmup analysis. Without numerous snaps, the averaging window cannot help you to see the warmup trends. See "Determining the number of snaps and snap length" on page 158.

4. Select **Normalize Values** if you want all values to be graphed between zero and one, instead of as their actual values.

Useful responses for determining warmup

Use WIP levels and queue time in front of highly utilized stations to determine whether WIP levels off at some point or whether it keeps building indefinitely. Looking at WIP for the entire model might mask a lack of a steady state; use WIP levels at various resources to get a more accurate picture of what is going on in the model.

Responses that are not useful for determining warmup

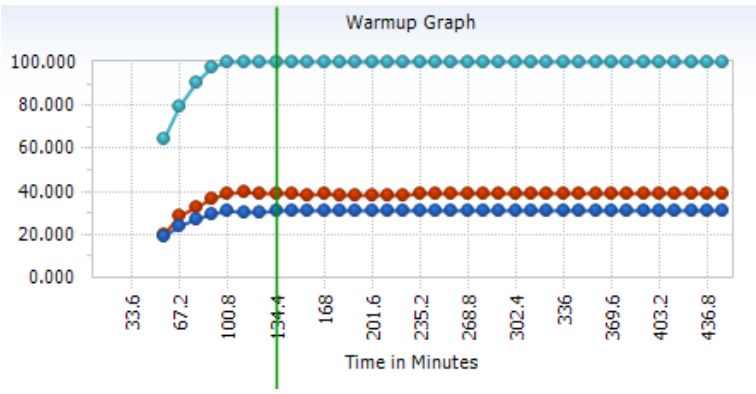
Utilization and *throughput* almost always reach steady state (and tend to reach it more quickly than queuing statistics), and therefore *are not good responses* to use for warmup determination.

Setting a warmup time for a model

Using the warmup graph, you can set the warmup time in the Model Properties dialog box. All analyses that use the default run control will use this warmup time when collecting statistics.

To set a warmup time

1. Using the warmup graph, determine the warmup time by deciding where all lines level off, indicating a steady state.



   In the graph, the responses level off after about 135 minutes.

   Tip When you have multiple responses, use the time it takes the longest response to warm up (135 minutes in this example).

2. Once you can clearly determine the longest warmup time, type that warmup time in the bottom of the graph and click **Set Default Warmup Time**.

   This changes the default used in the Model Properties dialog box and ensures that all future analyses use the correct warmup time (see "Setting model properties" on page 145).

   Note Changing the default warmup time in the Model Properties dialog box affects existing runs that used the default time, and those runs will need to be redone.

   ! *If your graph never levels off, rerun with longer warmup runs. If the model still never warms up, there is a problem with the model.* For example, if you do not have enough resources, WIP might build forever. You must fix the model so that it reaches steady state before conducting any analyses.

Determining when the model reaches a steady state

A model reflects a real system, and as such must eventually reach a steady state. If you run a warmup analysis and *the graph indicates that the model never levels off into a steady state, there is a problem with the model and you should not conduct any analyses until you fix the model.*

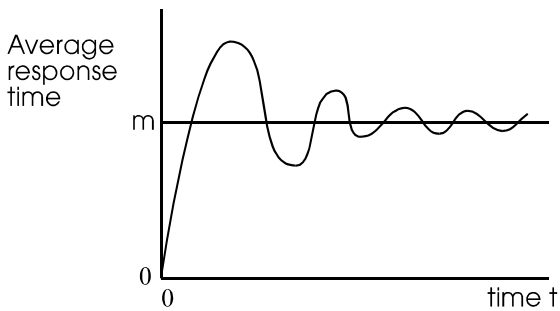
For example, if WIP builds indefinitely and a resource is 100 percent utilized and its queue is growing infinitely, it could indicate too little capacity, too many orders, and so on. Validate your model against the real world or against a realistic set of data (for new systems) to *find a good baseline that represents a realistic steady state.* Once you have a baseline, you can determine a warmup using a warmup graph, after which you can perform what-if scenarios.

Reducing the bias of initial conditions

The object of warmup determination is to eliminate initial response bias by starting statistics collection after the warmup period. There are two common strategies used to reduce initialization bias. The first is to delete the warmup period once it is identified and use only the statistics collected after that point. The second is to try to initialize the model as close to the steady state as possible from past experience or previous runs. Usually, it is better to use both strategies in combination.

It is often difficult to estimate realistic initial conditions. When examining new configurations, steady-state responses are usually unknown, and trying to find the steady state is often the reason for the simulation. One way to determine the warmup period is by viewing the warmup graph to visually determine when the system is operating at steady state. You can specify the period up to the steady state as the warmup period for future runs. Then you can eliminate the data from the warmup period in the analysis.

The graph that follows shows the cumulative average response time for a steady-state response. As illustrated, the warmup period biases the response as low in this case. As time goes to infinity, the cumulative average response asymptotically approaches the actual steady-state mean. However, in real-world analyses, an unlimited amount of time is not available, so the warmup time needs to be determined and discarded.



Equations used in determining warmup

AutoStat uses a graphical procedure developed by Welch (1981, 1983) to help identify the warmup period. Welch's method works as follows (see Law and Kelton, 1991):

1. Make n replications of the model, dividing each run into a number of adjacent non-overlapping time intervals or snaps. For each replication, the run length is T, where T = (# of snaps)∗(snap length).
2. Average the ith snap over all of the n replications to obtain an average response level, Y_i, for snap i.The mean response levels are graphed according to a moving average of size w. For each i, i = w,...,m, the moving average, $Y_i(w)$, is defined as:

$$\overline{Y}_i(w) = \sum_{s=1-w}^{0} \overline{Y}_{i+s} \Big/ w$$

The moving average helps smooth out the curve.

Vary one factor analysis

Using a vary one factor analysis, *you can analyze how the model behaves when you change the values of one factor*, such as a machine's processing time, while keeping other factors constant.

For information about other types of analyses, see "Analysis types" on page 147.

You define an analysis by using the Vary One Factor Analysis dialog box to:

- Name the analysis (see "Analysis information" on page 148).
- Edit the number of replications if your model is random (see "Analysis information" on page 148).
- Edit the run control (see "Analysis information" on page 148).
- Define a new factor (if necessary) and assign it values (see "Defining factors" on page 179).
- Define factors to hold constant (if necessary) and assign them values (see "Setting a constant value" on page 187).
- Make the runs for the analysis (see "Section 5: Making runs" on page 190).
- Display the results (see "Section 7: Displaying analysis results" on page 198).

Vary multiple factors analysis

Using a vary multiple factors analysis, you can *analyze how the model behaves when you change more than one factor*, such as a machine's processing time, number of operators, etc.

For information about other types of analyses, see "Analysis types" on page 147.

You define an analysis by using the Vary Multiple Factors Analysis dialog box to:

- Name the analysis (see "Analysis information" on page 148).
- Edit the number of replications if your model is random (see "Analysis information" on page 148).
- Edit the run control (see "Analysis information" on page 148).
- Define a new factor (if necessary) and assign it values (see "Defining factors" on page 179).
- Define factors to hold constant (if necessary) and assign them values (see "Setting a constant value" on page 187).
- Make the runs for the analysis (see "Section 5: Making runs" on page 190).
- Display the multiple factors graph (described in the following) or other results (see "Section 7: Displaying analysis results" on page 198).

Note When you define an analysis using the Vary Multiple Factors Analysis dialog box, at least two factors must be given values on the **Factors to Vary** tab. If you want to create an analysis that varies only one factor, use the Vary One Factor dialog box instead of the Vary Multiple Factors Analysis dialog box.

Multiple factors graph

To use the multiple factors graph

1. Select a confidence level (see "Confidence level" on page 201).
2. Select the response(s) you are interested in.

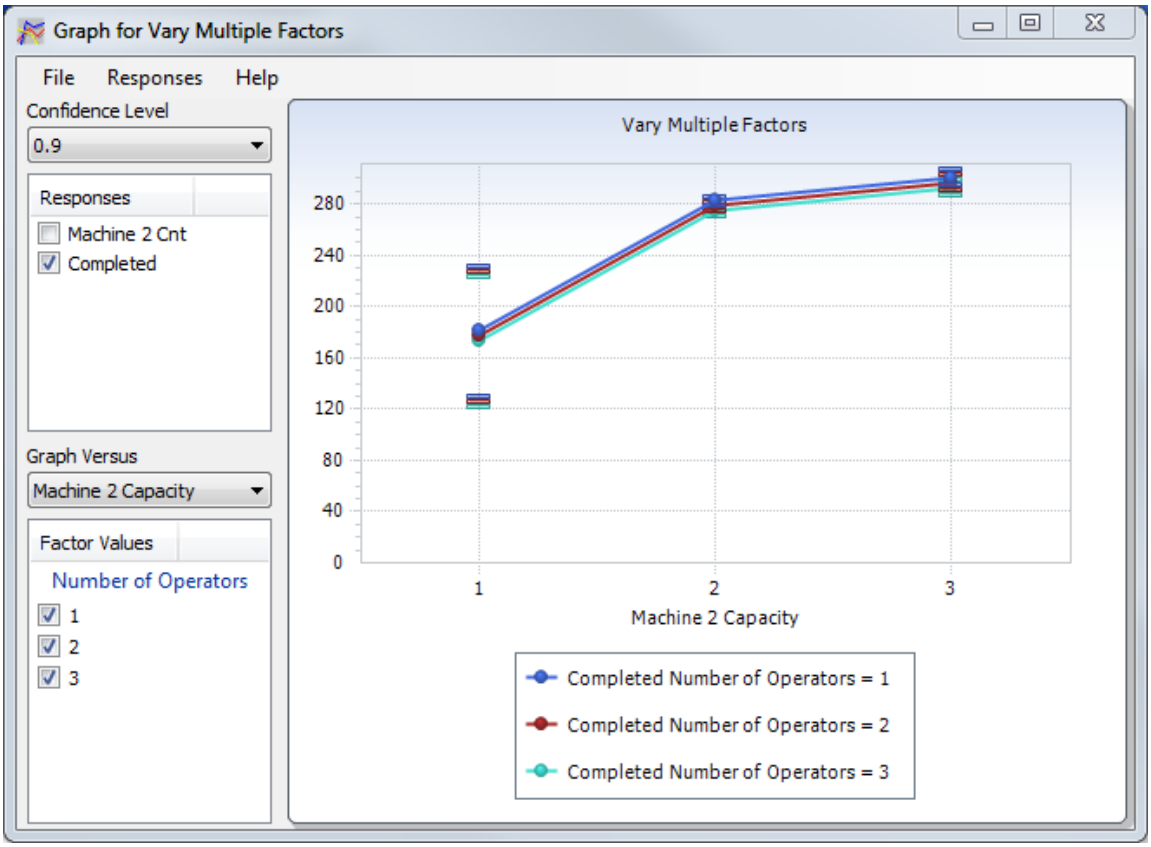
   The response values are graphed on the Y axis.
3. Select a factor from the **Graph versus** list to be used for the X axis.

   The remaining factors are graphed relative to this factor.
4. Select the desired value(s) for the other factor(s) in the enabled list(s) at the bottom of the dialog box.

Each factor value or value combination (if more than one factor is being graphed) creates a line in the graph. The graph shows the confidence level of the response in relation to the factors being graphed.

Multiple factors graph example

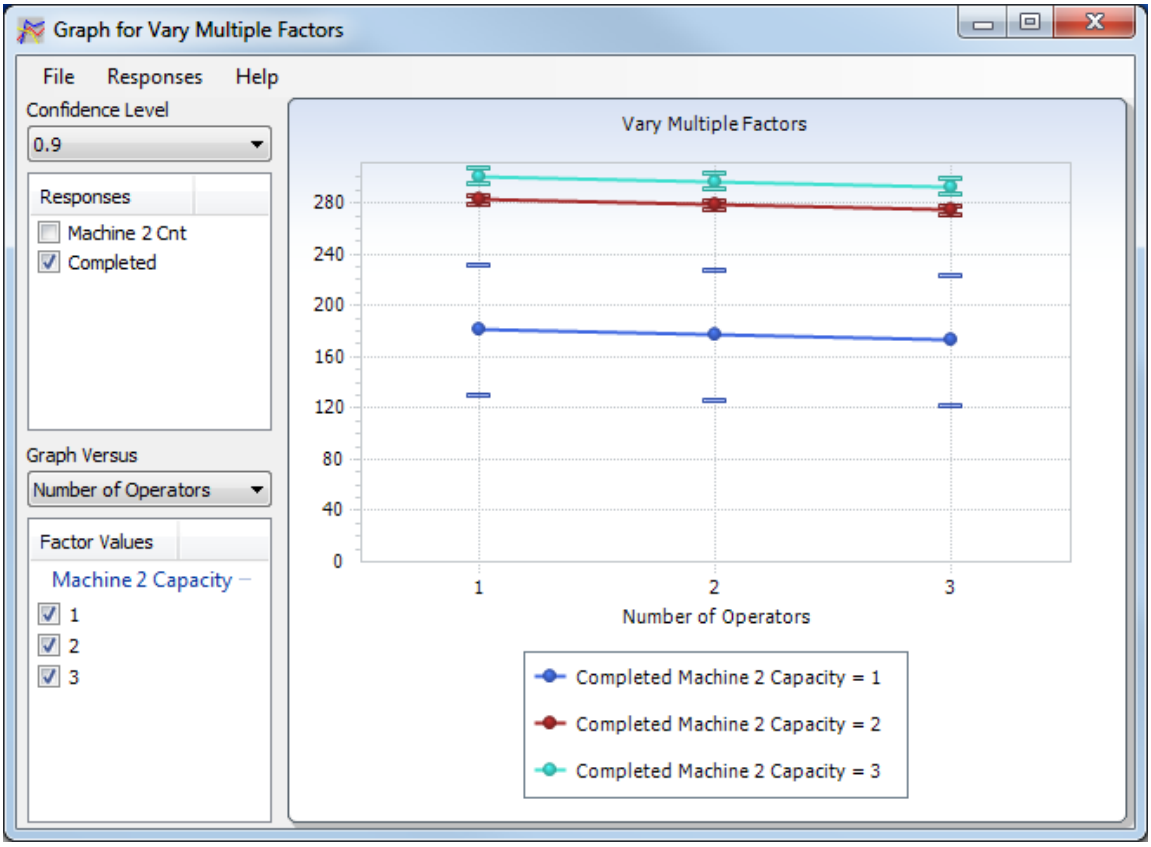
In the following graph, the response selected is the number of part type `part1s` finished. A machine's capacity is selected for the X axis. The response value is graphed in relation to the number of operators (each line shows a different number of operators: one, two, or three).



In this example, the graphed line represents the number of part1s finished in the model with the number of operators and with the defined machine capacity. A machine capacity of two produces more part1s than capacity one, but the

difference between a capacity of two and a capacity of three is small. The number of operators does not seem to affect the response (the response values are close together for each point).

You can also choose to view the same graph the opposite way by selecting the number of operators to be the X axis and the machine capacity to be graphed, as shown here:



This graph makes it more clear that the number of operators does not influence the response (all three points for any given line are almost the same). However, the response value varies between the capacity of one (the lower line) and a capacity of either two or three (which produce an almost equal numbers of part1s).

Single scenario analysis

Using a single scenario analysis, you can *determine confidence intervals on a single set of factor values*. This allows you to analyze how the model behaves when you vary the random numbers while holding all factors constant.

For information about other types of analyses, see "Analysis types" on page 147.

You define an analysis by using the Single Scenario Analysis dialog box to:

- Name the analysis (see "Analysis information" on page 148).
- Edit the number of replications if your model is random (see "Analysis information" on page 148).
- Edit the run control (see "Analysis information" on page 148).

- Define factors (if necessary) to hold constant and assign them values (see "Defining factors" on page 179 and "Setting a constant value" on page 187).
- Make the runs for the analysis (see "Section 5: Making runs" on page 190).
- Display the results (see "Section 7: Displaying analysis results" on page 198).

Optimization analysis

Using an optimization analysis, you can let AutoStat *determine the best set of factor values for your model*. Define a set of factors to vary and the values for each of the factors. Then define a fitness function for your model that lists factors (model parameters, such as the number of resources) and responses (a measure of the system, such as throughput or cycle time) to either maximize or minimize and their relative importance. For example, you can define a function that minimizes the number of operators (within a defined range) while maximizing throughput. AutoStat automatically manages the runs to find the best combination possible *in the given time or given number of runs* using its optimization algorithm.

» **Tip** Optimize independent parts of your system separately.

For a list of references about optimization, see "Before you begin" on page 133.

For information about other types of analyses, see "Analysis types" on page 147.

Understanding the optimization algorithm

AutoStat performs optimization using an optimization algorithm called an evolution strategies algorithm. Evolution strategies process a population of solutions during each iteration of the search. The optimization algorithm in AutoStat avoids finding a local optimum while seeking the global optimum (see "Local versus global optimum" on page 166).

Survival of the fittest

Evolution strategies are based on the theory of evolution. An initial population (made up of sets of factor values) combine to create the next generation of factor values (*children*). The children of that generation inherit traits from each of their *parents*, and they also have slight differences, called mutations (see "Mutation" on page 167). The fittest children of that generation (as defined by your fitness function) live to become the parents of the next generation, and so on.

AutoStat uses the following process for optimization.

For the first generation:

1. Randomly create the first generation of children. Each generation contains 7N number of children, where N is the number of parents per generation. For example, if you have defined the number of parents to be 3, AutoStat creates 21 children.

   Each child is randomly assigned factor values. For example, assume you have defined 3 factors, and each factor's values are being varied from 1 to 4. AutoStat randomly generates children such as:

child	Factor 1 value	Factor 2 value	Factor 3 value
child 1	1	3	4
child 2	4	1	3
child 3	1	4	2
.			
.			
.			

For each generation, including the first:

2. Make the runs for each child (see "Determining how many runs to make for each child" on page 167).
3. Based on the fitness score for each child, pick the best N children to use as parents for the next generation, where N is the number of parents per generation.
4. To create each child in the new generation, randomly pick two of the parents (selected in step 3), combine them (take some of the factor values from one parent and some values from the other), then mutate the factor values slightly within the factor's set of defined values. Create 7N number of children, where N is the number of parents per generation.

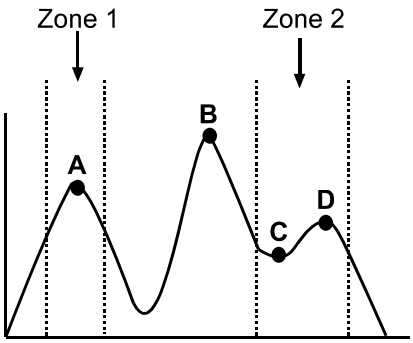
   Note Because parents are chosen randomly, it is possible that the two parents for a generation might occasionally be the same.

5. Repeat steps 2 - 4 until either the termination criteria are met or until the runs are stopped.

Local versus global optimum

The AutoStat search algorithm tries a wide variety of possible solutions before it narrows down its search. Some algorithms search a smaller area and find a solution that is not the best possible choice.

Suppose the following graph shows all possible response values for an optimization's fitness function, and that higher points represent a better solution.



A local search algorithm might only search zone 2, and it could determine the best value in that zone, that is, that D is better than C. But D is not as good as A or B, which a local algorithm would not find. Likewise, if the local search algorithm only searched zones 1 and 2, it could determine that A is better than D, but it would not find B, which is even better. AutoStat uses a globally-oriented search algorithm and does a wide search to find the best overall solution, not just the best solution in a limited area.

Mutation

A mutation is a change to a factor value within the factor's defined set of values. Each factor is mutated independently of other factors.

Note Integer factors are mutated and then rounded to the nearest integer value.

For a given factor, if further mutation is not helping the fitness score, AutoStat mutates it less and less until its optimal value is determined. Then the factor is set to the best value and is not changed any more (the factor has a mutation rate of zero). Other factors that are helping the fitness score continue to be mutated until the algorithm has focused in on the optimal combination of factor values.

Determining how many runs to make for each child

When generating children for each generation in an optimization, it is possible that the same set of values is generated more than once, such as child 3 and child 17 here.

child	Factor 1 value	Factor 2 value	Factor 3 value
.			
.			
child 3	1	3	4
.			
.			
child 17	1	3	4
.			
.			

It is also possible that AutoStat has made runs for that set of factor values before in a previous generation. Each run AutoStat does for the same set of factor values is called a replication. In order to control how many replications AutoStat makes for a child, AutoStat uses several optimization options.

How AutoStat makes replications

During each generation, AutoStat creates 7N number of children, where N is the number of parents per generation. When determining how many runs to make for each child, AutoStat uses the following information:

- Number of runs already completed for that child
- Number of replications per generation for a solution
- Maximum replications per solution (from the **Options** tab)

AutoStat always does the defined number of replications, unless that number will cause more than the maximum replications per solution to be made, in which case AutoStat makes only enough runs to reach the maximum replications per solution. *Once the maximum has been reached, no more runs are ever done for that child, even if the child is created in subsequent generations.*

Assume AutoStat has *already made two runs for a given child* (set of factor values), the maximum allowed is five, and the most allowed per generation is two. AutoStat would set up the following runs:

Generation	# completed for child	Max # for solution	Defined # of replications	# set up
4	2	5	2	2
5	4	5	2	1
8	5	5	2	0

In generation four, AutoStat sets up the most allowed within a generation (two). The next time the same child is created (in generation five), four runs have been made and the maximum possible is five, so AutoStat sets up one additional run for that child.

Defining an optimization analysis

You define an analysis by using the Optimization Analysis dialog box to:

- Name the analysis (see "Analysis information" on page 148).
- Edit the number of replications if your model is random (see "Analysis information" on page 148).
- Edit the run control (see "Analysis information" on page 148).
- Define new factors to vary (if necessary) and assign them values (see "Defining factors" on page 179).
- Define a fitness function, including a fitness response (see "Defining a fitness function" on page 169).
- Edit the options, if necessary (see "Setting optimization options" on page 171).

- Define factors (if necessary) to hold constant and assign them values (see "Defining factors" on page 179 and "Setting a constant value" on page 187).
- Make the runs for the analysis (see "Section 5: Making runs" on page 190).
- Display the progress graph (see *"Viewing the progress graph" on page 173*) or other results (see "Section 7: Displaying analysis results" on page 198).

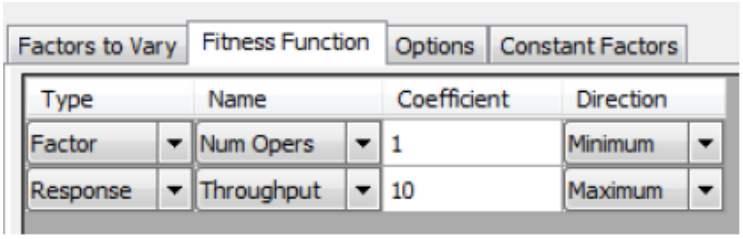
  » **Tip** You can see the scores of the top 30 children at any time by viewing the summary statistics, confidence intervals, compare all to all, compare all to one, run results, or runs used.

Defining a fitness function

A *fitness function* describes the factors and responses that you are trying to optimize for a model. Each factor or response in a fitness function is called a *term*. You can define each term's relative importance.

Note Some people refer to a fitness function as an objective function or a score function.

For example, you can define a function that minimizes the number of resources (within the range defined on the **Factors to Vary** tab) while maximizing throughput.



Note Only factors and responses that are numerical values can be used in a fitness function. You cannot use factors or responses that are names or other strings.

If a term is a distribution, the first parameter is used in the function. For example, if it is a normal distribution, the mean is used to calculate the fitness function and the standard deviation is ignored.

To define a fitness function

1. Double-click a cell.
2. Enter the following information:

Information	Definition
Type	Select whether the term is a factor or a response.

Information	Definition
Name	Select the factor or response you want to include in the fitness function. You do not need to include every factor you are varying in the fitness function. Only include factors that are important in the evaluation of overall system performance (for example, they cost money or strongly affect the responses you are interested in). If you include factors that are not relevant, the algorithm takes longer to arrive at a potential solution. » **Tip** You can define new factors and responses from the **Fitness Function** tab.
Coefficient	The relative importance (weight) of each term. One common way to use the coefficient is to convert each term to a monetary unit, such as a dollar. For example, perhaps each operator's monthly salary is $2,000, while each unit of throughput generates $200 that could offset the expense of a new operator. Therefore, it would take 10 units of increased throughput to pay for an additional operator. You can use either the dollar amounts or the ratio as the coefficients. Make sure you are using similar time periods for each term in the function. For example, if the throughput response is one month's total, but you want to consider throughput for a year, you can either use a coefficient to make them similar or change how the terms are defined (see "Defining factors" on page 179 or "Defining responses" on page 194). » **Tip** If you want to compare throughput to the time it takes to pay off a new piece of equipment, you need to modify the coefficients to equalize both the time and the cost.
Direction	Select whether to minimize or maximize the term. You generally want to increase, or maximize, things that generate money (like throughput), while minimizing things that cost money (like adding new resources).
Response Name	The name of the fitness function response. You must also *define whether you want high or low scores to indicate improvement* (whether the Progress graph values go up or down to show improvement).

Like other responses, the optimization response appears for all analyses. You can view the optimization response value for any run and for any analysis.

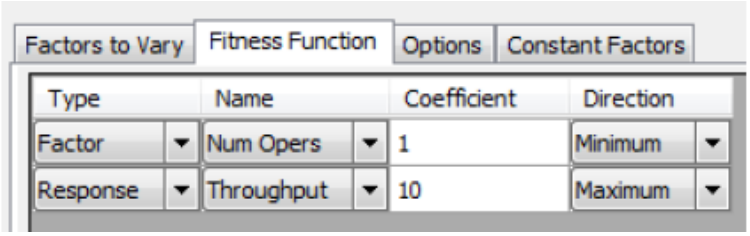
Fitness score

The score is the response value for an optimization's fitness function. The score is used in optimization results, including the progress graph.

Each term in the fitness function is multiplied by its coefficient and added to the other terms. A term's positive or negative sign is determined by whether you are maximizing or minimizing it in the fitness function and whether you selected higher or lower scores to be better.

$$\textbf{fitness score} = \pm \text{coefficient}_1 \bullet \text{term}_1 \pm \text{coefficient}_2 \bullet \text{term}_2 \ldots \pm \text{coefficient}_N \bullet \text{term}_N$$

For example, suppose you have the following fitness function:



Assume a scenario uses 2 operators and the response's value is 100. Assume also that higher scores are better. The fitness score is calculated as:

- 2(1) + 100(10) = 998

If a child has more than one completed replication, the fitness score for the child is the *average* of those replications.

Setting optimization options

The following options define how the optimization analysis works:

Note The minimum values allowed in the numeric controls are equal to the minimum valid values for the option being defined.

Optimization parameters

Option	Definition
Maximum replications per solution	AutoStat can make more than one replication for each child (set of factor values). As the scope of the search narrows, factor value combinations might be repeated (using different random numbers) to get more data on a potential solution. This parameter is the maximum number of times that AutoStat repeats runs for a set of values. For more information, see "Determining how many runs to make for each child" on page 167.
Number of parents per generation	The number of parents (sets of factor values) that are used to create the next generation of children. The higher the number of parents, the less chance of converging on a local optimum, but also the slower the search. The number of parents needed to avoid a local optimum varies by type of model. If you are varying fewer than four factors, you might be able to use two parents. For four or more factors, you probably need to use at least three parents. For more information about parents and children, see "Understanding the optimization algorithm" on page 165.

Termination criteria

AutoStat continues to make optimization runs until you stop the runs or until one of the termination criteria is met.

Note *AutoStat's optimization algorithm determines the best solution given the number of runs made.* Making too few runs might not provide enough data to find an optimal solution (although the answer might still be better than the base model). Make as many runs as you have time to make to ensure the quality of the solution.

If the mutation rate becomes so small that all new children are identical (no changes are occurring), AutoStat stops making runs automatically. For more information about mutation, see "Mutation" on page 167.

AutoStat also stops making runs when *either* of the following conditions occurs:

- **N generations = P% improvement -** This compares the best fitness score of the current generation to the best score of the previous Nth generation (for example, it compares the current generation to the 10th). If there is not the desired (P%) improvement in the fitness score between these generations, AutoStat stops making runs.
- **Maximum number of generations -** The maximum number of generations AutoStat can create. More generations and more runs increase the likelihood of the algorithm finding a better solution. For more information about generations, see "Understanding the optimization algorithm" on page 165.

Important *Once the algorithm stops searching, view the progress graph to make sure all of the lines have converged (see "Viewing the progress graph" on page 173). If all lines have not converged, you need to increase these settings to allow the algorithm to continue searching.*

Execution options

You can either include or exclude optimization runs from the **Do All Runs** option (for information about making runs, see "Section 5: Making runs" on page 190).

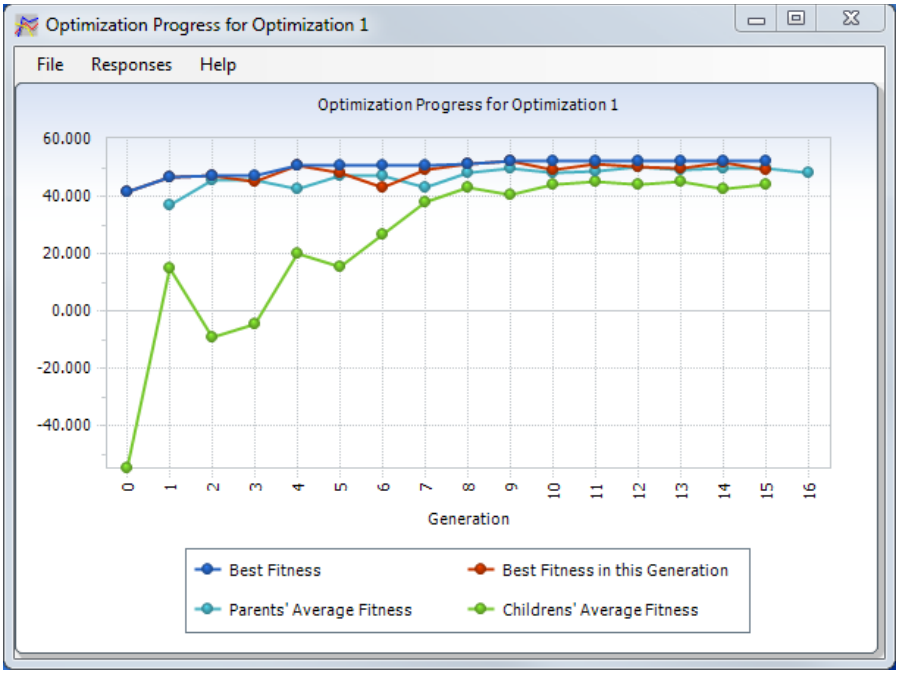
Tip Optimization analyses make runs continuously until the termination criteria are met, which can entail hundreds of long runs. Therefore, by default, you must select **Do Some Runs** in order to do the optimization runs.

Viewing the progress graph

When you run an optimization analysis, you can view a graph that shows the following measures of the optimization's progress:

- **Best fitness** - The best fitness score seen so far in any generation
- **Best fitness in this generation** - The best fitness score of a child in the generation
- **Parents' average fitness** - The average fitness of all the parents of the generation.

- **Children's average fitness** - The average fitness of all the children of the generation.



The graph shows that the first generation of children produced a fitness function response value that was negative (the terms being minimized in the fitness function were bigger than the terms being maximized). As the optimization made more runs, the response value became positive and all the lines converged, indicating that this is the best solution AutoStat could find.

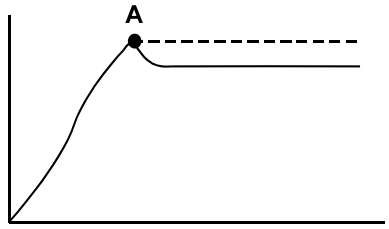
To get a list of all factor and response values for the optimal scenario, look at the summary statistics, confidence intervals, run results, or runs used for the optimization analysis.

Interpreting the graph

All the lines on the graph converge as the algorithm converges on the optimal solution, except in unusual cases. If the lines do not converge, the algorithm is still searching. For example, in the graph, the best solution is found in generation 8, but the lines do not converge until several generations later, after which AutoStat stops searching.

If values of your fitness response are not what you expected, make sure your fitness function is using coefficients that give you the intended results.

There are times when the best so far and the best of a generation might deviate due to an outlier (a response value that is very different than other values).



---- **Best so far**

—— **Best in generation**

For example, even though A is the best found so far, AutoStat has determined that it is an outlier and that the true best score is really below that. Therefore, these lines might never converge, but when the other lines (not shown) converge on the best in the generation and the lines flatten out for many generations, they are showing the best score AutoStat could find.

Sensitivity analysis

Using a sensitivity analysis, you can *estimate how sensitive the model is to changes in one or more factors*. Use this analysis when you have rough data and want to determine whether you need more exact data. If the model is *not* sensitive to the input, the existing data can be used. If the model *is* sensitive to the input, gather better data before drawing conclusions from the model.

This analysis works best when you want to vary three or more factors simultaneously.

Note A sensitivity analysis is based on a Design of Experiments.

For information about other types of analyses, see "Analysis types" on page 147.

You define an analysis by using the Sensitivity Analysis dialog box to:

- Name the analysis (see "Analysis information" on page 148).
- Edit the number of replications if your model is random (see "Analysis information" on page 148).
- Edit the run control (see "Analysis information" on page 148).
- Vary at least one numeric factor (define it, if necessary) and *assign it a value and a margin of error* (a plus/minus range, see "error range" in the following).
- Define factors (if necessary) to hold constant and assign them values (see "Defining factors" on page 179 and "Setting a constant value" on page 187).
- Make the runs for the analysis (see "Section 5: Making runs" on page 190).
- Display the results (see "Interpreting output for DOE and sensitivity analyses" on page 177).

Error range

For sensitivity analyses, you must define the factor's value and the margin of error, which is the largest possible value by which your data could be off. For example, suppose you had data for the time it takes an operator to perform an operation (2 minutes, or 120 seconds), but you thought your estimate could be off by as much 15 seconds either way. You would type 120 as the value and 15 as the margin of error.

To avoid misleading effects, make your range as narrow as possible.

Factorial DOE analysis

Using a factorial DOE analysis, you can perform a design of experiments (DOE) to *determine which factors or combination of factors have a statistically significant effect on the response you are measuring and which do not*.

A *statistical significance* is one that can be attributed to more than random variation (the factor has a systematic effect on the response).

» **Tip** You must compare statistical significance against practical or economic significance to determine whether the change is worth addressing in the real world.

Note A sensitivity analysis is a type of DOE. See "Sensitivity analysis" on page 175 for information about sensitivity analyses.

Use a DOE to eliminate the factors that have no effect and focus your analyses on the factors that do have an effect.

For information about other types of analyses, see "Analysis types" on page 147.

You define an analysis by using the Factorial DOE Analysis dialog box to:

- Name the analysis (see "Analysis information" on page 148).
- Edit the number of replications if your model is random (see "Analysis information" on page 148).
- Edit the run control (see "Analysis information" on page 148).
- Select the resolution for the experiment (see "Selecting a resolution for a DOE" on page 177).
- Define factors (if necessary) *assign them a low value and a high value (see "Setting high and low values for a DOE" on page 177)*.
- Define factors (if necessary) to hold constant and assign them values (see "Defining factors" on page 179 and "Setting a constant value" on page 187).
- Make the runs for the analysis (see "Section 5: Making runs" on page 190).
- Display the results (see "Interpreting output for DOE and sensitivity analyses" on page 177).

Selecting a resolution for a DOE

The design of experiments feature supports various levels of resolution (detail). The level of resolution determines the trade-off between accuracy and speed (fewer runs). *The lower the resolution, the less time required, but the less accurate the results, too.* The estimates of some or all interactions become less reliable as the resolution drops.

AutoStat chooses a default resolution for each analysis based on the number of factors you are varying.

» **Tip** Choose the highest resolution you have time to do (full factorial, V, IV, or III).

Design	Definition
Full Factorial	A full factorial design tests all combinations of low and high settings for all factors. For example, if you are varying two factors and are doing one replication, a full factorial design would require four runs. Three factors would require eight runs, and so on. If you are doing more than one replication of each scenario, multiply the preceding numbers by the number of replications. *The advantage of a full factorial design is that it is the most reliable estimate of both main and interaction effects,* including three-factor interactions (and higher). However, a full factorial design is time-consuming because it requires so many runs.
Resolution V	This design provides good estimates of both main effects and two-factor interactions. It assumes that three-factor interactions and higher are negligible.
Resolution IV	This design provides good estimates of main factor effects. Three-factor interactions and higher are assumed negligible; if they are not negligible, the two-factor interactions are unreliable.
Resolution III	The least desirable resolution. It assumes all interaction effects are negligible; if they are not negligible, the estimates of main effects are often unreliable.

Setting high and low values for a DOE

Choose the high and low values carefully.

Because AutoStat only tests for significance at the high and low value, if a value in the middle is significant, it could be missed. For example, suppose you experiment with the number of resources in a group, varying it from 1 to 5. Throughput might stay low with 1 and 5 resources, but 3 resources could maximize throughput. A DOE analysis would miss that. *To avoid missing significant values in the middle, make your range as narrow as possible.*

Interpreting output for DOE and sensitivity analyses

The output for DOE analyses shows the effects of individual factors and combinations of factors (if you have a high enough resolution selected).

Note Sensitivity analyses only show main effects; they do not show interactions.

Effects

Effects and interactions are shown in the following types of output:

- Bar graphs (see "DOE and sensitivity analysis bar graphs" on page 178)
- Summary statistics (see "Finding outliers" on page 214)
- Confidence intervals (see "Confidence intervals" on page 201)

Effect	Definition
Main effect	The effect of a factor on a response. For example, does throughput increase when you add a piece of equipment? What effect does the one factor (the piece of equipment) have on a response? The factor can have no effect, or can have a positive or negative effect (makes the response increase or decrease). Sensitivity analyses, which are a type of DOE, show main effects.
Interaction effects	Whether a combination of factors changes a response (only available for resolution V or full factorial resolutions). For example, it could be true that adding a faster lathe by itself has no effect on throughput, and neither does adding a faster grinder. However, adding both a faster lathe and a faster grinder significantly increases throughput. An interaction can have no effect, or can have a positive or negative effect (that is, make the response better or worse). Sensitivity analyses do *not* show interactions

Note For information about the equations used to calculate DOE results, see *Statistics for Experimenters: Design, Innovation, and Discovery, Second Edition*, Box, Hunter, and Hunter, Wiley-Interscience, 2005.

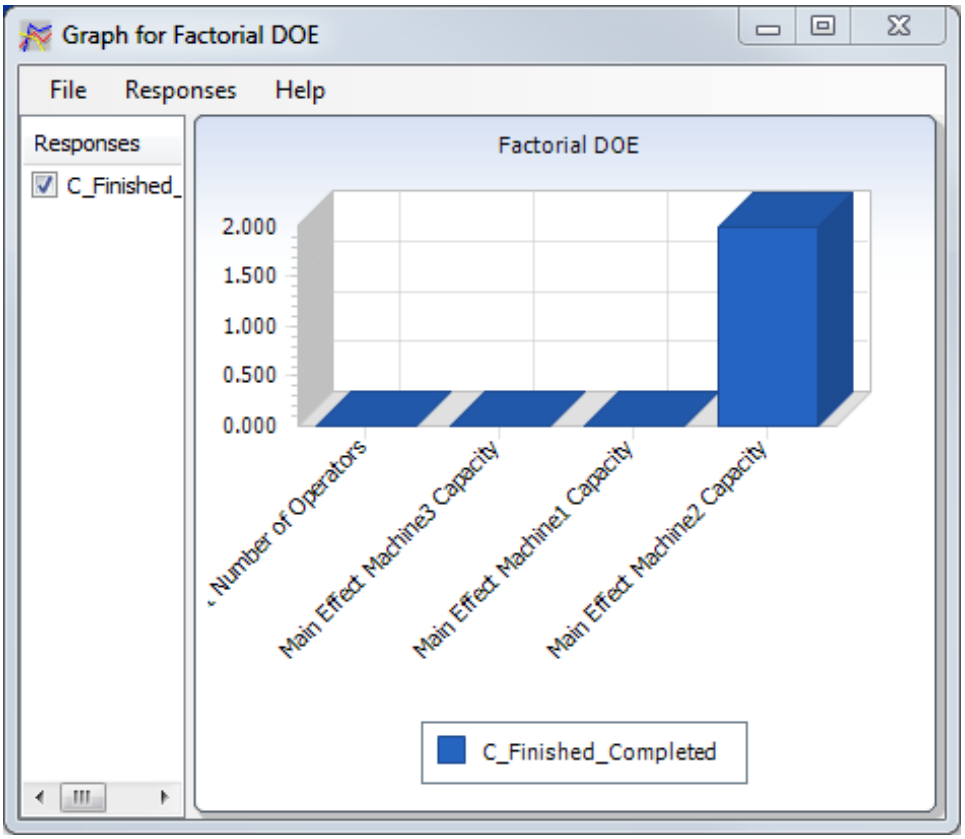
DOE and sensitivity analysis bar graphs

DOE and sensitivity analyses use bar graphs to show the average effect of each factor. DOE graphs also show factor interactions. To examine the significant effect of a factor, view confidence intervals.

To use a DOE bar graph

1. Select the desired response(s) in the list.
2. View the graph.

Each factor's main effect and factor interactions are graphed (the graph contains the average response from the summary statistics). **Lines that are close to zero show no effect. The farther a line extends away from zero in either direction, the larger the positive or negative effect on the response.**



In this graph, for the selected response, only one factor (machine 2's capacity) shows any significant effect on the response. All other factors and interactions show almost no effect.

Section 4: Defining factors

A *factor* is a model parameter, such as the number of vehicles, the speed of a conveyor section, or the processing time of a machine, that you want to vary in an analysis. You can vary one or more factors in a model to determine the effect they have on the model's output (responses).

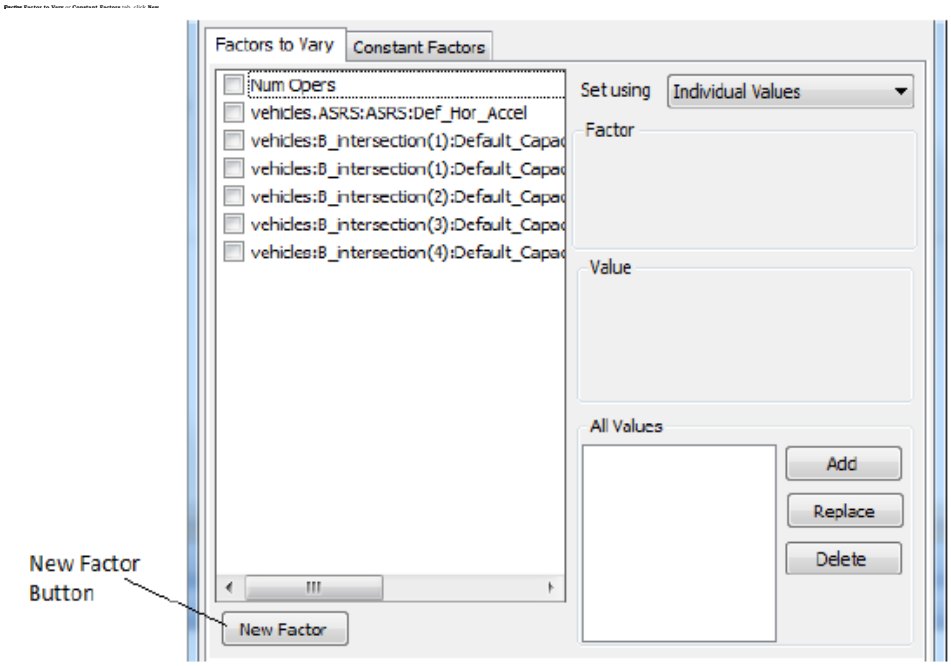
Defining factors

Factors are defined in one of two ways:

- Using an analysis dialog box
- Using the **Factors** tab

Using an analysis dialog box

To define an analysis from an analysis dialog box



3. Select the factor type you want to define and click **Create**. For information about factor types, see "Factor types" on page 181.
4. Name the factor (see "Analysis information" on page 148).
5. Define the factor.
6. Assign the factor values (see "Assigning values to factors" on page 187).

Using the Factors tab

To define an analysis from the Factors tab

1. Select the type of factor you want to define from the **Create New Factor of Type** list and click **New**. For information about factor types, see "Factor types" on page 181.
2. Name the factor (see "Analysis information" on page 148).
3. Define the factor.
4. Assign the factor values in an analysis (see "Assigning values to factors" on page 187).

Factor types

You can define the following types of factors:

Type	Function
AutoMod	Use to define attributes of entities from an AutoMod model, such as vehicles, processes, or conveyor sections, that you want to vary in an analysis.
Combination	Use to change more than one factor simultaneously.
Data file cell	Use to change a cell in an external data file.
Entire data file	Use to change an entire file being read by your model.

AutoMod factor

AutoMod factors are attributes of entities from an AutoMod model, such as vehicles, processes, or conveyor sections, that you want to vary in an analysis.

To define an AutoMod factor

1. From either the **Factors** tab or from an analysis dialog box, define a new AutoMod factor.
2. In the **System** list, click the AutoMod system containing the entity for which you want to define a factor.
3. From the **Entity** list, click the entity for which you want to define a factor.
4. In the **Attribute** list, click the statistic for which you want to define a factor.
5. Click **OK**. The program returns a default name.

   Notes

   - You can replace the default name with a user-defined name (see "Name" in "Analysis information" on page 148).
   - When you edit an AutoMod factor that was named using the default name, the name will be updated as the factor definition is changed.
6. Assign values to the factor in an analysis.

Tip You can **define multiple factors at a time** by selecting multiple entities (use the CTRL key to pick entities that are not consecutive in the list). The attribute must be the same for all selected entities. When you make multiple entity selections, the default name is displayed with a placeholder (<ENTITY>) for the entity name, for example, vehicles: <ENTITY>:Default_Capacity. When you click **OK** or **OK, New** this placeholder is replaced with the actual entity name for each factor that you create.

Combination factor

Use a *combination factor* to change more than one factor simultaneously.

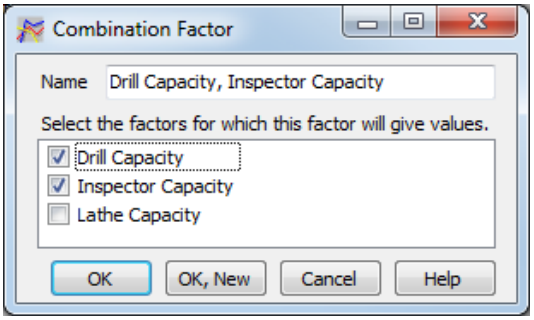
The most common use of a combination factor is to compare two types of entities that have more than one attribute. For example, suppose you want to compare two types of equipment that have different processing times and failure rates. You can combine the processing time and failure rate attributes together into one combination factor, then vary the attributes as a pair and compare the two equipment types to each other.

If you do not combine the attributes, but vary each attribute independently in an analysis, you could get some attributes from entity 1 with some attributes from entity 2, creating meaningless combinations. For example, you do not want to create an equipment type that uses the failure rate of equipment type 1 with the processing time of equipment type 2.

To use a combination factor

1. From either the **Factors** tab or from an analysis dialog box, define the individual factors.
2. From either the **Factors** tab or from an analysis dialog box, define a new combination factor.

   The program generates a default name. When you make one selection, the default name is Combo: followed by the selected factor name. When you make more than one selection, the default name is composed of the factor names separated by a comma, as shown in the following Combination Factor dialog.



   Notes

   - You can replace the default name with a user-defined name (see "Name" in "Analysis information" on page 148).
   - When you edit a combination factor that was named using the default name, the name will be updated as the factor definition is changed.

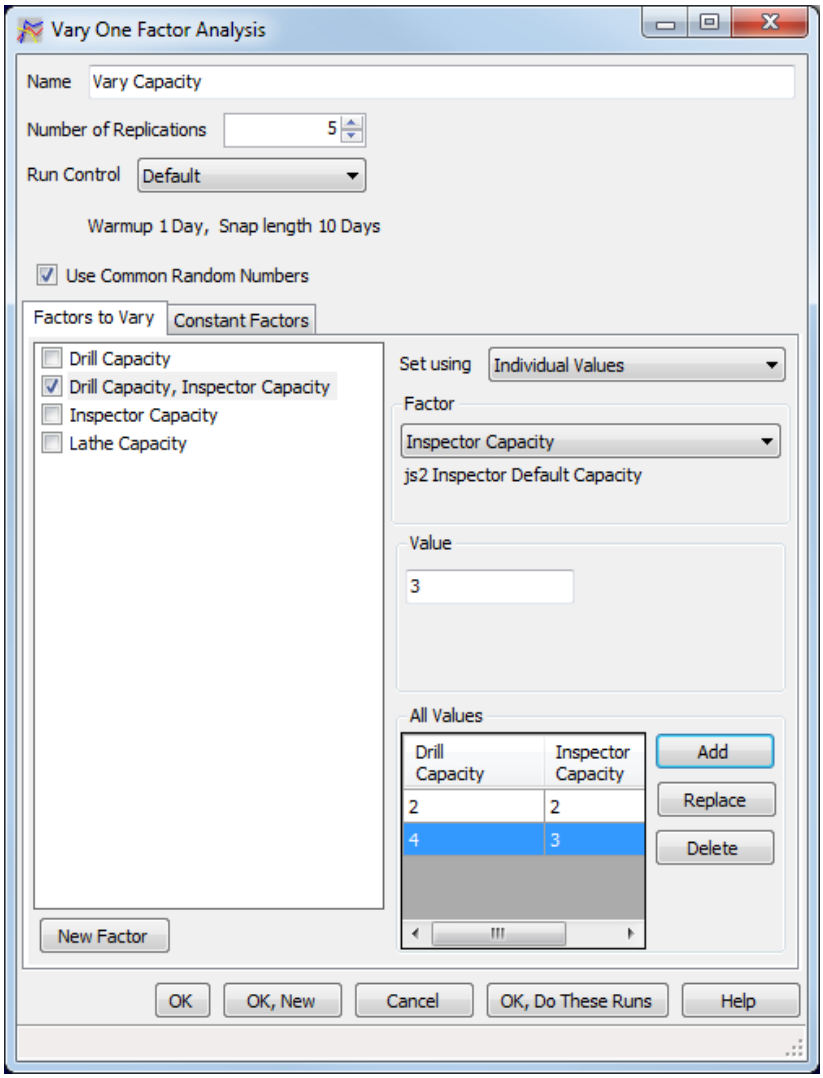
3. Click **OK.**
4. Define an analysis (usually vary one factor or vary multiple factors).
5. Click the combination factor.
6. Select **Individual Values** from the **Set using** list.

   Note You can set **Individual Values** for only the vary one factor, vary multiple factor, and optimization analyses.

7. In the **Factor** list, select one of the factors in the combination factor and enter its value in the **Value** box.

8. Repeat step 7 for each factor within the combination, until you have defined values for all of them.
9. Click **Add**.
10. Repeat steps 7 through 9 if you want to add another set of values to the **All Values** list.

In the following example, two sets of factor values have been added to the combination factor.



Notes

- Each set of values in the combination factor must be unique. Any attempt to add a set of values that is identical to a line already in the **All Values** list is ignored.
- Runs are made using the values in a combination factor in the order in which they appear in the **All Values** list.

» **Tip** To change factor values, highlight the line in the **All Values** list, repeat steps 7 and 8 to redefine all the values in that line, then click **Replace**.

11. Make the runs.
12. Display the results to compare the two scenarios.

Data file cell factor

You can change the data in any of your model's tab-delimited data files in an analysis. For example, suppose your model reads each product's routing from a data file, or reads information used in a custom function. You can change the value in a cell of that file and vary it in an analysis.

Notes

- If you need to change the entire file, you can define an entire data file factor (see "Entire data file factor" on page 185). If you define both an entire data file and a cell within that file as factors, AutoStat varies the entire file first, then the individual cell.
- Also, if your model reads external data files and you want to make runs remotely, you must add the data files to the list of data files to change, even if you do not vary the files.

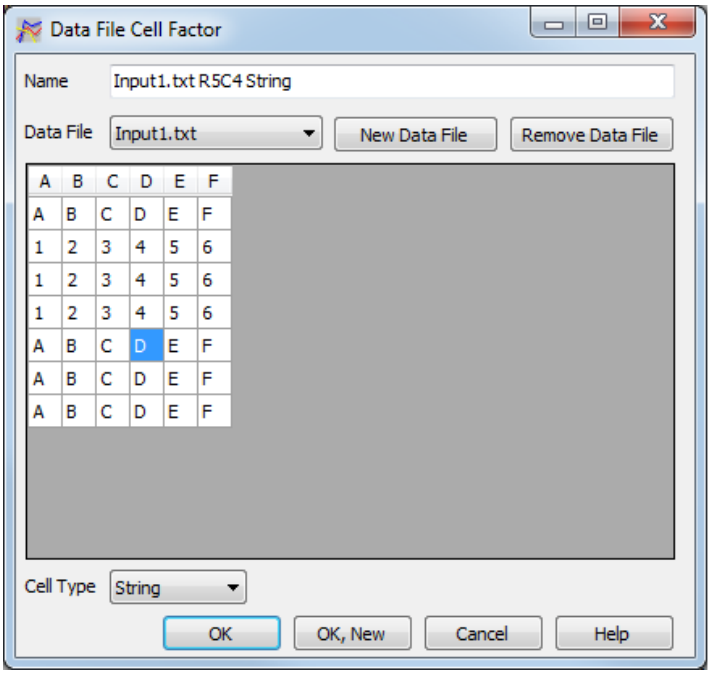
To define a data file cell as a factor

1. From either the **Factors** tab or from an analysis dialog box, define a new data file cell factor.
2. The program automatically selects the first data file in the list box as the default file name. If all data files have been deleted, the list box will be empty.

   Notes

   - You can replace the default name with a user-defined name (see "Name" in "Analysis information" on page 148).
   - When you edit a data file cell factor that was named using the default name, the name will be updated as the factor definition is changed.
   - The data files and alternate content files must be located within the model's base directory.
3. Select the data file that contains the cell you want to change. If necessary, click **New Data File** and navigate to the base version of the data file your model is reading (for example, Input1.txt). When you select a data file and cell, the default data file cell factor name is [filename] R[cell row]C[cell column] [cell type] (See the following example). Click **Open**.

4. This adds the file to the list of data files used in a model.



5. Click the cell to vary.
6. In the **Cell Type** list click the type of data the cell contains (**real**, **integer**, or **string**).
7. Click **OK**.

» **Tip** To create multiple factors select the cell then click **OK, New**. Repeat until all factors are created.

8. In an analysis, assign the factor values (see "Assigning values to factors" on page 187).

Entire data file factor

You can use a whole new version of your model's data file(s) in an analysis. For example, suppose your model reads each product's routing from a data file, or reads information used in a custom function. You can create several versions of the file and use specific versions in each analysis.

Notes

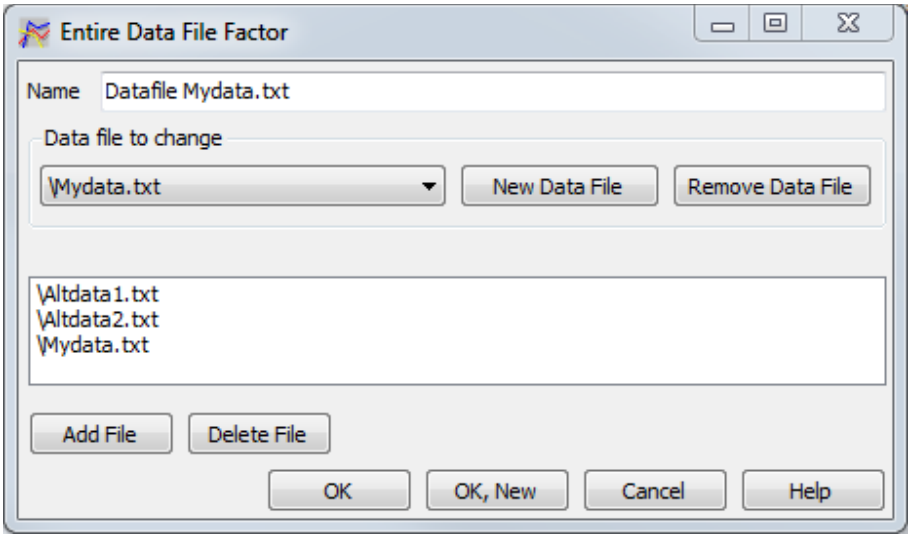
- If you only need to change a few cells in the file, you can define data file cell factors. If you define both an entire data file and a cell within that file as factors, AutoStat varies the entire file first, then the individual cell.
- Also, if your model reads external data files and you want to make runs remotely, you must add the data files to the **Data file to change** list, even if you do not vary the files.
- The desired versions of the data file must exist before defining the factor.

To define an entire data file as a factor

1. From either the **Factors** tab or from an analysis dialog box, define a new entire data file factor.
2. Click **New Data File** to select the data file you want to change. Once selected, a default name for the factor is generated. The default name is the key word, Datafile, followed by the name of the selected data file.

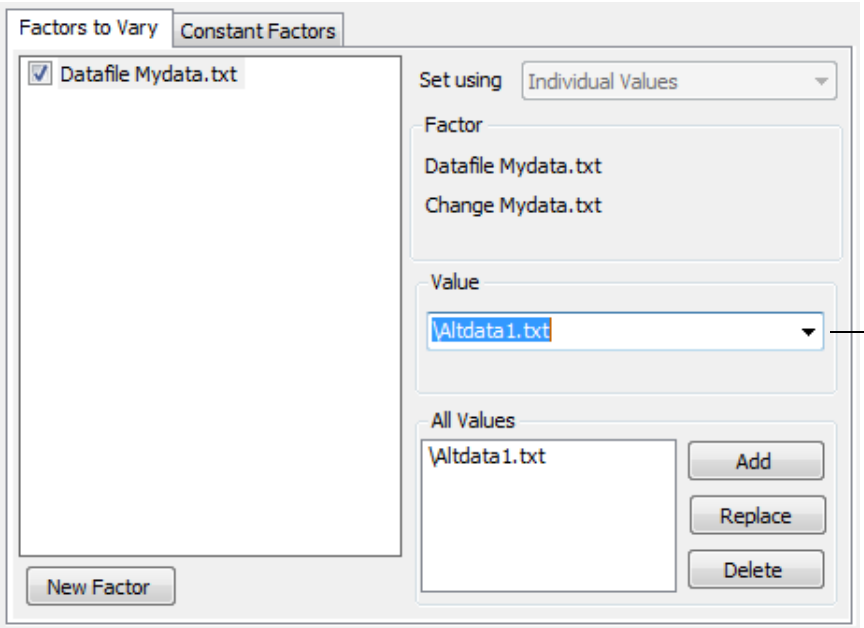
Notes

- You can replace the default name with a user-defined name (see "Name" in "Analysis information" on page 148).
- When you edit an entire data file factor that was named using the default name, the name will be updated as the factor definition is changed.
- If you edit an entire data file factor and delete the data file it depends on, the dialog box from which you delete the data file will automatically close when you delete the entire data file factor.
- The data files and alternate content files must be located within the model's base directory.



3. Click **Add File**, then open the copy of the file that you want to use. The alternate copy of your data file appears in the list after the base file's name. When you switch data files, all of the alternate file(s) are updated, using the last selections that were made for the data file in the dialog form (if any).
4. Repeat step 3 until you have defined all the possible versions of the file that you want to use.
5. Click **OK**.

6. In an analysis, vary the data file factor by selecting the desired version(s) of the file from the **Value** list. In this example, Altdata1.txt is substituted for Datafile Mydata.txt:



Select the version(s) of the file you want to use and click **Add** or **Replace.**

Assigning values to factors

There are several ways to assign values for a factor in an analysis:

- Setting a constant value
- Varying values over a range using begin, end, and increment
- Setting individual (specific) values (not consecutive in a range)

Setting a constant value

You can set a specific value for one or more factors in an analysis. For example, suppose in your base model, the number of resources is four, but through an analysis you determine that six resources are required. For analyses you perform after that, you can set the number of resources to six while experimenting with other factors.

To set a constant factor value

1. Select the **Constant Factors** tab in an analysis.
2. Select the factor you want to hold constant and assign it the desired value.

Varying values over a range

You might want to vary factor values over a range during an experiment. For example, suppose you want to vary the number of resources from 7 to 11, each time using one additional resource (7, 8, 9, 10, and 11 resources).

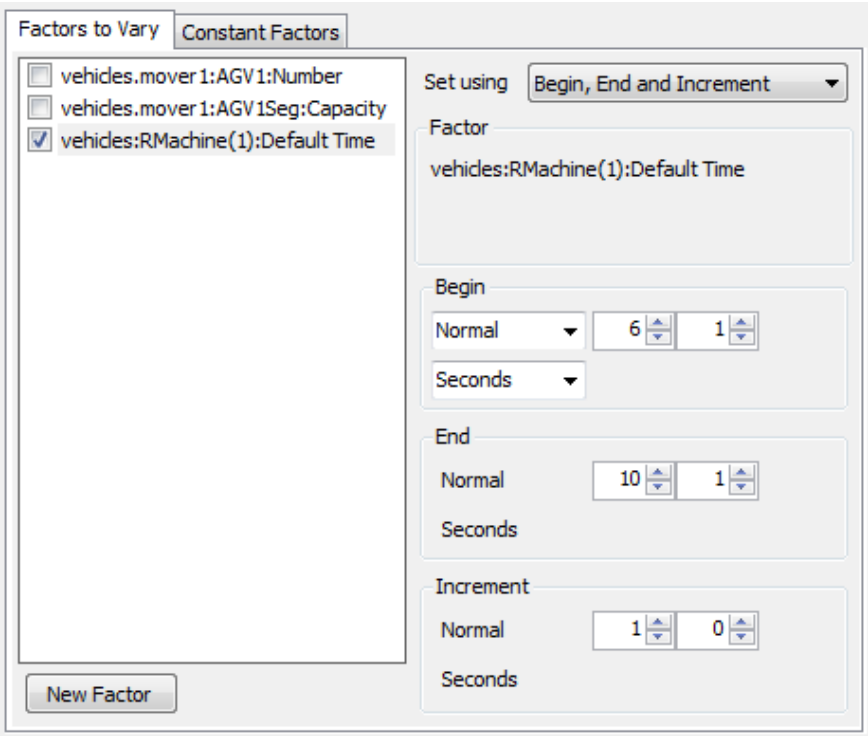
To vary the value of a factor over a range

1. In an analysis dialog box, click the **Factors to Vary** tab.
2. Select the factor in the list or define a new factor.
3. Select **Begin, End, and Increment** from the **Set values using** list.
4. Type the range of numbers in the edit fields, including the beginning value, the ending value, and the incremental increase in value. In the example of varying resources, you would type **7**, **11**, and **1**.

Note When using distributions, all distributions and units must be the same.

To increment distribution values

1. Select **Set values using Begin, End, and Increment**.
2. Edit the beginning, ending, and increment values.



In the preceding example, the default process time is varied from a normal 6, 1 seconds to normal 10, 1 seconds by 1 second increments (6, 1 seconds, 7, 1 seconds, 8, 1 seconds, and so on).

Guidelines for incrementing distributions

- All distribution types (such as normal or exponential) must be the same.
- All time units are same.
- Type an increment of 0 for arguments that you want to hold constant.
- If the values in the increment differ, AutoStat creates the minimum number of combinations until one of the arguments has reached its end value.

Setting individual (specific) values

If you want to set specific values for a factor and they are not numerical or are not consecutive over a range, use the **Individual Values** option in an analysis dialog box.

To set specific values with the Individual Values option in an analysis dialog box

1. In an analysis dialog box, click the **Factors to Vary** tab.
2. Select the factor in the list, or define a new factor.
3. Select **Individual Values** from the **Set values using** list.

   Note For sensitivity analyses, you do not need to select individual values; simply type a value and a margin of error number (the value plus or minus the margin of error).

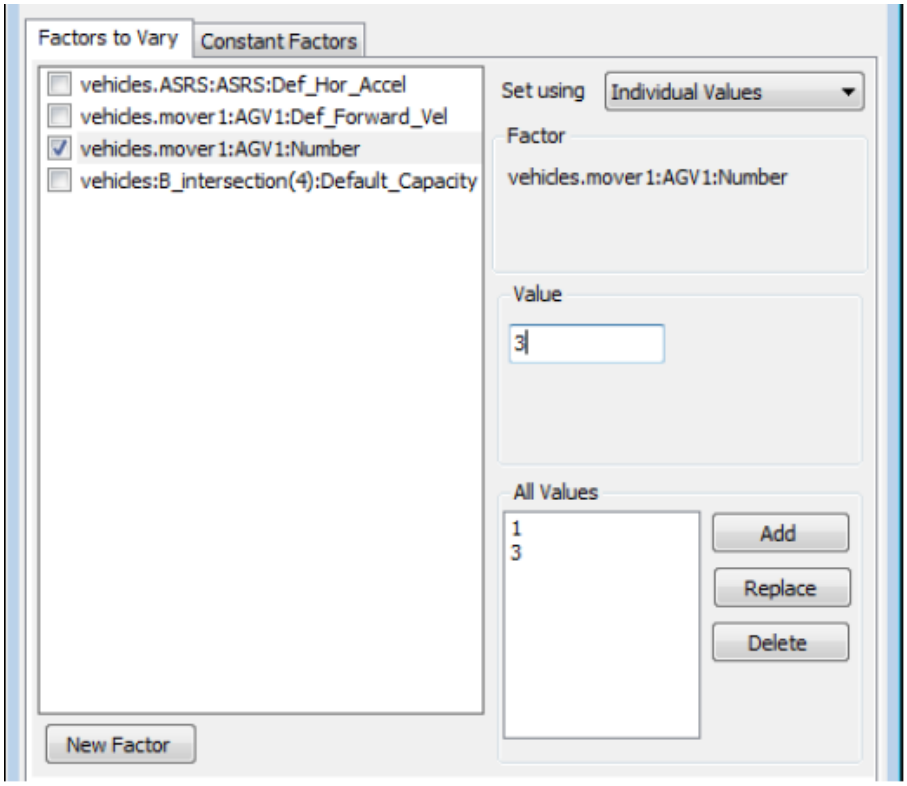
4. Type a value in the **Value** field below the **All Values** list and click **Add**.
5. Enter all desired values using the **Add**, **Replace**, and **Delete** buttons.

   Note When the factor is a data file, pick alternate file names from the list.

Example—Setting individual values

In the following example, the number of AGV1 vehicles is set to the specific values 1 and 3.

1. In the list, select the **vehicles.mover1:AGV1:Number** factor**.**
2. In the **Set using** list, select **Individual Values**.
3. Type **1** in the **Value** field and click **Add**.
4. Type **3** in the **Value** field and click **Add**.



Section 5: Making runs

AutoStat makes runs for each of your experiments, varying the random numbers and/or any factors that you have defined to vary.

The **Runs** tab shows a list of all runs that have been made for defined analyses, and a list of error runs (runs that could not finish due to an error in the model).

There are several ways to make runs for an analysis:

- From an analysis tab
- Using the **Execution** menu
- From the **Runs** tab

You can make runs in parallel on your local machine or on remote machines or both. See "Making parallel runs on local or remote computers" on page 192.

Making runs from an analysis dialog box

You can make runs from an analysis tab, such as the Single Factor Analysis dialog box.

To make runs from an analysis tab

1. Click **OK, Do These Runs**.
2. Display the results (see "Section 7: Displaying analysis results" on page 198).
3. Define a response (see "Defining responses" on page 194).

Note When you click **OK, Do These Runs**, a dialog asks you to confirm the number and estimated duration of the runs. If you click **Yes**, the runs will be made. If you click **No**, you will be returned to the analysis form. Any edits will be saved, no matter the response.

Making runs using the Execution menu or Runs tab

The **Execution** menu and **Runs** tab have the following options:

Option	Definition
Do All Runs	Make all runs that have been defined for all analyses. By default, this does not include optimization runs, but you can include optimization runs using optimization options (see "Setting optimization options" on page 171).
Do Some Runs	Select the analyses for which you want to make runs. The displayed status of each analysis will be one of the following: • *No Runs Required* - The analysis is finished. • *Requires Runs* - The analysis is not finished; the number of runs still needed is unknown. • *Requires n Runs* - An additional *n* runs are needed to finish the analysis.

Option	Definition
Do Runs Until	Define a time to stop making runs, for example, to continue making runs until 8:00 A.M. tomorrow morning (which is the default stop time). Optimization analysis runs can be made with the Do Runs Until option at either of two times: 1. When all required runs are being done. 2. After all required runs have completed (if there is still time remaining before the stop time). Extra runs for an analysis can be made only if all the following criteria have been met: • The model is random (not deterministic). • The analysis is included among the required runs. • All the required runs and optimization runs have been done first. • There is still time remaining before the stop time.
Parallel Execution	Set up AutoStat to make runs on more than one machine.

Once you have made runs, display the results (see "Section 7: Displaying analysis results" on page 198).

All completed runs and error runs are listed in the **Runs** tab. For information about error runs, see the Help.

Viewing run reports

▶ **To view the report files for any run**

From the Runs Used, Failed Audits, or Outlier Runs dialog box, click **View Reports**.

Report files are opened using the standard AutoStat report viewer.

▶ **To view the message file for any run**

From the Runs Used, Failed Audits, or Outlier Runs dialog box, click **View Messages**.

The message file is opened using the application listed in the Model Properties dialog box.

Making parallel runs on local or remote computers

To speed up model analysis in AutoStat, you can make multiple parallel runs on the local computer or on remote computers, or both. Parallel runs can be executed across multiple processors. For example, if you have a machine with three processors, you can specify three parallel runs and the operating system keeps three runs executing concurrently until all scheduled runs are completed. The operating system balances the runs across the three processors as necessary.

Making parallel runs on the local machine

▶ **To execute multiple concurrent runs on the local machine**

1. In AutoStat, click **Execution > Parallel Execution Setup**.
2. Select **Make Local Runs**.
3. In the Number of Parallel Local Runs field, type the number of runs to be executed concurrently. The valid range is 1 to 100 and the default value is 1.

   If the machine is a multiprocessor machine, the operating system manages the distribution of tasks among the processors.
4. Click **OK**.
5. Do the model runs.

Making parallel runs on remote machines

■ **Note** The only software the remote machines need to have installed and running is the Process Server application. (The Process Server is a separate application available on the installation CD. For information about using the Process Server, see the Help.) The remote machines also need to have disk space available (approximately 5 MB plus the space required for one AutoStat run of your model).

▶ **To execute parallel runs on remote machines**

1. Set up each remote machine.
2. Set up remote runs in AutoStat on the local machine.

See the subsequent sections for details.

Setting up remote machines

▶ **To set up remote machines**

On each remote machine, open the Process Server. (If the Process Server is not installed on the remote machines, install it using the same CD from which you installed AutoStat.)

For details about configuring the Process Server on the remote machines, see the Help on the remote machines.

» **Tip** ***Write down*** the port number you define in the Process Server; all machines on which you want to run must use the same port number.

Once all remote machines are configured correctly, set up your remote runs in AutoStat.

Setting up remote runs for AutoStat on the local machine

To set up remote runs in AutoStat on the local machine

1. On the local machine, select **Execution > Parallel Execution Setup** in AutoStat.
2. If you want to make runs on your local machine while making remote runs, select the **Make Local Runs** check box. If the local machine has more than one processor, type the number of processors in the **Number of Parallel Local Runs** field.

   The operating system manages the distribution of the tasks between the processors and writes temporary files.
3. Use **Insert** or **Append** to add machines on your network to the list of remote machines.

   You can define remote machines using the following options:

Option	Definition
Use	A flag to denote whether to use the defined machine for the next set of runs.
Machine Name	The name of a machine on your network (as it appears in the Windows Network Neighborhood).
Number of Runs	The number of processors you want to use on the remote machine (if it has more than one processor). The operating system manages the distribution of the tasks between the processors. The valid range is 1 to 100 and the default value is 1.
Port	A number greater than 1000 that identifies a unique process (application) running on a machine. The port number listed in AutoStat must match the port number defined in the Process Server on the remote machines you want to use. Use the same port number for the Process Server for all machines you are using for remote runs. There is no default port value.

4. Make all defined runs.

Note If your model reads external data files, you must add them to the list of files to change for either data file factors or cell file factors in order to run on remote machines. Otherwise, the files will not be copied to the remote machines and the runs will fail.

Section 6: Defining responses

A response is a statistic of a model, found in either a standard report or in a user-defined report. Responses are used to measure the performance of a system, such as throughput, machine utilization, or vehicle congestion. A response's value can change when you vary model input (factors).

Defining responses

Responses are defined in one of two ways:

- Using an output dialog box
- Using the **Responses** tab

Using an output dialog box

When viewing output for an analysis, you can define new responses.

▶ **To define responses using an output dialog box**

1. Select and display an output type, such as a graph (see "Section 7: Displaying analysis results" on page 198).
2. Click **Responses > New Response**.
3. Select the type of response you want to define and click **Create**.

   For information about response types, see "Response types" on page 195.
4. Name the response (see "Analysis information" on page 148).
5. Define the response.
6. View the analysis results to see the response's values (see "Section 7: Displaying analysis results" on page 198).

Using the Responses tab

▶ **To define responses from the Responses tab**

1. In the **Create New Response of Type** list, select the type of response you want to define and click **New**.

   For information about response types, see "Response types" on page 195.
2. Name the response (see "Analysis information" on page 148).
3. Define the response.
4. View the analysis results to see the response's values (see "Section 7: Displaying analysis results" on page 198).

Response types

You can define several types of responses:

Type	Definition
AutoMod	Statistics from the AutoMod report.
Combination	A combination response takes one or more already-defined factors or responses and combines them to define a new response. Combination responses are useful when there are several characteristics you want to measure for an entity in order to compare the overall effect to another entity.
User	Statistics that are included in a user-defined report (as opposed to a standard report).

AutoMod response

AutoMod responses are statistics from the AutoMod report.

! **Important** When you create an AutoMod response, AutoStat uses the values in the AutoMod report to generate the response statistics. Consequently, before creating an AutoMod response for an entity, you must be sure that entity is configured to generate statistics in the AutoMod report.

▶ **To define an AutoMod response**

1. From either the **Responses** tab or from an output dialog box, define an AutoMod response.
2. In the **System** list, click the system containing the entity for which you want to create a response.
3. In the **Entity** list, click the response's entity type, then click the entity for which you want to create a response.
4. In the **Statistic** list, click the statistic type (such as Average Claim Count, Average Wait Time, etc.) for which you want to create a response. All available statistics are listed in the **Statistic** list box.
5. For total statistics, click in the **Report as** list to determine whether the statistic is the raw total or the total over a length of time. For example, if the model generates 800 widgets in an 8-hour snap, do you want the response to show `800` total or `100 per hour`?
6. Click **OK**.

■ **Notes**

- You can replace the default name with a user-defined name (see "Name" in "Analysis information" on page 148).
- When you edit an existing AutoMod response that was named using the default name, the name will be updated as the response definition is changed.

» **Tip** You can define multiple responses at a time by selecting multiple entities (use the CTRL key to pick entities that are not consecutive in the list). The statistic must be the same for all selected entities. When you

make multiple entity selections, the default name is displayed with a placeholder(<ENTITY>) for the entity name, for example, job shop:<ENTITY>:Average Claim Count. When you click the **OK** or **OK, New** button this placeholder is replaced with the actual entity name for each response that you create.

Like all responses, this response is calculated for all runs and for all analyses.

Combination response

A combination response takes one or more already-defined factors or responses and combines them to define a new response. Combination responses are useful when there are several characteristics you want to measure for an entity in order to compare the overall effect with another entity. You can also average several machines' utilization into one statistic, or sum several different WIP levels to get total WIP.

For example, suppose you want to determine the effect of buying one type of equipment versus another. You can define a combination response that includes each machine's processing time, failure rate, throughput rate, and so on. Then you can compare one type of equipment to the other with a single response.

Note Only factors and responses that are numerical values can be used in a combination response. You cannot use factors or responses that are names or other strings.

To define a combination response

1. Define all the necessary factors and responses separately.
2. From either the **Responses** tab or from an output dialog box, define a combination response.
3. The system generates a default name, Combo Response. If the default name is already used, new combination response names are appended with an index number such as Combo Response 1. To create multiple combination responses, click **OK, New**. (See "Name" in "Analysis information" on page 148.)
4. Insert and append the desired terms using the following information:

Type	Whether the term is a factor or response.
Name	The name of the term.
Weight	The relative importance (weight) of each term. For example, perhaps failure rate is two times more important than processing time

Notes

- When factors and responses are added, renamed, or deleted the list boxes are updated, any invalid selections are cleared, and the affected cell is left blank.
- The formula used to calculate the response is shown at the bottom of the dialog box.

5. Click **OK**.

   Like all responses, this response is calculated for all runs and for all analyses.

6. View the analysis results to see the response's values (see "Section 7: Displaying analysis results" on page 198).

User response

User responses are statistics that are included in a user-defined report (as opposed to a standard report).

! **Important** Be sure to use the **model snap function** to create your user-defined reports and include a snap separator. For information on using the model snap function, see "model snap function" in the AutoMod Help.

Note The number of snaps in the user report must match the number of snaps being used in the AutoStat run control.

To define a statistic in a user-defined report as a response

1. From either the **Responses** tab or from an output dialog box, define a user response.
2. In the **User Reports** list box, click the user-defined report file that contains the desired response. The user report is automatically displayed. If necessary, click **New User Report** to locate the report.

Notes

- You may need to complete a run to generate the user report file if one does not exist.
- Any user report used to create a user response MUST be tab delimited.

3. Define the snap separator.

   The *snap separator* is a unique string that identifies the beginning of each snap in a report. The separator cannot appear anywhere else in the reports, that is, it cannot be a statistic name.

4. Click in the cell for the desired statistic (response). You can select only one cell at a time. To create multiple responses select the cell you want to change, click **OK, New** then repeat until all responses are created.
5. Select whether the response is an average, minimum, maximum, or total statistic from the **Statistic Type** list. (Confidence intervals can only be calculated for average and total statistics.)
6. Click **OK**.
7. When you select a user report, snap separator, and valid cell, the default user response name is *[filename] +[cell row offset from the snap separator]C[cell column] [statistic type]*, for example, UserReport1.tx+3C4 Average.

Notes

- You can replace the default name with a user-defined name.
- When you edit an existing user response that was named using the default name, the name will be updated as the response definition is changed.
- If you edit a user response and delete the user report file it depends on, the dialog box from which you delete the data file will automatically close when you delete the user response.

- The data files and alternate content files must be located within the model's base directory.

8. View the analysis results to see the response's values (see "Section 7: Displaying analysis results" on page 198).

Like all responses, this response is calculated for all runs and for all analyses.

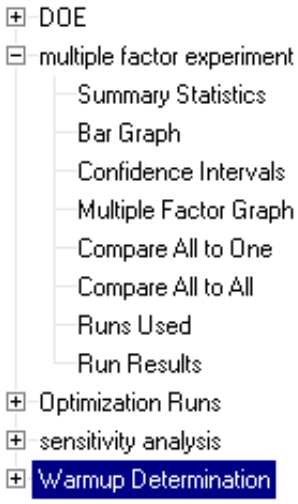
Section 7: Displaying analysis results

After you have made runs for an analysis, you can display the results.

Displaying analysis results

To display analysis results

1. In the analysis tab, click the plus (+) sign to expand an analysis' list.



2. Select the type of output you want to view (output varies by analysis type) and click **Display Results**:
   - **Line graph** (see "Line graphs" on page 199)
   - **Bar graph** (see "Bar graphs" on page 199)
   - **Warmup graph** (see "Viewing the warmup graph" on page 159)
   - **Progress graph** for optimization (see *"Viewing the progress graph" on page 173*)
   - **History** for optimization (see the Help)
   - **Multiple factors graph** (see "Multiple factors graph" on page 163)
   - **Summary statistics** (see "Summary statistics" on page 199)
   - **Confidence intervals** (see "Confidence intervals" on page 201)
   - **Compare all to all** (see "Compare all to all" on page 203)
   - **Compare all to one** (see "Compare all to one" on page 209)
   - **Run results** (see "Run results" on page 210)

- **Runs used** (see "Runs used" on page 210)

Line graphs

Line graphs are available for certain analyses.

To use a line graph

1. Display the graph for the desired analysis.
2. Select the response(s) for which you want to view data. Define a new response if necessary (see "Defining responses" on page 194).
3. Select the desired confidence level. Bars denoting the upper and lower values of the confidence interval appear. For more information, see "Confidence intervals" on page 201.

Bar graphs

Bar graphs are available for certain analyses.

To use a bar graph

1. Display the graph for the desired analysis.
2. Select the response(s) for which you want to view data. Define a new response if necessary (see "Defining responses" on page 194).

Tip See "Interpreting output for DOE and sensitivity analyses" on page 177 if you are looking at a DOE or sensitivity analysis graph.

Summary statistics

Summary statistics are the response values for each set of factor combinations in an analysis.

To display summary statistics

1. In the analysis tab, click the plus (+) sign to expand an analysis' list.
2. Double-click **Summary Statistics**.
3. If desired, save the data to a text file to use in another application or copy and paste the data into another application.

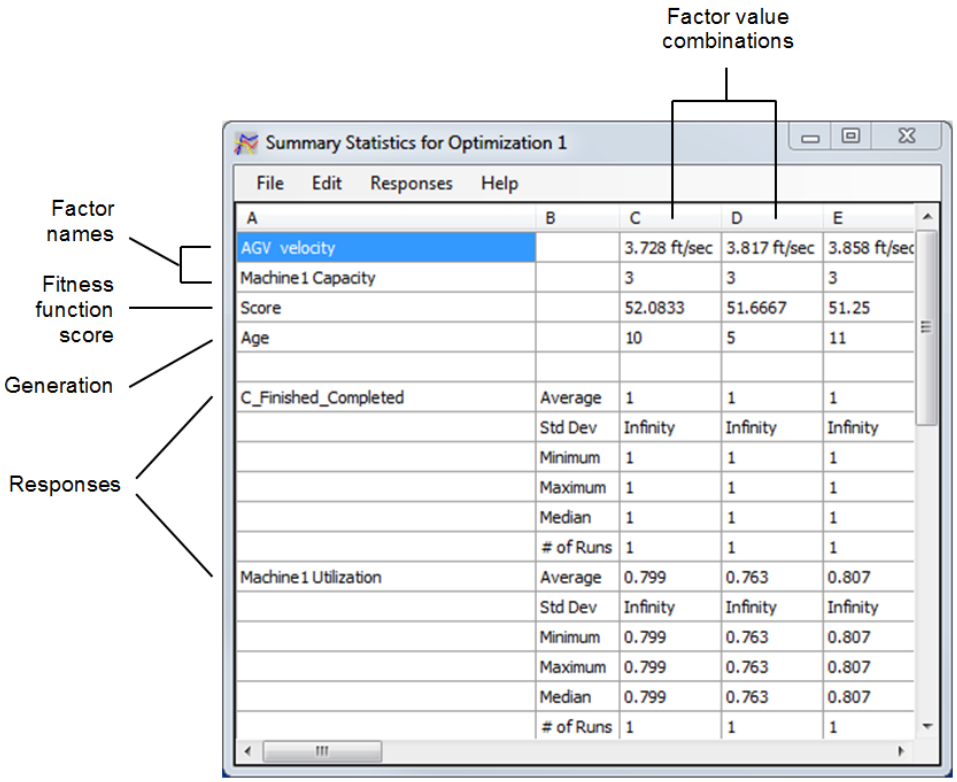
Statistics

The columns show combinations of factor values across the top, with each combination's response values in rows.

For each response, the following information is displayed (summary statistics for optimization, DOE, and sensitivity analyses are slightly different):

Value	Definition
Average	The average value of the response over the number of runs listed in the **# of Runs** field.
Std. Dev.	The sample standard deviation (Std. Dev.) for the average.
Minimum	The minimum value for the response over the # of Runs listed.
Maximum	The maximum value for the response over the # of Runs listed.
Median	The median (middle value) for the response over the # of Runs listed.
# of Runs	The number of runs used to calculate the average response (determined by the number of replications defined for the analysis).

Example of summary statistics



A	B	C	D	E
AGV velocity		3.728 ft/sec	3.817 ft/sec	3.858 ft/sec
Machine1 Capacity		3	3	3
Score		52.0833	51.6667	51.25
Age		10	5	11
C_Finished_Completed	Average	1	1	1
	Std Dev	Infinity	Infinity	Infinity
	Minimum	1	1	1
	Maximum	1	1	1
	Median	1	1	1
	# of Runs	1	1	1
Machine1 Utilization	Average	0.799	0.763	0.807
	Std Dev	Infinity	Infinity	Infinity
	Minimum	0.799	0.763	0.807
	Maximum	0.799	0.763	0.807
	Median	0.799	0.763	0.807
	# of Runs	1	1	1

Optimization statistics

For optimization analyses, *the best scenario is listed first* (closest to the left) in the summary statistics. The fitness score response is shown, along with:

Score	The average score for the fitness function response. The score allows you to quickly see differences between scenarios.
Age	The number of generations that a child (set of factor values) has been one of the best 30 solutions.

For more information about optimization analyses, see "Optimization analysis" on page 165.

DOE statistics

Instead of factor values, each column in the summary statistics for a DOE shows a main effect or factor interaction effects. For each response you can see whether the effect is significant or not.

For more information about DOE analyses, see "Factorial DOE analysis" on page 176.

Sensitivity analysis statistics

Instead of factor values, each column in the summary statistics for a sensitivity analysis shows a main effect. For each response you can see whether the effect is significant or not to determine how sensitive the model is to that data.

For more information about sensitivity analyses, see "Sensitivity analysis" on page 175.

Confidence intervals

A confidence interval is a range of numbers such that there is a given confidence that the true average value of the response will be within the interval. For example, before performing an analysis, you could be 95 percent confident that the average throughput response will be between 1100 units per week and 1150 units per week. If the range shown is too large, you can make more runs to reduce it.

» **Tip** To decrease the width of a confidence interval by approximately half, you must increase the sample size by 4 (either make 4 times as many replications or increase the run length by 4 times).

AutoStat only computes confidence intervals for responses that are an average or total statistic. *If your response is a maximum or minimum value, no confidence intervals are displayed.*

▶ **To view confidence intervals**

1. From the analysis tab, display **Confidence Intervals** for an analysis.

   ■ **Note** You can also see confidence intervals for the difference between compare all to all, compare all to one, and line graphs.

2. Select the confidence level from the **Confidence Level** list at the top of the dialog box (see "Confidence level" on page 201).
3. Read the top of the columns to find the desired set of factor values or effects, then read down the column to the desired response to view the statistics.
4. If desired, save the data to a text file to use in another application or copy and paste the data into another application.

Confidence level

The confidence level is the level of assurance that the response value being viewed will fall within the range shown.

Note If you specify `none` as the level, the average of the response is shown.

The following table illustrates a response's possible ranges at various confidence intervals:

Confidence Level	Range's Low Value	Range's High Value
.8 (80 percent)	795.2	820.8
.9 (90 percent)	790.2	825.8
.95 (95 percent)	784.8	831.2

In the preceding example, the response's *range of possible values is smaller when the level of confidence is lower, and the range is larger when the level is higher.* For example, you can be 80 percent certain that the value falls between 795.2 and 820.8, but if you want to be 95 percent certain of the range, the range is 784.8 to 831.2.

Statistics

Statistic	Definition
CI Low	The lower value in the confidence interval range.
CI High	The upper value in the confidence interval range.
# of Runs	The number of runs used to calculate the confidence interval (determined by the number of replications).

Note If there has only been one run, the run's value is shown in CI Low, and CI High is blank.

Equations used to calculate a confidence interval

AutoStat uses the replication/deletion method to construct confidence intervals for means of steady-state responses. The Replication/Deletion method forms a confidence interval on the mean response. The initial warmup period (if any) is not included.

The point estimate for the mean response is defined as:

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Where X_i is the response from run i, and n is the number of replications. The X_i's are independently and identically distributed approximately normal random variables. Due to deletion of warmup statistics, this results in an approximately unbiased estimator of the mean steady-state response.

A 100 (1-α) percent confidence interval for the mean response can be constructed as:

$$\bar{X} \pm t_{(n-1,\ 1-\alpha/2)} \sqrt{\frac{s^2}{n}}$$

where $\sqrt{\frac{s^2}{n}}$ is the standard error, $t_{(n-1,\ 1-\alpha/2)}$ is the 100 (1−α/2) percentage point of a t-distribution with n−1 degrees of freedom, and the variance estimate is:

$$s^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1}$$

Compare all to all

One of the greatest benefits of simulation is the ability to compare different system designs. These different designs are called scenarios or configurations. It is more convenient and inexpensive to answer "What if..." questions using computer simulation than it is to experiment on the shop floor or in the warehouse.

Comparison confidence intervals in a non-CRN analysis

When an analysis does not use common random numbers, AutoStat uses *Welch's confidence interval* to estimate the magnitude of differences between two configurations for a chosen response. Welch's confidence interval is used for both "compare all to all" and "compare all to one" results.

Welch's confidence interval method compares the mean response from two configurations with the assumption that the output obtained from one configuration is independent of output from the other configuration. Because this method compares mean responses, it does not restrict the sample sizes to being equal (that is, AutoStat can calculate confidence intervals even if the same number of runs have not been completed for each configuration). Tables of pairwise comparisons provide a good reference for the relative efficiency of each configuration under consideration (see the table on page 1.83). Confidence intervals are calculated for the difference in response between each pair of configurations.

Suppose you have four configurations you want to compare on the basis of throughput/hour. The output from each replication, as well as the sample mean and sample variance, is as follows:

	Response Config 1	**Response Config 2**	**Response Config 3**	**Response Config 4**
Replication 1	10.1	12.0	9.8	14.1
Replication 2	9.5	13.1	10.4	13.5
Replication 3	11.0	13.1	10.5	15.0
Replication 4	10.2	—	10.1	14.5
Replication 5	11.1	—	10.6	—

	Response Config 1	**Response Config 2**	**Response Config 3**	**Response Config 4**
Mean $(\overline{X}_i)$:	10.38	12.73	10.28	14.28
Variance (s_i^2):	0.447	0.403	0.107	0.403
Replications (n_i):	5	3	5	4

From this data, AutoStat can compare the mean for each possible pair of configurations to create comparison confidence intervals. For more information, see "Confidence interval equations (non-CRN analysis)" on page 206.

Comparison confidence intervals in a CRN analysis

When an analysis uses common random numbers, AutoStat uses the *paired-t method* of calculating confidence intervals for both "compare all to all" and "compare all to one" results.

AutoStat uses the paired-t method to compare two configurations, with the assumption that the responses from both configurations are based on the same random numbers (that is, the responses are correlated rather than independent). Subtracting one configuration's response from the other configuration's response for the same run provides the independent difference between the configurations.

To calculate the confidence intervals for a CRN analysis, AutoStat first calculates the difference in responses for each run in a given replication. Because of this comparison, the sample sizes must be equal (that is, AutoStat must have completed the same number of runs for each configuration). Confidence intervals are calculated for the difference in response between each pair of runs for each configuration.

Suppose you have two configurations you want to compare on the basis of throughput/hour. The output from each replication and differences in response value for each replication are shown in the following. The table also shows the sample mean and sample variance for the differences:

	Response Config 1 $(\mathbf{X_{1j}})$	**Response Config 2** $(\mathbf{X_{2j}})$	**Response Difference Between Configs 1 & 2** $(\mathbf{D_j})$
Replication 1	60.3	54.3	6.0
Replication 2	64.4	53.1	11.3
Replication 3	66.2	55.5	10.7
Replication 4	60.7	56.2	4.5
Replication 5	62.1	56.0	6.1
Mean $(\overline{D})$:	—	—	7.72
Variance (s_D^2):	—	—	9.41

From this data, AutoStat can use the mean and the variance of the differences to create comparison confidence intervals. For more information, see "Confidence interval equations (CRN analysis)" on page 207.

Viewing a comparison

The Compare All to All dialog box displays confidence intervals for the average difference between scenarios or configurations. The factor combinations defining a scenario are displayed across the top of each column.

Note AutoStat only computes confidence intervals for responses that are defined as either an average or total statistic. If your response is a maximum or minimum statistic, no confidence intervals are displayed. For more information, see"Defining responses" on page 194.

To compare two scenarios

1. Find the combination of sets of factor values across the top of the columns.
2. Select the confidence level from the **Confidence Level** list at the top of the dialog box (see "Confidence level" on page 201).
3. Read down the column to the desired response and view the desired statistics.
4. If desired, save the data to a text file to use in another application or copy and paste the data into another application.

Statistics

Statistic	Definition
Ave. Diff.	The average difference between the selected scenario and the scenario listed at the top of the column.
CI Low	The lower value in the confidence interval range.
CI High	The upper value in the confidence interval range.

Interpreting the numbers

The smaller the range, the more accurate the estimate.

Tip To decrease the width of a confidence interval by approximately half, you must increase the sample size by four (either make four times as many replications or increase the run length by four times).

When using confidence intervals to compare two systems (compare all to all or compare all to one), the confidence interval (CI) can be used in the following ways:

Usage	Definition
CI spans 0	If the interval between the low and high values spans zero (for example, from -3.2 to 1.6), you cannot tell statistically which scenario is better; make more runs to reduce the range.
CI negative (<0)	A's mean response is < B's. If the response is throughput, for example, and you are trying to increase throughput, B's mean response is higher and therefore better than A's (within the confidence level selected, such as 95 percent).

Usage	Definition
CI positive (>0)	A > B. If the response is throughput, for example, and you are trying to increase throughput, A's mean response is higher and therefore better than B's (within the confidence level selected, such as 95 percent).

Confidence interval equations (non-CRN analysis)

When comparing two alternative configurations in an analysis that does not use common random numbers, the Welch confidence interval (Welch, 1938) is a flexible approach. The Welch method can be used to compare configurations of equal or different sample sizes. The general procedure that AutoStat uses to calculate Welch's confidence intervals is as follows:

1. Make n_i replications of configuration i (the number of replications must be greater than or equal to 2 for each configuration, but the number of replications per configuration does not need to be the same).
2. For each configuration, calculate the sample mean response over the n_i replications:

$$\bar{X}_i = \sum_{j=1}^{n_i} \frac{X_{ij}}{n_i}$$

   where X_{ij} is the response for replication j of configuration i.
3. Calculate the sample variance for configuration i:

$$s_i^2 = \sum_{j=1}^{n_i} \frac{[X_{ij} - \bar{X}_i]^2}{n_i - 1}$$

Note For comparing configurations 1 and 2, the estimated degrees of freedom for the t-statistic is:

$$\hat{f} = \frac{[s_1^2/n_1 + s_2^2/n_2]^2}{[s_1^2/n_1]^2/(n_1 - 1) + [s_2^2/n_2]^2/(n_2 - 1)}$$

4. Combine the preceding statistics to form a 100(1–α) percent confidence interval on the difference in mean response of configurations 1 and 2.

$$\bar{X}_1 - \bar{X}_2 \pm t_{(\hat{f},\, 1-\alpha/2)} \sqrt{(s_1^2/n_1) + (s_2^2/n_2)}$$

   where $\sqrt{(s_1^2/n_1) + (s_2^2/n_2)}$ is the standard error.

   You can use this confidence interval to determine if there is any evidence of a difference in the mean response of the two configurations. If the confidence interval contains zero, there is no evidence of a difference based on the available data.

   If the sample sizes become large enough, eventually there will be a statistically significant difference in the two configurations (see step 3 in the preceding). This test tells you whether a statistically significant difference is found based on a limited sample size, and in addition, the

confidence interval's lower and upper bounds provide an estimate of the magnitude of the difference with the pre-specified (100[1−α] percent) level of confidence.

Example—Method for calculating Welch's confidence intervals

To demonstrate how to use the Welch confidence interval equations, this example uses the data provided in the table in "Comparison confidence intervals in a non-CRN analysis" on page 203 to compare configuration 2 with configuration 1 and calculate a 95 percent confidence interval:

The sample mean response of configuration 2 is: $\overline{X}_2 = 12.73$

The sample mean response of configuration 1 is: $\overline{X}_1 = 10.38$

The difference between the sample mean responses is: $12.73 - 10.38 = 2.35$

The estimated degrees of freedom are:

$$\hat{f} = \frac{[0.403/3 + 0.447/5]^2}{[0.403/3]^2/2 + [0.447/5]^2/4} = 4.54 = 5 \text{ (rounded to the nearest integer)}$$

Consequently, $t_{(5,\,0.975)} = 2.57$ and the standard error is:

$$\sqrt{(0.403/3) + (0.447/5)} = 0.473$$

Combining these statistics yields the 95 percent confidence interval:

$$2.35 \pm (2.57)(0.473) = 2.35 \pm 1.22$$

Using the Welch method, you can calculate comparison confidence intervals for any number of configurations. A convenient way to represent the differences between multiple configurations is to enumerate all possible pairs of configurations. The following 95 percent confidence intervals compare each of the configurations shown in the table in "Comparison confidence intervals in a non-CRN analysis" on page 203:

	Config2	**Config3**	**Config4**
Config1	2.35±1.22	-0.10±0.82	3.90±1.03
Config2	—	-2.45±1.26	1.55±1.35
Config3	—	—	4.00±0.97

When you use AutoStat to conduct a Compare All to All analysis, the software automatically calculates confidence intervals for each pair of configurations in the analysis.

Confidence interval equations (CRN analysis)

When comparing alternative configurations in an analysis that uses common random numbers, AutoStat uses the paired-t method of calculating confidence intervals. This method first calculates the difference between response values for each pair of correlated runs (because of this comparison, you must have completed the same number of runs for each configuration in the analysis). The general procedure that AutoStat uses to calculate paired-t confidence intervals is:

1. Make n replications of each configuration.
2. Calculate the differences in the response X for each replication j in the configurations being compared:

   $$D_j = X_{1j} - X_{2j}$$
3. Calculate the response's sample mean difference between configurations 1 and 2:

   $$\bar{D} = \frac{1}{n}\sum_{j=1}^{n} D_j$$

   where n is the total number of replications of each configuration (that is, $\bar{D} = \bar{X}_1 - \bar{X}_2$).
4. Calculate the sample variance of the differences D_j:

   $$s_D^2 = \frac{1}{n-1}\sum_{j=1}^{n} (D_j - \bar{D})^2$$
5. Combine the preceding statistics to form a 100(1−α) percent confidence interval on the difference in mean response of configurations 1 and 2.

   $$\bar{D} \pm t_{(n-1,\, 1-\alpha/2)}\sqrt{\frac{s_D^2}{n}}$$

   where $\sqrt{\frac{s_D^2}{n}}$ is the standard error.

You can use this confidence interval to determine if there is any evidence of a difference in the mean response of the two configurations. If the confidence interval contains zero, there is no evidence of a difference based on the available data.

Example—Method for calculating paired-t confidence intervals

To demonstrate how to use the paired-t confidence interval equations, this example uses the data provided in the table on page 1.79 to compare configuration 1 with configuration 2 and calculate a 95 percent confidence interval:

The response's sample mean difference between the two configurations is: $\bar{D} = 7.72$

The sample variance of the differences is: $s_D^2 = 9.41$

The degrees of freedom are: $n - 1 = 4$

Consequently, $t_{(4,\, 0.975)} = 2.78$ and the standard error is:

$$\sqrt{\frac{9.41}{5}} = 1.37$$

Combining these statistics yields the 95 percent confidence interval:

$$7.72 \pm (2.78)(1.37) = 7.72 \pm 3.81$$

As with the Welch method of calculating confidence intervals, you can use the paired-t method to calculate comparison confidence intervals for any number of configurations. When you use AutoStat to conduct a Compare All to All analysis, the software automatically calculates confidence intervals for each pair of configurations in the analysis.

Compare all to one

The Compare All to One dialog box displays confidence intervals for the average difference between two scenarios (sets of factor values). The factor combinations are displayed in the bottom half of the dialog box. AutoStat uses either Welch's confidence interval or the
paired-t method to calculate comparison intervals. For more information, see "Compare all to all" on page 203.

Note AutoStat only computes confidence intervals for responses that are defined as either an average or total statistic. If your response is a maximum or minimum statistic, no confidence intervals are displayed. For more information, see "Defining responses" on page 194.

To compare two scenarios

1. In the lower half of the dialog box, select the column containing the base scenario (set of factor values) to which you want to compare all other scenarios.
2. Select the confidence level from the **Confidence Level** list at the top of the dialog box (see "Confidence level" on page 201).
3. In the top half of the dialog box, read the columns to find the desired combination of scenarios. View the statistics.
4. If desired, save the data to a text file to use in another application or copy and paste the data into another application.

Statistics

Statistic	Definition
Ave. Diff.	The average difference between the selected scenario and the scenario listed at the top of the column.
CI Low	The lower value in the confidence interval range.
CI High	The upper value in the confidence interval range.

Interpreting the numbers

The smaller the range, the more accurate the estimate.

Tip To decrease the width of a confidence interval by approximately half, you must increase the sample size by four (either make four times as many replications or increase the run length by four times).

When using confidence intervals to compare two systems (compare all to all or compare all to one), the confidence interval (CI) can be used in the following ways:

Usage	Definition
CI spans 0	If the interval between the low and high values spans zero (for example, from -3.2 to 1.6), you cannot tell statistically which scenario is better; make more runs to reduce the range.
CI negative (<0)	A's mean response is less than B's. If the response is throughput, for example, and you are trying to increase throughput, B's mean response is higher and therefore better than A's (within the confidence level selected, such as 95 percent).
CI positive (>0)	A's mean response is greater than B's. If the response is throughput, for example, and you are trying to increase throughput, A's mean response is higher and therefore better than B's (within the confidence level selected, such as 95 percent).

For the equations used to compute the comparison confidence intervals, see "Compare all to all" on page 203.

Run results

Run results show the response values for each scenario (combination of factor values) for each run for the analysis.

You can click a specific run to display run results. By default the average of all runs is displayed.

For optimization analyses, the run results are only displayed for the 30 best scenarios. For all other analyses, the run results include response values for the warmup snap (if defined) and the sample snap.

- **Note** If desired, save the data to a text file to use in another application or copy and paste the data into another application.

Runs used

The Runs Used dialog box shows a list of all of the runs that were made for an analysis.

- **Note** For optimization analyses, the run results are only displayed for the 30 best scenarios. All other analyses show the results of all runs.

▶ **To view the Runs Used dialog box**

1. In the analysis tab, click the plus (+) sign to expand an analysis' list.
2. Double-click **Runs Used**.

Use the Runs Used dialog box to:

- View the run details, including:

Detail	Definition
Audit Results	A list of any audit conditions failed by the run, and also information about the machine and software version. For information about auditing a model, see "Auditing a model" on page 212.
Run Control details	The first value is the length of the warmup snap, and the second value is the length of the sample time. In warmup analyses, snaps are defined by number and duration; for example, 20 snaps of 8 hours.
Factor values	A list of what the factor values were for the run.
Response values	A list of what the response values were for the run.
Random number set	The set of random numbers used by each random stream in the model.

- View the run's reports.
- Delete runs.
- Make a run an error run (see "Making error runs" on page 215).
-

Note The **View Reports, Message File, Delete Run, Make Error Run**, and **Run with Animation** buttons are enabled when a run is selected.

Section 8: Debugging a model

AutoStat contains several useful tools for finding problems in your model and helping you solve them. These tools include:

- Auditing a model for strange values
- Finding outliers
- Making error runs
- Debugging runs

Auditing a model

Audits **check each run for output values that might indicate a model problem**, such as resources that never get used, unusual response values, and so on. Audits:

- Check an AutoMod model for strange values.
- Check user-defined responses for extreme values.

Once you have defined and run your audit, you can display audit results either for a particular analysis or for all runs made.

Checking an AutoMod model for strange values

An AutoMod model audit checks each run's reports for output values that might indicate a model problem, such as resources that never get used, vehicles that park the entire simulation, and so on. You can also audit responses for a model (see "Setting up a response audit" on page 213).

Note Warmup runs and error runs are not audited.

To define a model audit

1. Select **Diagnostics > Set Up AutoMod Auditing**. There are several default audit conditions that you can edit or delete if desired. For example, by default, AutoStat looks for all queues, resources, and processes with a total equal to zero.

   Note The default audit definitions are always displayed when there are no other AutoMod audit definitions.

2. Select **Rows > Append Row**.
3. Define your audit using the following columns:

- System
- Compare
- Type
- Entity Type
- Value Type
- Entities
- Statistic
- Value

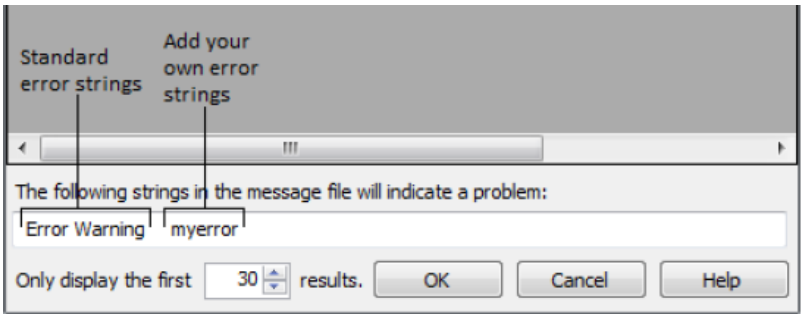
Tip See the Help for information about using these columns.

For example, you could define the following:

System	Entity Type	Statistic	Compare	Value Type	Value	Type	Entities
asrs_demo	Queue	Total Claim Count	>	the number	200	Except for	Space

In this case, any runs in which any queue except the queue "Space" has a total claim count of greater than 200 loads fails the audit and the reason is listed in the audit results.

4. If necessary, add any additional keywords that you are printing to the message file to indicate a problem. The standard software warning and error flags are already included. The strings in the message file are not case-sensitive.



Whenever one of these strings appears in the message file, the run fails the audit.

5. Click **OK**.

The audits are saved and executed, and the audit results appear in a new window.

Deleting an audit condition

▶ **To delete an audit condition**

1. Click in the row you want to delete.
2. Select **Rows > Delete Rows**.

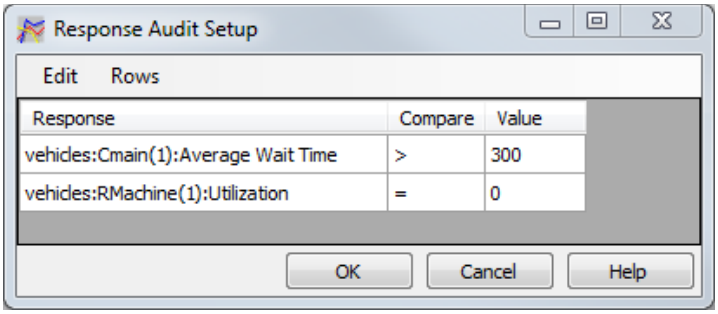
Setting up a response audit

A response audit checks each run's responses for values that might indicate a model problem, such as throughput that is below a certain value, cycle time that is too long, and so on.

▶ **To set up a response audit**

1. Click **Diagnostics > Set Up Response Auditing**.
2. Enter the responses and values that you want to compare. Use the Rows menu to add and delete items to the audit list.

For example, the audit below finds any run in which the response vehicles:Cmain(1):Average Wait Time is greater than 300 or the vehicles:RMachine(1):Utilization = 0.



AutoStat flags any run that meets *any* of the criteria listed.

3. Click **OK**.

   The audits are saved and executed, and the audit results appear in a new window.

Viewing audit results

Audit results are automatically displayed whenever you execute an audit.

▶ **To view audit results without running an audit**

Click **Diagnostics > Display Failed Audits**.

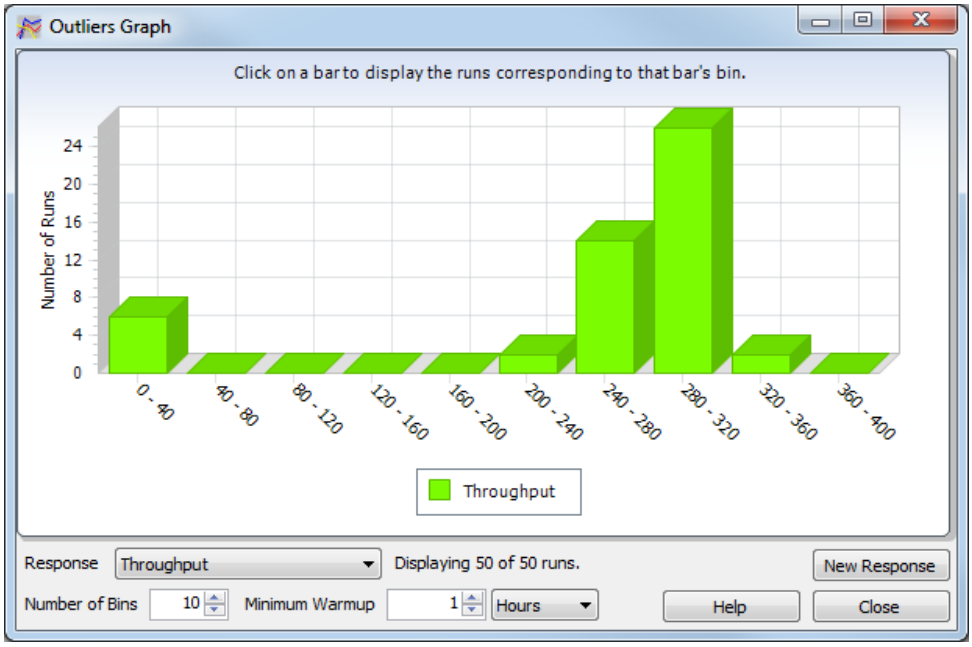
Finding outliers

Outliers are response values from a run that are very different from most other runs' response values.

▶ **To show outliers for a response**

1. Click **Diagnostics > Outliers**.

   The Outliers graph opens.



2. Select the response you want to view from the **Responses** list. You can define new responses if necessary.
3. If desired, edit the number of ranges (bins) for the response value by editing the number of bins. Generally, the more categories there are, the lower the number of runs per category.
4. If desired, edit the warmup length (see "Warmup analysis" on page 156).

▶ **To view the runs that make up a bar in the Outliers graph**

Click in the desired bar in the graph.

A list of runs used appears.

Interpreting the graph

The bars represent the number of runs for which the response was within a range of values. Most of the runs should be grouped around a set of similar values.

» **Tip** Any runs for which the response value is much lower or higher than the majority of the runs might represent a run of the model that you need to debug to determine whether there was a problem with the model.

Making error runs

Error runs are runs that failed for some reason. For example, if the model became stuck in an infinite loop or ended with an error condition, AutoStat flags the run as an error run automatically. You can also manually define one or more runs as error runs.

■ **Note** When setting the model properties, you can define how many error runs are allowed before AutoStat automatically stops making runs. For more information, see "Setting model properties" on page 145.

Error runs are saved so you can debug them to figure out the problem, but are not used to calculate any output graphs or statistics.

Displaying error runs

▶ **To display error runs**

1. Select the **Runs** tab in the AutoStat window.
2. Select **Error Runs** from the **Display** list.
3. Expand the run's list and view the **Error Reasons**.

Making a run an error run

You can make a run an error run if you determine that there is something wrong with it (for example, it is an extreme outlier). Making the run an error run moves it to the error runs list and excludes it from reports and graphs.

▶ **To designate a run as an error run**

1. Select the **Runs** tab in the AutoStat window.
2. Select **Completed Runs** from the **Display** list.
3. Highlight a run and click **Make Error Run**.

» **Tip** You can also make a run an error run from the Runs Used dialog box.

A The AutoMod File System

Model files

Each model that you build creates several files and directories, which are discussed in this section.

.amodviews

The .amodviews file is created when you select **Create View** while running your model. This file stores the saved views.

By default, the .amodviews file is saved in the current directory. The file's location determines which models use it.

If you want the configuration file to affect:	Place the file in:
One model	The model directory
Your models	Your HOME directory
All models for all users	The /lib directory in the installation directory

.am2rc

The .am2rc file is created when you click **File** > **Save Startup Config**. This file saves dialog box size and placement, grid spacing, and so on.

By default, the .am2rc file is saved in the current directory. The file's location determines which models use it.

If you want the configuration file to affect:	Place the file in:
One model	The model directory
Your models	Your HOME directory
All models for all users	The /lib directory in the installation directory

.arc

The archive database is an ASCII representation of the entire model. The <modelname.arc> directory is created when you export. Use the archive to:

- Upgrade to a new version of the software.

- Back up models.

The archive directory contains two types of files:

Type	Definition
.amo	A text file that defines the model defaults and systems used in the model.
.asy	A text file that defines a process or movement system.

.asilibrc

The .asilibrc file defines the model background color. The location of this file determines whether it will affect a specific model, or multiple models (the .asilibrc file is like the .am2rc file described in the preceding). If a model directory contains both a .asilibrc file and an asi.env file, settings in the asi.env file take precedence.

Note You can change the background color of AutoMod using the ASI Configuration Editor. To open the ASI configuration Editor select **Start > Utilities > ASI Configuration Editor**. For more information, click **Help** in the ASI Configuration Editor.

.dat

Files ending with a .dat extension are output files that can be used for debugging your model.

.dir

The <modelname.dir> is a binary version of your model that is version-specific; it is created when you save your model. The .dir directory contains several types of files necessary to run your model. Most of the files are used internally by AutoMod; two of the files can be opened:

File	Definition
.mod	A binary representation of the model.
<modelname>_ver.txt	A text file that can be opened in any text editor; the file provides the version and build number of the AutoMod software that was used to save the model. If you are using a later version of the AutoMod software than what is listed in this file, you need to import the model from the .arc directory instead of opening the model in the .dir directory.

! **Important** Never delete the .dir directory when building or running a model. Doing so irrevocably corrupts your model.

.exe

The <modelname.exe> file is the executable for the model.

.m

The .m file is created when you do a listing of your model.

.message

The .message text file contains all of the text that appeared in the Messages dialog box during the last run. This file can be helpful when debugging your model.

.mviewrc

The .mviewrc file is a legacy file that stored saved views in AutoMod 12.6 and earlier versions; it has been replaced by the .amodviews file. The .mviewrc file is still read by the AutoMod software (to support backward compatibility with older models), but only if the .amodviews file cannot be found.

.print

The print file is a text file that is created if you have `print` statements in the model that are not directed to specific places, for example, the Messages dialog box or a specific file.

.rep

Tab-delimited report files for the model. Generated automatically with the SimController or with the **-B**, **-b**, **-R**, or **-r** command line options.

.report

Comprehensive report file for the model. Automatically generated according to model settings.

.sta

The .sta directory is where AutoStat saves runs and experiment information.

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